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Master thesis

Energy-Guided Sampling for Controllable Text Generation: From Langevin Dynamics to Amortized Inference

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Abstract

Controllable text generation asks a language model to produce text satisfying a global property that cannot be checked until a sequence is complete. One prominent family pursues this without retraining: it treats the frozen likelihood of a pretrained autoregressive language model as an *energy function* over sequences and samples from it with gradient-guided Langevin dynamics, adding an arbitrary constraint at inference time. This thesis tests the premise that construction rests on, that the gradient of the frozen likelihood is a usable local search direction on discrete text.

The study covers five energy functions across 145 configurations on masked-token recovery on WikiText-2 validation, with a token-space and an embedding-space sampler. The central result is negative and consistent: the input-embedding gradient provided no reliable proposal advantage over a norm-matched random direction, exact recovery was 0.0% in 139 of the 145 configurations, and a 5×5 sweep over step size and proposal temperature, spanning proposal entropies from uniform to deterministic, finds no configuration at which the gradient helps.

Two measurements locate the failure. First, a *linearization failure*: the first-order surrogate by which the discrete sampler ranks candidates is uncorrelated with the true change in sequence energy ($|\rho| < 0.06$ on all five energies). Second, and decisively, the deficiency belongs to *which* derivative is taken. The one-hot input gradient, which is what the discrete samplers this family adapts actually use, correlates at $\rho = 0.60$ to 0.73 at admissible distances, and in the same sampler recovers 40% of tokens on GPT-2 and 41% on Llama-3, where the input-embedding gradient recovers at most 2%. The usable direction exists in closed form on the model’s output side; taking the input-embedding derivative is what discards it.

Amortizing the search with a GFlowNet did not escape the failure: sampling on the tuned energy was indistinguishable from sampling on the untuned base. A ladder of proposals inside one exact-energy chain locates what actually matters. A uniform draw recovers 0.5%, matching the gradient-guided sampler; the model’s own left-to-right conditional recovers 23.5%; a score-entropy diffusion model 39%; and a bidirectional masked language model, which is *not* score-trained, 44.5%. The operative variable is what a proposal may condition on, not the objective it was trained with. The practical implication is direct: on a frozen autoregressive model, search the energy with output-side quantities rather than differentiating it with respect to its input embeddings.

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1 Introduction

The dominant paradigm behind modern language models is **autoregressive generation**: a model factorizes the probability of a sequence into a product of next-token conditionals and produces text one token at a time, from left to right (Radford et al., 2019; Brown et al., 2020). This factorization is what makes these models trainable at scale and fast at inference, and it is the reason they work as well as they do. It is also, as this thesis argues, the reason one popular approach to controlling them does not.

That same factorization is the source of a practical limitation. Because generation proceeds strictly left to right and each token is fixed the moment it is emitted, the model has no mechanism for revising an earlier decision in light of a later one: if the second half of a sentence reveals that the first half has drifted in logic or tone, the model cannot go back, only continue or discard and restart. This is no obstacle when the goal is a fluent continuation, which is what these models are trained to produce, but it becomes one the moment we want to *control* what they produce.

The reason control is hard has a precise structure. Many of the properties we care about most, such as the overall sentiment of a passage, its topical focus, its adherence to a factual constraint, or the absence of toxic content, are **global properties** of a completed sequence rather than local properties of any single token. Whether a paragraph is positive in sentiment cannot be read off its first word; it is a property of the whole. A greedy, left-to-right decision about the next token cannot see such a property, for the simple reason that the object which would be evaluated against it, the finished text, does not yet exist at the moment the decision is made. Enforcing a global constraint autoregressively therefore requires something like generating many complete sequences and keeping the ones that happen to satisfy the constraint, which is wasteful when the constraint is easy to satisfy and effectively impossible when it is not. The problem of steering a language model toward such global objectives, while keeping its output fluent, is the problem of **controllable text generation**, and it is the subject of this thesis.

1.1 Established Routes to Control and Their Costs

There are, broadly, two established ways of imposing control on a language model, and each carries a cost that motivates the search for a third.

The first is to bake the desired behaviour into the parameters of the model itself, through supervised fine-tuning on curated data or, more commonly today, through reinforcement learning from human feedback (Ouyang et al., 2022). This approach is effective and now standard; the aligned assistants in wide use are its product. But the control it provides is *static* in a specific sense: a model tuned to be helpful and non-toxic embodies exactly those constraints and no others, and introducing a genuinely new constraint at inference time, a new attribute the user cares about today but did not anticipate at training time, requires collecting new preference data and running a new and expensive training procedure. The control is fixed at training time, and adapting it is not cheap.

The second is to leave the model untouched and instead filter or re-rank its outputs, generating many independent candidates and discarding those that violate the constraint. This is conceptually simple and it preserves the model, but it inherits precisely the intractability described above: for a constraint that a fluent generation satisfies only rarely by chance, the number of candidates one must draw to find a satisfying one grows without practical bound.

What both approaches lack, and what would be genuinely valuable, is the ability to introduce a fresh constraint at the moment of generation, cheaply, without touching the underlying model and without generating a haystack to find a needle. This is the capability that the energy-based approach promises, and it is the promise that this thesis sets out to examine.

1.2 Energy-Based Control and Its Central Assumption

An elegant alternative reframes decoding itself. Rather than generating a sequence token by token, one can define a probability distribution over the space of all possible sequences and *sample* from it directly. Following the standard formulation of

energy-based models (LeCun et al., 2006; Hinton, 2002), the distribution is written in the Boltzmann form

$$(1) \quad p(\mathbf{x}) = \frac{1}{Z} \exp(-E(\mathbf{x})),$$

where $E(\mathbf{x})$ is the **energy** of a sequence $\mathbf{x}$, a scalar that is low for desirable sequences and high for undesirable ones, and Z is a normalizing constant. The constant Z is intractable to compute, because it sums over every possible sequence, but the great advantage of the sampling framework is that it need never be computed: the methods used to draw samples from (1) depend only on *ratios* of probabilities or on gradients of the log-probability, in both of which the constant cancels.

The attraction of this view for controllable generation is that the energy can be assembled additively from independent pieces. One writes

$$(2) \quad E(\mathbf{x}) = -\log p_{\text{LM}}(\mathbf{x}) + \lambda C(\mathbf{x}),$$

where the first term is the negative log-likelihood of a frozen, pretrained language model and supplies *fluency*, pulling samples toward text the model considers natural, and the second term $C(\mathbf{x})$ is an arbitrary differentiable *constraint*, weighted by a coefficient λ that trades off fluency against constraint satisfaction. Because the constraint is simply added on, it can be specified at the moment of generation and swapped freely. A practitioner who wants positive sentiment today writes C as the negative log-probability of the positive class under a sentiment classifier; one who wants factual consistency tomorrow writes a different C ; and in both cases the language model is untouched. This is the promise of plug-and-play, inference-time controllable generation, and it is an appealing one. A substantial and growing body of recent work pursues it, through gradient-guided decoders such as PPLM (Dathathri et al., 2020), FUDGE (Yang and Klein, 2021), and GeDi (Krause et al., 2021), and, most directly related to this thesis, through sampling-based methods such as COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022), which draw samples from the energy in (2) using continuous relaxations and dynamics derived from Langevin sampling.

Every method in this family rests on one assumption, which is worth isolating. To sample from the energy in (2) using a gradient-guided method, two things must

be true. First, one must be able to compute a gradient of $-\log p_{\text{LM}}(\mathbf{x})$ that points, at least on average, in the direction of better sequences. Second, one must be able to *follow* that gradient across the space of possible texts, moving from a worse sequence to a better one by steps the gradient recommends. Taken together, these amount to the presupposition that the frozen likelihood of an autoregressive language model, viewed as a function over whole sequences, defines a *landscape that a sampler can actually navigate*: a surface with meaningful slopes, whose downhill direction can be computed locally and trusted to lead somewhere good. This presupposition is intuitive, and it is almost never tested directly in the literature that relies on it. Testing it, and explaining what is found, is the subject of this thesis. In the settings tested here that presupposition does not hold: following the input-embedding gradient proves no more effective than following a norm-matched random direction, the finding stated precisely in Section 5.5. This outcome is stated here, before the research questions are posed, because the thesis is organized as an explanation of that result rather than as a reveal.

1.3 From Sampling to Amortization

The natural response to a gradient that will not cooperate is to stop using the gradient. There is a version of the energy-based idea that does not differentiate the energy at all. If a per-sequence energy can still be *evaluated*, that is, if one can take a finished sequence and compute its score, even when one cannot usefully *differentiate* that score to decide how to improve a sequence, then one can treat the language model as a black box that ranks finished sequences and learn a separate policy that generates in proportion to that ranking. This is the idea of **amortized inference**: rather than paying the cost of a search procedure afresh for every sample, one pays a one-time training cost to learn a generator that produces good samples directly.

The most principled recent realization of this idea for sequence generation is the **Generative Flow Network**, or GFlowNet (Bengio et al., 2021; 2023), which trains a generative policy so that the probability of producing a complete sequence is proportional to its reward. Applied to language models, GFlowNets have been

proposed as a way to sample diverse, high-reward text without the mode-collapse that afflicts reward-maximizing reinforcement learning (Hu et al., 2024). The second half of this thesis asks a specific question about this approach: does amortizing the search, and thereby never touching the gradient, escape the difficulties that defeat the gradient-guided samplers? The answer, established in Section 5, is that it does not. Because the initial sampling experiments failed, the study developed into a diagnostic investigation of the underlying mechanisms.

1.4 Research Questions and Contributions

The work is organized around four research questions, which the results section answers in turn and the discussion section revisits explicitly.

RQ1. Can the frozen likelihood of an autoregressive language model be sampled effectively with theoretically faithful Langevin dynamics on the discrete text manifold, and if not, why not?

RQ2. What is the role of the Metropolis–Hastings correction in each of the discrete and continuous samplers, and does making the samplers theoretically correct also make them work?

RQ3. Does amortizing the energy with a GFlowNet, which never differentiates the reward, escape the failures observed for the gradient-guided samplers?

RQ4. Can an additive constraint term steer generation toward a target attribute on this energy landscape, as the plug-and-play framework requires?

These four questions are the operationalized forms of the questions posed in the thesis proposal, refined once the study became diagnostic rather than constructive. The proposal’s RQ1 and RQ1a, on whether faithful Langevin dynamics can sample the frozen likelihood and how the discrete and continuous samplers compare, are consolidated into RQ1 above. The proposal’s RQ1b, on the role of the Metropolis–Hastings correction, becomes RQ2. The proposal’s RQ3a, on amortizing

the energy with a GFlowNet, becomes RQ3. The proposal’s RQ2a, on whether an additive constraint can steer generation, becomes RQ4. The proposal’s RQ2b, which asked whether energy-based recovery improves on ordinary autoregressive decoding, is answered by the gradient-free baselines of Section 5.5.2.

In answering these questions the thesis makes five contributions.

1. A controlled ablation that separates the direction of the *input-embedding* gradient from its magnitude, run across five energy functions, establishing that in the tested token-substitution settings that gradient’s direction gave no reliable advantage over a norm-matched random one, together with a sweep over step size and proposal temperature showing that the null is not an artifact of a proposal too flat to express the gradient.
2. A measurement of the *linearization failure* behind that result. The region over which the sampler’s first-order approximation holds is smaller than the smallest admissible token substitution, so the input-embedding gradient is inapplicable rather than merely imprecise.
3. An identification of what is actually discarded. The one-hot gradient of the same frozen likelihood contains the candidate’s own conditional log-probability, which the input-embedding derivative cannot see; re-analysing the identical measurements with that surrogate recovers a strong rank correlation with the true energy change where the embedding surrogate has none. The usable direction lives on the model’s output side.
4. A decomposition of the Metropolis–Hastings acceptance ratio in continuous space that attributes its collapse to the proposal term rather than the target term, together with a measurement of the *likelihood trap* on the study’s own models, in which the lowest-energy text is the most degenerate.
5. A documented taxonomy of three failure modes of GFlowNet fine-tuning on a frozen-likelihood reward, and a unifying experiment showing that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base.

6. A diffusion positive control, run rather than proposed, in which a score-trained model on the same tokenizer supplies the local direction the autoregressive gradient lacks and performs the recovery the gradient-guided sampler could not.

1.5 Structure of the Thesis

Section 2 develops the technical background: the energy-based view of generation, Langevin dynamics, the Metropolis–Hastings correction, the two samplers studied here, and GFlowNets. Section 3 reviews the literature on controllable and energy-based text generation and states the gap this work addresses. Section 4 describes the task, the five energy functions, the sampler configurations, the evaluation metrics, and the diagnostic experiments. Section 5 reports the sampling results, the diagnostics that explain them, the GFlowNet experiments, the constrained-generation extension, and the diffusion positive control. Section 6 answers each research question, synthesizes the mechanism, and states the limitations and implications. Section 7 concludes.

2 Background

This section develops the technical background the thesis relies on: the energy-based view of sequence generation and the way a frozen language model is repurposed as an energy function, Langevin dynamics as a gradient-guided sampling method, the Metropolis–Hastings correction that makes such samplers asymptotically exact, the two concrete samplers studied in the experiments, score matching, and GFlowNets as an amortized alternative. The aim throughout is to expose the assumption each tool makes rather than to survey the tools exhaustively.

2.1 Autoregressive Language Models as Energy Functions

An autoregressive language model defines a probability distribution over a sequence of tokens $\mathbf{x} = (x_1, \dots, x_T)$ by the chain rule of probability,

$$(3) \quad \log p_{\text{LM}}(\mathbf{x}) = \sum_{t=1}^T \log p_{\text{LM}}(x_t \mid x_{<t}),$$

where each conditional $p_{\text{LM}}(x_t \mid x_{<t})$ is produced by a neural network, in the models studied here a Transformer (Vaswani et al., 2017), applied to the prefix $x_{<t}$. The network reads the prefix, produces a vector of logits over the vocabulary, and a softmax turns those logits into the conditional distribution for the next token. The model is trained by **teacher forcing**: on a large corpus, with the true prefix supplied at every position, it is asked to maximize the log-likelihood in (3), which decomposes into a sum of independent next-token prediction problems, one per position.

It is essential for everything that follows to be precise about what this training objective does and does not shape, because the entire diagnosis of this thesis turns on the difference. The objective shapes each conditional $p_{\text{LM}}(x_t \mid x_{<t})$ to be an accurate predictor of the next token given a *correct* history drawn from the data distribution. That is what it optimizes, and modern models optimize it extraordinarily well. What the objective never does is ask the joint quantity $\log p_{\text{LM}}(\mathbf{x})$ to behave in any particular way as a function of $\mathbf{x}$ when $\mathbf{x}$ is varied away from the data, for instance by substituting one token for another in the middle of an otherwise fixed sequence. The model is never trained on such counterfactual sequences, never asked to assign them calibrated scores, and never penalized for the shape of the surface it induces over them. There is, in other words, no reason internal to the training objective for the joint log-likelihood to be a smooth function of the sequence, or a well-calibrated measure of sequence quality, or a function whose local gradient points toward better sequences. Section 6 returns to this distinction between a well-trained local conditional and a well-behaved global energy, once the evidence is in hand.

With that caveat noted rather than resolved, the energy-based reformulation is immediate. Identifying the Boltzmann form of (1) with the language model, one

simply sets

$$(4) \quad E_{\text{LM}}(\mathbf{x}) = -\log p_{\text{LM}}(\mathbf{x}),$$

so that sequences of high probability are sequences of low energy and vice versa. Sampling text from the language model is then, formally, sampling from the Boltzmann distribution of this energy, and the two descriptions are interchangeable. For controllable generation, an additive constraint term $\lambda C(\mathbf{x})$ is appended as in (2), where C is typically the negative log-probability that an attribute classifier assigns to the desired label, so that low energy now means both fluent and on-attribute. The task of generation becomes the task of drawing low-energy samples from the combined energy, and the central computational question, to which the next two sections are devoted, is how one draws such samples at all.

2.2 Langevin Dynamics

Suppose, for a moment, that the state $\mathbf{s}$ lived in a continuous space $\mathbb{R}^D$ and that the energy were differentiable everywhere. Then one could sample from the distribution $p(\mathbf{s}) \propto \exp(-E(\mathbf{s}))$ using **Langevin dynamics**, a gradient-guided Markov chain Monte Carlo method with a long history in statistics and physics (Roberts and Tweedie, 1996; Welling and Teh, 2011). The discretized update, which is what one actually runs, is

$$(5) \quad \mathbf{s}_{t+1} = \mathbf{s}_t + \frac{\epsilon_t}{2} \nabla_{\mathbf{s}} \log p(\mathbf{s}_t) + \sqrt{\epsilon_t} \boldsymbol{\xi}_t, \quad \boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where ϵ_t is a step size at iteration t . The update has two parts, and their interplay is what distinguishes sampling from optimization.

The first part, the **drift**, is a step of size $\epsilon_t/2$ in the direction of the gradient of the log-density. Taken on its own, iterating the drift alone is gradient ascent on $\log p$, and it drives the state monotonically toward the nearest mode of the distribution, where it comes to rest. If one wanted to *find* the single most probable point, the drift alone would suffice. But finding the mode is not the goal; the goal is to draw a *representative sample*, and a distribution is more than its mode. This is where the

second part enters. The **diffusion** term injects fresh Gaussian noise at every step, scaled by $\sqrt{\epsilon_t}$, and this noise is what prevents the chain from simply collapsing onto the mode. It knocks the state off the deterministic ascent path, allowing it to climb away from a mode, cross into a neighbouring region, and in the limit visit every part of the distribution with the correct frequency.

For intuition, the drift acts like gravity pulling a marble downhill on a landscape whose valleys are the high-probability regions, while the diffusion vibrates the surface so the marble does not settle in the first valley it finds. **Annealing** the step size ϵ_t toward zero on a suitable schedule makes the distribution of $\mathbf{s}_t$ converge to the target $p(\mathbf{s})$ rather than to its mode (Welling and Teh, 2011), but this fair-draw reading holds only under the technical conditions of that result, in particular a suitable schedule and a well-behaved landscape, and does not follow in general simply because the vibration was switched off.

The guarantees behind the method, however, rest on regularity conditions that are easy to pass over and that will matter a great deal later. The discretization error of (5) grows with the step size, so the finite-step chain is only an approximate sampler and its bias increases with ϵ_t (Roberts and Tweedie, 1996). The Metropolis-adjusted variant of the next section removes that bias in principle, but its acceptance rate degrades sharply once the drift loses smoothness, because the correction depends on the reverse move being predictable from the forward one. Non-asymptotic convergence guarantees for Langevin samplers are established under smoothness of the target, and typically under log-concavity or a related condition (Dalalyan, 2017; Durmus and Moulines, 2017), and in particular require the drift to be **Lipschitz continuous** (Roberts and Tweedie, 1996), so that a small step lands where the gradient at the starting point predicted. The continuous sampler of Section 2.4 violates these assumptions: the drift it uses is computed from a differentiable pathway whose prediction target changes discontinuously at each Voronoi cell boundary, so the drift is not Lipschitz across a boundary and neither the smoothness nor the log-concavity the guarantees assume holds there. Section 5.3 measures the consequence.

2.3 The Metropolis–Hastings Correction

The update in (5) is a discretization of a continuous-time process, and any finite step size introduces a bias: the chain does not sample exactly from the intended distribution $p(\mathbf{s})$ but from a perturbed distribution that depends on the step size. For many purposes this bias is tolerated. If one wants asymptotic correctness, however, it can be removed exactly by the **Metropolis–Hastings** (MH) correction (Metropolis et al., 1953; Hastings, 1970), which turns a biased proposal into an exact sampler by accepting or rejecting each proposed move. Having proposed a move from the current state $\mathbf{s}$ to a new state $\mathbf{s}'$ according to a proposal density $q(\mathbf{s}' | \mathbf{s})$, one accepts the move with probability

$$(6) \quad \alpha(\mathbf{s}' | \mathbf{s}) = \min \left\{ 1, \frac{p(\mathbf{s}') q(\mathbf{s} | \mathbf{s}')}{p(\mathbf{s}) q(\mathbf{s}' | \mathbf{s})} \right\},$$

and otherwise rejects it and remains at $\mathbf{s}$, counting the current state again. When Langevin dynamics is equipped with this correction the method is known as the Metropolis-adjusted Langevin algorithm, or MALA.

The acceptance ratio factors into two parts. The **target ratio** $p(\mathbf{s}')/p(\mathbf{s})$ compares the probability of the proposed state to that of the current state under the distribution being sampled; it is greater than one when the move is toward higher probability, and it is the part of the ratio that expresses the sampler’s preference for good states. The **proposal ratio** $q(\mathbf{s} | \mathbf{s}')/q(\mathbf{s}' | \mathbf{s})$ compares how likely the proposal mechanism was to suggest the reverse move against the forward move; it is what enforces **detailed balance**, the condition that guarantees the chain leaves p invariant and therefore samples from it. Intuitively, the proposal ratio penalizes moves that are easy to make but hard to undo, because such moves, if accepted freely, would let the chain drift in a preferred direction and distort the distribution.

Two facts about the correction deserve emphasis, because both recur in the results. First, a great many energy-based decoders in the literature omit the correction entirely. Without it, the noise-augmented gradient update of (5) is not a sampler at all but a biased optimizer with a particular kind of stochasticity, and calling its output a sample from the energy is, strictly, incorrect. The distinction is not merely

pedantic: a biased optimizer stopped early and a correct sampler run to convergence can produce very different distributions of outputs, and much of the apparent success of correction-free methods is, on inspection, the success of optimization under early stopping rather than of sampling. Second, and more subtly, the proposal ratio is delicate to compute correctly for a Langevin proposal. The reverse proposal density $q(\mathbf{s} \mid \mathbf{s}')$ must be evaluated using the drift computed *at the proposed point $\mathbf{s}'$* , not at the starting point, because the reverse move would start from $\mathbf{s}'$. If the drift at $\mathbf{s}'$ differs wildly from the drift at $\mathbf{s}$, which is what happens when the two points fall on opposite sides of a discontinuity in the drift, then the reverse proposal can place the original state $\mathbf{s}$ deep in the tail of a Gaussian centred far away, driving $q(\mathbf{s} \mid \mathbf{s}')$ toward zero and with it the entire acceptance probability. Section 5.3 decomposes the acceptance ratio into its two terms and reports which of them is responsible.

2.4 Sampling in a Discrete Space: DLS and CLS

The difficulty that motivates the entire thesis is that text is not continuous. A sequence is a list of discrete tokens drawn from a finite vocabulary V , and there is no state *between* two tokens: one cannot be halfway between the word *cat* and the word *dog*. Langevin dynamics, which assumes a continuous differentiable space in which infinitesimal steps are meaningful, cannot be applied to such a state without modification. Two strategies for adapting it are studied in this thesis, and they differ precisely in where they confront the discreteness, which is why studying both, rather than either alone, is what allows the results to be attributed to the energy rather than to one sampler’s idiosyncrasies.

The **Discrete Langevin Sampler** (DLS), following the discrete gradient-based samplers of Grathwohl et al. (2021) and Zhang et al. (2022), never leaves token space at all. Its state is always a genuine sequence of tokens. It uses the gradient of the log-likelihood, taken with respect to the continuous input embedding of a masked position, only as a *score* for constructing a categorical proposal distribution over which token to place at that position next. Concretely, let $\mathbf{e}(x_i)$ denote the embedding of the current token at a masked position i , and let $\mathbf{g} = \nabla_{\mathbf{e}_i} \log p_{\text{LM}}(\mathbf{x})$

denote the gradient of the sequence log-likelihood with respect to that embedding. For a candidate token v with embedding $\mathbf{e}(v)$, the proposal assigns a logit that combines a term penalizing the embedding displacement $\|\mathbf{e}(v) - \mathbf{e}(x_i)\|$, which keeps proposals local, with a term proportional to $\frac{1}{2}\mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$, which biases the proposal toward tokens the gradient suggests will lower the energy. This second term is a **first-order Taylor surrogate**: it approximates the true change in log-likelihood that would result from substituting v for x_i by the inner product of the gradient with the embedding displacement, which is exactly the first-order term in the Taylor expansion of the log-likelihood around the current embedding.

Written out in full, the discrete sampler adapts the continuous Langevin proposal of (5) to a categorical distribution over the vocabulary by the construction of Zhang et al. (2022). In the continuous case, a Gaussian proposal centred at $\mathbf{s} + \frac{\alpha}{2}\nabla \log p(\mathbf{s})$ assigns to a candidate $\mathbf{s}'$ a log-density proportional to $-\frac{1}{2\alpha} \|\mathbf{s}' - \mathbf{s} - \frac{\alpha}{2}\nabla \log p(\mathbf{s})\|^2$. Expanding this and discarding terms constant in $\mathbf{s}'$ leaves a term linear in the displacement, $\frac{1}{2}\nabla \log p(\mathbf{s})^\top(\mathbf{s}' - \mathbf{s})$, plus a quadratic penalty on the displacement itself. Transplanting this to the discrete setting, where the candidate states are token embeddings $\mathbf{e}(v)$ and the current state is $\mathbf{e}(x_i)$, gives a categorical proposal whose logit for token v is

$$(7) \quad \log q(v \mid x_i) \propto \frac{1}{2}\mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i)) - \frac{1}{2\alpha} \|\mathbf{e}(v) - \mathbf{e}(x_i)\|^2,$$

where $\mathbf{g} = \nabla_{\mathbf{e}_i} \log p_{\text{LM}}(\mathbf{x})$. The first term, the **gradient-alignment term**, is the surrogate: it rewards candidates whose displacement from the current embedding aligns with the gradient, and it is precisely the first-order Taylor estimate of how much the log-likelihood would change if the substitution were made, since to first order $\log p(\mathbf{x}_{i \rightarrow v}) - \log p(\mathbf{x}) \approx \mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$. The second term, the **distance term**, keeps the proposal local, penalizing distant candidates.² Everything the sampler knows about which token is good, as opposed to merely near, is contained in that gradient-alignment term, and if it is uninformative the proposal reduces to a

²The two terms of (7) come from expanding the Gaussian log-density $-\frac{1}{2\alpha} \|\mathbf{e}(v) - \mathbf{e}(x_i) - \frac{\alpha}{2}\mathbf{g}\|^2$, which yields the gradient-alignment term $\frac{1}{2}\mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$ and the distance term $-\frac{1}{2\alpha} \|\mathbf{e}(v) - \mathbf{e}(x_i)\|^2$, together with a term $-\frac{\alpha}{8} \|\mathbf{g}\|^2$ that is constant in the candidate v and is therefore discarded.

distance-penalized random choice. The surrogate is therefore the sampler’s entire claim to be gradient-guided rather than random, and Section 5.7 tests it directly.

The **Continuous Langevin Sampler** (CLS), following the continuous-relaxation approach of methods such as MuCoLa (Kumar et al., 2022), takes the opposite route. It relaxes the state out of token space and into the continuous embedding space $\mathbb{R}^D$, where it runs genuine Langevin dynamics according to (5), and it projects the continuous state back onto the nearest token embedding whenever a discrete sequence is required, for instance to evaluate the energy. The target density it navigates is therefore defined *through* a **nearest-neighbour projection** into the vocabulary, and this construction has a consequential geometric structure. The embedding space is partitioned into **Voronoi cells**, one per token, where a cell is the region of space closer to that token’s embedding than to any other. Within a cell the projected token is constant, so the projected energy, which depends on the sequence only through its projected tokens, is also constant: the cell is a flat plateau. Across a cell boundary the projected token changes abruptly and the projected energy jumps. The target is thus not the smooth surface the Langevin theory assumes but a collection of flat plateaus separated by cliffs.

Writing $\text{proj}_V(\mathbf{s})$ for the nearest-neighbour map that sends the continuous state at each masked position to the token whose embedding is closest in Euclidean distance, that target energy is

$$(8) \quad E_{\text{CLS}}(\mathbf{s}) = -\log p_{\text{LM}}(\text{proj}_V(\mathbf{s})),$$

the negative log-likelihood of the discrete sequence obtained by projecting the state back to tokens, to which the additive constraint λC of (2) is appended when a constraint is imposed.

Two objects must be kept apart here, and the rest of the thesis depends on the distinction. The energy of (8) is piecewise constant: it possesses no classical gradient, its derivative being zero in the interior of every cell and undefined on the boundaries, so no sampler can differentiate it and none does. What the implemented sampler differentiates is a distinct, *differentiable pathway*: the log-likelihood evaluated with the continuous state supplied as the input embedding at the masked positions, while

the projected token $\text{proj}_V(\mathbf{s})$ supplies the prediction target. Inside a single cell the target token is fixed, so this pathway is a smooth function of $\mathbf{s}$ and its gradient is well defined; it is also, for the same reason, nearly but not exactly flat there, unlike the projected energy, which is exactly flat. Crossing a cell boundary changes the prediction target, so the pathway itself jumps, and the drift computed from it jumps with it. Every claim in this thesis about a discontinuous or non-Lipschitz drift attaches to this differentiable pathway *across cell boundaries*, not to a classical gradient of the piecewise-constant energy of (8), which does not exist. It is this arrangement, a pathway that is smooth within a cell and discontinuous across one, sampling a target that is flat within a cell and jumps across one, that puts the correction of Section 2.3 under strain.

The arrangement has a direct consequence for the acceptance probability, derived once here. Consider a proposed move that stays within a single Voronoi cell, so that $\text{proj}_V(\mathbf{s}') = \text{proj}_V(\mathbf{s})$. The projected sequence, and therefore the energy of (8), is unchanged, so the target ratio $p(\mathbf{s}')/p(\mathbf{s})$ is at or near unity and the acceptance probability of (6) is governed almost entirely by the proposal ratio $q(\mathbf{s} \mid \mathbf{s}')/q(\mathbf{s}' \mid \mathbf{s})$. For the discrete sampler a within-cell move is the proposal of the same token, which is its own reverse and for which the forward and backward proposal probabilities coincide, so the proposal ratio is one and the move is accepted with certainty. For the continuous sampler a within-cell move is still a continuous displacement in embedding space, and its reverse proposal density is evaluated from the drift at the proposed point, which even inside the same cell need not make the return move likely, so the proposal ratio is not one and within-cell acceptance is not protected. Section 5.3 reports the measured acceptance rates on both sides of this split.

2.5 Score Matching and the Score Function

One further notion is needed before the experiments: the **score function**. For a differentiable density $p(\mathbf{s})$, the score is the gradient of its log-density, $\nabla_{\mathbf{s}} \log p(\mathbf{s})$, and it is exactly the quantity the Langevin drift of (5) requires. A model that provides an accurate score at every point can be sampled by Langevin dynamics; a

model that does not cannot, however the sampling is arranged. The question running through this thesis can therefore be restated compactly: does a frozen autoregressive language model provide a usable score over sequences?

Some models are trained specifically to provide one, and the theory connecting them to Langevin sampling is well established. **Score matching** (Vincent, 2011) is a training objective that fits a model directly to the score of the data distribution, and the observation behind modern score-based and diffusion generative models (Song and Ermon, 2019; Ho et al., 2020) is that learning to denoise a corrupted input is equivalent to learning the score. A diffusion model, trained to reverse a gradual noising process, therefore learns by construction the gradient field that Langevin dynamics needs. This makes such a model a natural instrument for a controlled comparison, since it differs from an autoregressive model in the training objective while sharing the sampling machinery, and Section 5.14 uses it that way.

2.6 Discrete Diffusion and the Concrete Score

The score of the previous subsection is a gradient, and text is discrete, so the construction does not transfer unchanged: there is no gradient of a log-density with respect to a token index. The discrete analogue is the **concrete score** (Meng et al., 2022), which replaces the gradient by the collection of ratios $p(\mathbf{y})/p(\mathbf{x})$ between a sequence $\mathbf{x}$ and each neighbour $\mathbf{y}$ that differs from it in a single position. These ratios carry the information the gradient carries in the continuous case: for every single-token move they state how much more or less probable the data distribution makes it, which is precisely the local directional signal a sampler needs to choose a substitution.

Discrete diffusion models are trained to supply those ratios. Their noising process corrupts a sequence through a Markov chain over the vocabulary; in the **absorbing** variant (Austin et al., 2021) each token is independently replaced, with a probability that grows along the schedule, by a distinguished absorbing (mask) state, and the model is trained to reverse the process by predicting the clean token at each masked position conditioned on the surviving context on both sides. That reverse

step is a bidirectional denoising query, so an absorbing diffusion model, unlike an autoregressive one, is trained to answer exactly the question the revision task poses: given a sequence with one position held out, which token belongs there.

Score-entropy discrete diffusion (SEDD) (Lou et al., 2024), the model used for the positive control of Section 5.14, learns the concrete score directly, fitting the ratios of the data distribution over the discrete state space with a loss adapted to non-negative ratio targets. What matters here is the consequence: SEDD provides, in a single forward pass, a trained estimate of the per-token probability ratios, that is, a usable local direction over discrete sequences at exactly the positions a substitution could act on. This is the property the results of this thesis find the frozen autoregressive likelihood to lack, and it is what makes a score-trained model on the same tokenizer the natural instrument for isolating the training objective, rather than the sampler or the energy, as the cause of that lack.

2.7 Amortized Inference with GFlowNets

Both samplers above require the gradient of the energy: the discrete sampler to score its proposals, the continuous sampler to compute its drift. An entirely different strategy is to give up on the gradient and instead *learn* to sample. A **Generative Flow Network** (Bengio et al., 2021; 2023) treats generation as a sequential process of constructing an object one piece at a time, and it trains a policy so that the probability of terminating at a complete object is proportional to a non-negative reward $R(\mathbf{x})$ assigned to that object. The name carries a useful metaphor: a unit of flow, like water, enters at the empty state and is split among the branches at each step by the policy, which is trained so that the flow arriving at each complete object matches its reward, and generation releases a unit of flow and follows where it goes. Because the policy distributes flow in proportion to reward rather than concentrating it all on the single highest-reward object, the objects it produces are *diverse*, and this is the property that distinguishes a GFlowNet from a reward-maximizing reinforcement-learning agent, which would learn to produce only the one best object and would therefore be useless as a sampler.

For text, the object under construction is a sequence and the construction proceeds from left to right, one token at a time, so the space of partial constructions is a tree: there is exactly one path from the empty state to any given complete sequence. This tree structure simplifies the training objectives, which enforce a consistency condition on the flow along trajectories; the objectives used in practice include trajectory balance (Malkin et al., 2022) and its sub-trajectory variants (Madan et al., 2023), which improve the assignment of credit along long generative trajectories and are what the reference implementation this thesis builds on employs. The single feature of the whole approach that matters most for the present investigation is this: the reward $R(\mathbf{x})$ is only ever *evaluated*, never *differentiated*. The policy learns from scalar scores of complete sequences, so the discontinuity and non-smoothness of the underlying energy landscape are invisible to it. That property is what makes the GFlowNet useful here: if the difficulty with energy-guided sampling were purely a property of the gradient, a method that never uses the gradient ought to escape it. Section 5.11 reports whether it does. Following Hu et al. (2024), the GFlowNet is realized here as a parameter-efficient low-rank adapter (Hu et al., 2022) on the very same frozen base model that supplies the energy for the Langevin samplers, so that the amortized and gradient-guided halves of the study operate on one shared model and the comparison between them is not confounded by any difference in the underlying network.

3 Related Work

The work in this thesis sits at the intersection of three lines of research: controllable text generation, energy-based and sampling-based decoding, and the study of the geometry and behaviour of language-model representations. This section reviews each in turn. Rather than cataloguing methods, it tells the story of how the field arrived at energy-based sampling as an approach to control, because understanding that trajectory is what makes the unexamined assumption at its heart visible, and it is that assumption which the thesis tests. The section closes by stating precisely the gap in the literature that the work addresses.

3.1 Controllable Text Generation

Approaches to controllable generation can be organized by how much they change the underlying model, and reviewing them in that order reveals a progression toward ever lighter-touch intervention, of which energy-based sampling is the logical endpoint.

At one extreme are the **fine-tuning methods**, which adjust the model’s parameters so that the desired behaviour is internalized. The earliest of these simply continued training on attribute-specific data, and the modern and dominant form is reinforcement learning from human feedback (Ouyang et al., 2022), which optimizes the model against a learned reward that encodes human preferences. These methods work, and the widely used aligned assistants are their fruit, but the control they produce is fixed once training is complete. A new attribute requires new data and a new training run, and the cost of that adaptation is precisely what motivates the search for methods that impose control at inference time. Conditioning methods such as CTRL (Keskar et al., 2019) occupy a middle ground, prepending control codes to the input so that a single model can be steered among a fixed set of attributes anticipated during training, but they too cannot accommodate a genuinely novel constraint after the fact.

At the other extreme are the **decoding-time methods**, which leave the model’s parameters entirely untouched and intervene only during generation. These are the methods most relevant to this thesis, because they are the ones that promise the inference-time flexibility that fine-tuning cannot offer. Plug and Play Language Models (Dathathri et al., 2020) perturb the hidden states of a frozen model at each step in the direction indicated by an attribute classifier’s gradient, nudging the generation toward the desired attribute without altering any weight. FUDGE (Yang and Klein, 2021) and GeDi (Krause et al., 2021) take a complementary approach, reweighting the next-token distribution using a discriminator that predicts, from a prefix, whether the eventual completion will bear the desired attribute. These methods are ingenious and effective within their scope, but they share a structural feature that limits them: they remain autoregressive. They modify the local, per-token decision,

but they retain the strictly left-to-right generation order and therefore inherit its fundamental inability to revise an earlier token in light of a later one. A method that could edit any part of the sequence in light of the whole would be strictly more powerful, and it is the pursuit of that capability that leads to the sampling-based methods.

3.2 Energy-Based and Sampling-Based Decoding

The decisive conceptual move, and the one that defines the family of methods this thesis studies, is to abandon left-to-right generation entirely and to cast decoding as sampling from an energy defined over whole sequences. Once a sequence is treated as a single object drawn from a distribution, rather than as a stream of tokens produced in order, a global constraint can be imposed on the completed object directly, and any position in the sequence can be revised in light of any other. This is the promise that makes the approach attractive, and several influential systems realize it.

COLD decoding (Qin et al., 2022) performs energy-based constrained generation by running Langevin dynamics on a continuous relaxation of the sequence, combining a fluency energy derived from a language model with task-specific constraint energies, and it demonstrated the approach on tasks such as constrained generation and abductive reasoning. MuCoLa (Kumar et al., 2022) samples in the continuous embedding space and projects back to tokens, and it is the closest published antecedent of the continuous sampler studied in this thesis; its formulation of the sampling problem, and its use of a projection to return to discrete text, are exactly the ingredients whose geometric consequences Section 5 examines. The continuous sampler of this thesis implements that COLD and MuCoLa mechanism faithfully, and the constrained *mucola* arm of the experiments is precisely this comparison, run at the level of the shared energy rather than at the level of the original papers’ tasks and metrics, which use different benchmarks and are out of scope here. Related efforts include gradient-based inference for lexically constrained generation and infilling (Sha, 2020), and a body of work on discrete samplers adapted from the energy-based-model literature, including the gradient-informed discrete proposals of

Grathwohl et al. (2021) and the discrete Langevin proposals of Zhang et al. (2022) on which the discrete sampler of this thesis is directly based. A third strategy for confronting the discreteness, distinct from both the continuous relaxation of the state and the gradient-informed discrete proposal, relaxes the categorical choice itself with a straight-through estimator (Bengio et al., 2013) or the Gumbel-softmax reparameterization (Jang et al., 2017), passing gradients through a differentiable surrogate of the discrete sample. It is out of scope here, because it alters the model’s forward computation rather than leaving the frozen likelihood untouched, and this thesis studies sampling from that frozen likelihood without a differentiable relaxation of the categorical.

Reading across this literature, a feature recurs that is rarely brought to the foreground, and it is the feature this thesis makes central. The Metropolis–Hastings correction, which as Section 2.3 explained is what separates genuine sampling from biased optimization, is frequently omitted. Methods are described in the language of sampling from an energy while in practice running a noise-augmented gradient descent, and where the correction is included, the difficulty of computing its reverse-proposal term on a landscape defined by projection is seldom examined. This omission is understandable, because including the correction faithfully often makes these methods stop working, but the reason it makes them stop working is itself informative, and recovering that reason is one of the contributions of this thesis. Where these methods report success without an exact correction, it typically comes from running the update as an early-stopped biased optimizer and filtering the resulting samples afterward, rather than from sampling the energy correctly. Taking the correction seriously, implementing it faithfully for both a discrete and a continuous sampler, and measuring exactly what its inclusion reveals is a lens through which the failure of the whole approach comes into focus.

Two further lines bear directly on the target these methods sample from. The machine-translation literature established, more sharply than anywhere else, that the mode of a well-trained sequence model is not where the quality is. Stahlberg and Byrne (2019) showed that for a large fraction of inputs the globally most probable translation under a neural model is the empty string, and Eikema and Aziz (2020)

argued from this that maximum-a-posteriori decoding is the wrong criterion and that the model’s useful information lies in the bulk of the distribution rather than at its peak. Meister et al. (2023) give a complementary account, that high-probability continuations are dispreferred by human readers because they carry atypically little information. These results are the published form of the likelihood trap this thesis measures in Section 5.8, and they matter here because a method that succeeds at concentrating mass on the lowest-energy sequences of a frozen likelihood succeeds at reaching exactly the region these works identify as degenerate. Deng et al. (2020) take the opposite route of learning a residual energy on top of a language model rather than treating the likelihood itself as the energy, which is one of the few constructive responses to the same problem.

A distinct branch of the same literature deserves to be isolated here, because it anticipates the diagnosis this thesis reaches. Not every method that samples from a frozen-language-model energy uses the *gradient* of that energy. Mix-and-Match (Miresghallah et al., 2022) performs controllable generation by Metropolis–Hastings over a product-of-experts energy, proposing token substitutions from an auxiliary masked language model and scoring them with the frozen energy, so that the energy is only ever evaluated, never differentiated. The twisted sequential Monte Carlo methods of Lew et al. (2023) and Zhao et al. (2024) likewise sample from an intractable posterior defined by a frozen model using importance weights and resampling rather than energy gradients. That these gradient-free samplers operate successfully on essentially the same frozen-LM energies that defeat the gradient-guided methods is, in retrospect, a strong hint about where the difficulty lies, and this thesis makes the hint concrete: a gradient-free Gibbs sampler run on exactly the energy the Langevin samplers navigate, reported in Section 5.5.2, recovers coherent text on the same benchmark. The interpretation of that separation, between evaluating the energy and differentiating it, is developed where the evidence for it is presented, in Sections 5.5.2 and 6.3. This branch has a direct antecedent that the present work’s most successful baseline reproduces: Wang and Cho (2019) sample text from a frozen masked language model by Gibbs sampling over token positions, without any gradient, and Miao et al. (2019) perform constrained sentence genera-

tion by Metropolis–Hastings over word-level insert, delete and replace proposals on a frozen model. The gradient-free Gibbs sampler of Section 5.5.2, which matches the best Langevin configuration on the same energy while never differentiating it, is the direct descendant of these two, and their existence is part of why the negative result of this thesis is scoped to the gradient rather than to the training-free premise.

3.3 Amortized Sampling and GFlowNets

Generative Flow Networks were introduced by Bengio et al. (2021) as a method for sampling discrete, compositional objects in proportion to a reward, with the explicit motivation of producing diverse high-reward samples rather than the single mode that reward-maximizing reinforcement learning tends to collapse onto. The framework was subsequently formalized and given a theoretical foundation (Bengio et al., 2023), and a family of training objectives was developed to address the practical difficulty of assigning credit along long generative trajectories, including trajectory balance (Malkin et al., 2022) and its partial-episode sub-trajectory refinement (Madan et al., 2023). The application most directly relevant here is that of Hu et al. (2024), who framed a range of language tasks, including infilling and reasoning, as sampling from an intractable posterior defined by a frozen language model, and trained a GFlowNet to perform that sampling, reporting that the resulting policies yield diverse samples that transfer better to downstream use than those obtained by reward maximization.

The GFlowNet component of this thesis builds directly on that formulation and on the accompanying codebase, using the same modified sub-trajectory objective and the same parameter-efficient adaptation of a frozen base model (Hu et al., 2022). The question posed here, however, is narrower and more critical than the one those works set out to answer. Their concern is whether amortized sampling is useful, and they show that it is, on tasks where the reward defines a well-behaved target. The concern here is whether amortized sampling *escapes a specific pathology*, namely that of the frozen autoregressive likelihood used as an energy over free-form sequences, which the first half of this thesis shows defeats the gradient-guided

samplers. These questions are compatible, and Section 5 takes up the sharper one by direct experiment, with the interpretation of why amortization inherits rather than escapes the problem developed in Section 6.1.

3.4 The Geometry and Behaviour of Language-Model Representations

A complementary line of work concerns the representations in which these samplers operate and the behaviour of the likelihood they optimize, and it supplies part of the explanation for the results. Two phenomena from this literature are directly relevant.

The first is **anisotropy**. A well-documented property of the embeddings produced by Transformer language models, both contextual and static, is that they do not spread uniformly through the representation space but instead occupy a narrow cone, so that two randomly chosen token embeddings have high average cosine similarity rather than being roughly orthogonal as one might naively expect (Gao et al., 2019; Ethayarajh, 2019). This has a direct consequence for any method that treats Euclidean or cosine distance in embedding space as a meaningful measure of similarity, because anisotropy compresses the dynamic range of those distances: the difference between a good and a bad token, measured in the embedding metric, is smaller than the geometry would suggest. This thesis measures the anisotropy of both models it studies and shows that it renders Euclidean distance an unreliable evaluation metric, which is one reason the experiments adopt a divergence-based measure of contextual fit instead, and it shows in addition that the anisotropy scale differs enough between models to make a single step-size schedule non-transferable, a consequence the prior work did not draw.

The second is the failure of high-probability text to be high-quality text. Holtzman et al. (2020) demonstrated that the most probable sequences under a neural language model are frequently degenerate, repetitive, and unnatural, and that maximization-based decoding such as beam search produces markedly worse text

than sampling from the same model. This observation is central to the interpretation of the present results, because it means that any procedure which succeeds in concentrating probability mass on the lowest-energy regions of the frozen-likelihood energy is thereby concentrating it on degenerate text. This thesis reproduces the phenomenon quantitatively on its own models, terms it the *likelihood trap* in the context of energy-based sampling, and uses it to argue that the objective the samplers pursue is itself undesirable, quite apart from whether they can pursue it effectively.

3.5 Diffusion Models for Text and the Score Function

A final body of work is directly relevant not to the methods this thesis evaluates but to the positive control it proposes, and so it is reviewed here to prepare the discussion of Section 6. Diffusion models, which generate by gradually reversing a noising process, have become the dominant approach to image generation and have been adapted to text in two broad families. The first embeds tokens into a continuous space and runs a continuous diffusion there, as in Diffusion-LM (Li et al., 2022), which demonstrated controllable generation by applying classifier guidance to the continuous latents at each denoising step. The second operates directly on discrete tokens, as in the score-entropy discrete diffusion of Lou et al. (2024), which learns the ratios of the data distribution over a discrete state space and has narrowed much of the quality gap between diffusion and autoregressive language models.

What makes these models pertinent to this thesis is not their generation quality but their training objective. The foundational result connecting them to the samplers studied here is that of Vincent (2011): learning to denoise a corrupted input is mathematically equivalent to learning the score of the data distribution, the gradient of its log-density, and it is this equivalence that gives score-based generative modeling its name and its mechanism (Song and Ermon, 2019; Ho et al., 2020). A diffusion model, unlike an autoregressive one, is therefore trained precisely to provide the gradient field that Langevin dynamics requires, at every noise level and across the whole space rather than only on the data manifold. This is exactly the property that the results of this thesis find the autoregressive likelihood to lack, which is why,

as Section 6 develops, a diffusion language model is the natural instrument for a positive control: it holds the sampling method fixed and changes only the training objective that shapes the energy, and so it can isolate that objective as the cause of the failure this thesis diagnoses.

3.6 The Gap Addressed by This Thesis

The literature reviewed above establishes energy-based sampling as an attractive and actively pursued route to controllable generation, and it supplies both the samplers this thesis studies and the amortized alternative it examines. What the literature does not supply is a direct, controlled test of the assumption on which the entire enterprise rests, namely that the gradient of a frozen autoregressive likelihood is a usable search direction on the discrete text manifold, together with a mechanistic account of what happens when that assumption fails. Negative observations, where they appear, tend to be reported in passing, as brittleness, as sensitivity to hyperparameters, or as a need for careful tuning, rather than isolated as a phenomenon and explained. This thesis addresses that gap directly. It designs an ablation that cleanly separates the direction of the gradient from its magnitude, and thereby tests the assumption rather than a proxy for it; it constructs diagnostic experiments that identify the geometric, statistical, and analytic mechanisms behind the result; and it shows that the same mechanism survives amortization, which locates the cause in the training objective of the language model rather than in any particular sampling algorithm. The result is not merely the observation that these methods fail, which is implicit in the difficulty the field has had with them, but an explanation of why, supported at every step by measurement.

4 Methodology

This section describes the experimental platform in the detail needed to reproduce the study, and it does so in a particular order: it first establishes the task and the models, then the samplers and the metrics, and only then the diagnostic experiments,

because the diagnostics were designed after the main experiments and in response to what those experiments revealed. Where a choice was made, the reasoning behind it is given, because in a diagnostic study the design of each measurement is part of the argument. The section sets out the masked-recovery task and why it was chosen, the five energy functions, the two samplers and the factorial space of configurations, the evaluation metrics and the justification for each, the GFlowNet training setup, the constrained-generation extension, and finally the five diagnostic experiments, each introduced together with the question it was built to answer.

4.1 Task: Masked Token Recovery

A sampler can only be studied if there is a way to tell whether it is being guided toward good sequences or is merely wandering. An open-ended generation task provides no such handle, because there is no reference against which to judge the output: a fluent but arbitrary continuation is neither clearly a success nor clearly a failure of the sampler, and one cannot separate the sampler’s contribution from the difficulty of the underlying generation. The study therefore adopts **masked token recovery**, also called infilling, as its diagnostic bench. A sequence is drawn from the corpus, one or more of its tokens are replaced with random tokens, and the sampler is asked to restore a coherent sequence by resampling only the corrupted positions while the rest of the sequence is held fixed. Because the original sequence is known, the recovery can be scored; because the corruption and the surrounding context are controlled, the behaviour of the sampler can be attributed to the method rather than to the open-endedness of the task. This is the standard reason infilling is used as a diagnostic for controllable generation, and it is the reason it is used here.

An important subtlety shapes the choice of evaluation metric and is worth stating at the outset. The primary metric, defined in Section 4.4, compares the next-token distribution that the recovered fill induces against the distribution induced by the ground-truth fill, so it measures how well the recovered token fits its context relative to a reference, and a coherent synonym of the original token, such as *dorm* for *roommate*, scores well even though it does not match. This is a measure of relative-

to-reference contextual fit, not of absolute sequence likelihood used as a proxy for quality. It is therefore not in tension with the likelihood-trap finding of Section 5.8, which concerns the absolute sequence likelihood as a maximization objective in open-ended generation, a different quantity serving a different purpose. The full justification for preferring this metric over exact-match accuracy is given once, with the metric definition, in Section 4.4.

Two corpora are used, and it matters which experiment ran on which. The masked-recovery sequences of the main sampling grid are drawn from the **WikiText-2** validation set (Merity et al., 2017), a standard held-out language-modelling corpus of encyclopedic prose, chosen because its held-out status keeps the recovery task independent of any fine-tuning data and its sentences give a masked position a locally checkable correct recovery. The base GPT-2 Large is instead supervised fine-tuned on **ROCStories** (Mostafazadeh et al., 2016), a collection of short five-sentence everyday narratives and the corpus of the amortized-inference reference implementation of Hu et al. (2024), so the GFlowNet half of the study trains and generates on ROCStories and operates on the same data distribution as the work it builds on. The main sampling grid therefore evaluates recovery *out of domain* relative to the fine-tuning corpus, while the diagnostic experiments are run in domain on ROCStories. Table 1 maps each experiment family to its corpus.

Experiment family	Corpus
Masked-recovery grid (145 configurations)	WikiText-2 validation
Gradient-free baselines, continuation, external judge	WikiText-2 validation
Constrained generation	WikiText-2 validation
Diffusion positive control (SEDD)	WikiText-2 validation
GPT-2 Large fine-tuning; GFlowNet training and variants	ROCStories
Diagnostics: linearization, likelihood trap, anisotropy, trajectory	ROCStories

Table 1: Corpus used by each experiment family. The base model is fine-tuned on ROCStories; the main sampling grid recovers out-of-domain WikiText-2 sequences, while the diagnostics run in domain on ROCStories.

The domain shift is not innocent, so it is addressed directly. If the gradient null were an artifact of evaluating out-of-domain text it should weaken or vanish in domain, and it does not: the in-domain ROCStories diagnostics of Section 5.7 and Section 5.8 reproduce both the gradient null, with the linearization correlation below 0.06 throughout, and the likelihood trap. The central findings are therefore properties of the frozen likelihood rather than of the domain shift.

4.2 Energy Functions

The study evaluates five energy functions, chosen to let the central findings be tested for robustness across training regimes and across architectures.

The reference energy is a **GPT-2 Large** model (Radford et al., 2019), of 774 million parameters and embedding dimension $D = 1280$, supervised fine-tuned on ROCStories. This same fine-tuned model serves as the base for the amortized experiments, so that the Langevin and GFlowNet halves of the study share weights and are not confounded by any difference in the underlying network, an identity that is essential to the unifying experiment of Section 5.11. Three **GFlowNet-tuned variants** of this base model are included, produced as described in Section 4.5: two trained with the unnormalized-length reward setting, at 500 and 2000 optimization steps respectively, so that the effect of longer training can be observed, and one trained with the length-normalized setting at 500 steps. These tuned variants are included to test whether amortization repairs the energy landscape: the GFlowNet objective demonstrably moves the energy, as Section 5.11 quantifies, and running the same samplers on the tuned models asks whether that movement made the landscape navigable by gradient or merely relocated it. Finally, a **Llama-3 8B** model (Dubey et al., 2024), with embedding dimension $D = 4096$, serves as a cross-architecture control. Because Llama-3 uses a different tokenizer and a markedly different embedding geometry, and therefore a different step-size scale, it is never placed on a shared numerical axis with the GPT-2 family; it is used to establish that the qualitative findings, such as the gradient null of Section 5.5 and the Metropolis–Hastings breakdown, hold across architectures rather than being artifacts of one model. The four

GPT-2-based models, by contrast, share the tokenizer, the corruption seeds, and the step-size schedule, and their numbers are therefore directly comparable without qualification.

4.3 Samplers and Configurations

The two samplers of Section 2.4, the Discrete Langevin Sampler (DLS) and the Continuous Langevin Sampler (CLS), are each run under a factorial combination of design choices. The two are studied together, rather than either alone, because they represent the two opposite strategies for confronting the discreteness of text and therefore isolate different failure modes: the DLS, following Grathwohl et al. (2021) and Zhang et al. (2022), keeps the state in token space and uses the gradient only to build a proposal, which tests whether the gradient is an informative ranking signal; the CLS, following Kumar et al. (2022), relaxes into continuous space and runs genuine Langevin dynamics, which tests whether continuous Markov chain Monte Carlo is compatible with a projected discrete energy. A finding that holds for both is a finding about the energy rather than about one sampler’s idiosyncrasies, which is why both are needed.

The continuous sampler’s update as implemented deserves to be stated precisely, because Section 5.4 quantifies how far its state drifts from the token manifold and the update is what bounds that drift. At step t with step size ϵ_t , the sampler first forms an interim continuous state by the Langevin drift, $\mathbf{s}_{\text{interim}} = \mathbf{s}_t + \frac{\epsilon_t}{2} \nabla_{\mathbf{s}} \log p$, then projects it to the nearest token embedding, $\text{proj}_V(\mathbf{s}_{\text{interim}})$, and centres the proposal not at the interim state but at the interpolated mean of the two,

$$(9) \quad \mathbf{s}_{t+1} \sim \mathcal{N}\left(\frac{1}{2}(\mathbf{s}_{\text{interim}} + \text{proj}_V(\mathbf{s}_{\text{interim}})), \epsilon_t \mathbf{I}\right),$$

after which the Metropolis–Hastings correction of Section 2.3 accepts or rejects it, with the reverse proposal density evaluated from the drift at the proposed point. Averaging the interim state with its projection pulls the proposal halfway back toward the token manifold at every step, which bounds the off-manifold drift a pure continuous update would accumulate. This interpolation is an implementation

choice adapted from the continuous-relaxation samplers of Kumar et al. (2022), not a property of Langevin dynamics itself; it is load-bearing for the off-manifold distances of Section 5.4, since even with this pull toward the manifold the state settles far from any token.

One asymmetry in how the correction was applied across the compared arms has to be declared, because it falls on exactly the contrast this thesis rests on. In the main grid the reverse-proposal term of Equation (6) is computed exactly for the *policy* arm and set to zero for the two random-direction arms, on the reasoning that a random-direction proposal is a symmetric random walk whose proposal ratio cancels. That reasoning is not sound. For the continuous sampler the proposal mean of Equation (9) carries both a drift and a nearest-neighbour projection, neither of which is symmetric, whatever the direction; for the discrete sampler the distance term is symmetric but the softmax normalizers at the current and proposed tokens are not. The consequence is that in every correction-enabled comparison the policy arm is charged a reversibility penalty from which its comparators are exempted, and Section 5.3 measures that penalty reaching -1325 nats in the continuous case. This works against the policy arm, so it cannot manufacture the null reported in Section 5.5, but it could inflate the cases in which the gradient appears *worse* than a random direction.

The samplers therefore carry a corrected mode, enabled by `--mh_exact_all_arms`, in which the reverse term is computed exactly for every arm. Because a random direction is drawn independently of the current state, holding that draw fixed across the forward and reverse evaluation of a single step yields a kernel q_g that is exactly computable and π -invariant, and a mixture of π -invariant kernels is itself π -invariant, so the corrected mode is a valid sampler rather than an approximation. Section 5.6 reports the arms rerun under it. The archived grid is left as it was run, and the legacy treatment remains the default, so that every number reported from it remains reproducible.

The design axes are as follows. The **proposal method** is one of three. The *policy* method uses the true gradient of the energy. The *grad-norm-preserved random* method replaces the gradient with a random vector rescaled to the exact norm of

the true gradient, so that it differs from the policy only in direction. The *random* method uses a random vector of random magnitude. The comparison among these three is the central ablation of the thesis, isolating respectively the full gradient, the effect of its direction alone, and a pure baseline, and the reasoning behind this three-way design is developed where it is used, in Section 5.5. The **Metropolis–Hastings correction** is either enabled or disabled, so that the effect of theoretical correctness can be observed directly. **Gradient normalization**, which rescales the gradient to unit norm for numerical stability, is either enabled or disabled; this choice interacts with the proposal-method ablation in a way that turns out to be critical and is explained in Section 5.5. The **annealing schedule** runs for either 50 or 100 steps, with the step size decreasing linearly; for the GPT-2 family the schedule runs from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$, a scale established by the calibration described in Section 5.1. An *oracle* variant, in which the ideal step size at each iteration is selected using knowledge of the ground-truth token, is included as an upper-bound reference; its purpose is to rule out the possibility that the samplers fail merely for want of a well-tuned schedule, since if even an oracle-tuned schedule does not let the gradient outperform a random direction, the failure cannot be one of calibration. The oracle sweep is run independently for each proposal method, with a separate per-method oracle run rather than a single schedule shared across methods, so the schedule selection cannot advantage the policy gradient over the random comparators. In total, across the five energy functions, the study comprises 145 sampling configurations, each evaluated on 200 sequences; the arithmetic of that count is given in Appendix A.2.

4.4 Evaluation Metrics

Three quantities are recorded, and the choice among them is itself informative.

The first is the **Euclidean distance** in embedding space between the recovered token and the ground-truth token. This is the most obvious measure of recovery, and one might expect it to be the natural criterion. It proves to be unreliable, for a reason established quantitatively in Section 5.9: because language-model embeddings

are anisotropic, occupying a narrow cone in the representation space (Ethayarajh, 2019), Euclidean distance compresses the distinction between good and bad tokens, so that a token can be geometrically close to the target while being grammatically inadmissible, and conversely a perfectly good synonym can lie far away. It is reported for completeness, and studied as an object in its own right, but it is not relied upon as the primary criterion.

The primary criterion is the **KL divergence** (Kullback and Leibler, 1951) between the model’s predictive distribution at the recovered position and a reference distribution, which measures how well the recovered token fits its context, that is, its *contextual coherence*. Concretely, let $\mathcal{M}$ be the set of masked positions in a sequence of length L , and let $\mathcal{M}' = \{m \in \mathcal{M} : m < L - 1\}$ be those masked positions that have a right neighbour, the only ones at which the metric is defined. For a position $m \in \mathcal{M}'$, write $p_{\text{ref}}^{(m)}$ for the model’s next-token distribution at m when every masked position holds its *ground-truth* token, and $p_{\text{pred}}^{(m)}$ for the same distribution when the masked positions instead hold the *recovered* fill. The metric is the mean divergence over these positions,

$$(10) \quad \overline{\text{KL}} = \frac{1}{|\mathcal{M}'|} \sum_{m \in \mathcal{M}'} \text{KL}\left(p_{\text{ref}}^{(m)} \parallel p_{\text{pred}}^{(m)}\right),$$

which is exactly the quantity logged by the sampler in `core/base_sampler.py` and recomputed in the revision diagnostics. By construction the ground-truth fill gives $\overline{\text{KL}} = 0$, since then $p_{\text{pred}}^{(m)} = p_{\text{ref}}^{(m)}$ at every position, so the metric has a hard floor of zero that a perfect recovery attains. This metric is chosen over exact-match accuracy for the reason given in Section 4.1: a recovered token that differs from the reference may still be perfectly coherent, and a metric that penalized it would confound the sampler’s quality with the arbitrariness of the reference. A divergence between predictive distributions measures contextual fit directly and is insensitive to which particular admissible token was chosen. One limitation of the divergence measure must be acknowledged, because it shapes how the results are argued: its magnitude depends on the position of the corrupted token within the sequence, since corrupting an early token perturbs the conditional probability of a long suffix and yields a large divergence, whereas corrupting a final token yields almost none. For

this reason no model is ranked on small differences in KL divergence; the central results are argued from the *absence* of a gap between conditions rather than from a small measured gap, and a null gap is robust to this positional dependence in a way that a narrow measured win would not be. To make “absence of a gap” a decision rather than an impression, the analysis fixes an equivalence margin in advance, at five percent of the policy method’s mean KL for each configuration, and treats two methods as practically equivalent when their paired mean difference falls inside that margin. The margin is calibrated to the pipeline’s own noise rather than chosen to fit the result: as Section 4.8 reports, the across-seed standard deviation of the final KL for the flagship configuration is 0.183, comfortably inside the corresponding margin of about 0.327, so the threshold below which a difference is deemed negligible is smaller than the study’s stated ceiling on it yet still larger than a single standard deviation of seed noise.

Finally, for the generation-quality analyses, the study records **repetition rate**, the fraction of repeated n -grams, and **distinct- n** . These are the standard instruments for detecting the degeneration studied by Holtzman et al. (2020), and they are therefore the appropriate measures for the likelihood trap. For the constrained-generation extension the study records classifier accuracy and **BERTScore** (Zhang et al., 2020), the latter chosen because it measures semantic similarity through contextual embeddings rather than surface overlap, and so does not penalize a valid paraphrase of the reference.

4.5 GFlowNet Training

The GFlowNet-tuned variants are produced following the amortized-inference formulation of Hu et al. (2024), using their modified sub-trajectory-balance objective (Malkin et al., 2022; Madan et al., 2023). The GFlowNet is chosen as the amortized alternative, over reward-maximizing reinforcement learning, for a reason specific to the question of this thesis: a reward-maximizing agent would converge on the single highest-reward sequence and so could not distinguish a sampler that explores the energy from one that merely finds its mode, whereas a GFlowNet is trained to

sample in proportion to reward and therefore tests amortized *sampling* rather than amortized optimization. It only ever evaluates the reward and never differentiates it, so the gradient behaviour that defeats the Langevin samplers is invisible to it, which is what makes it a suitable instrument for asking whether that behaviour can be sidestepped.

The policy is a low-rank adapter (Hu et al., 2022) on the frozen GPT-2 Large base, and the reward is the log-likelihood assigned by that frozen base to the completed sequence, so that the GFlowNet is trained to sample in proportion to the model’s own likelihood, exactly the energy the Langevin samplers navigate. The task is a story-infilling formulation in which, given a preceding context and a known ending, the policy generates a connecting span, and the reward scores the concatenation of context, generated span, and ending under the frozen model. A single hyperparameter, denoted `len_beta`, controls the length normalization of the reward: at `len_beta = 0` the reward is the raw, unnormalized log-likelihood, and at `len_beta = 1` it is fully normalized by sequence length. The consequences of this single knob, which turn out to be dramatic and to produce two opposite degeneracies, are one of the central findings of the GFlowNet study and are developed in Section 5.11.

Using the frozen log-likelihood as the reward may seem to contradict the thesis’s own argument that this likelihood is a poor objective, so the choice is worth defending. First, that the gradient of the likelihood is uninformative does not make the likelihood value-uninformative: the gradient-free baselines of Section 5.5.2 recover coherent text by evaluating exactly this quantity, so the reward carries usable signal even where its gradient does not. Second, the likelihood trap concerns the behaviour of the likelihood at its extreme, the degenerate text that globally maximizes it, not its ranking of fluent candidates near the data manifold, which is the regime the reward operates in during training. Third, and decisively, training a policy to sample in proportion to the base model’s own likelihood is the standard amortized-inference formulation of Hu et al. (2024), and its premise, that this likelihood defines a target worth sampling from, is exactly the premise this thesis sets out to test; using the same reward is therefore the condition under which the test is meaningful, not an

oversight.

One difference in task formulation must be made explicit, because it governs how the GFlowNet and the samplers can be compared. The GFlowNet reformulates in-filling as left-to-right generation of the blank span under a restructured prompt that places the context and the known ending around it, whereas the Langevin samplers perform in-place substitution at the masked positions with bidirectional context. The two therefore solve differently shaped problems, and the comparison between them is run at the level of the energy, the final KL divergence under the same frozen base model, rather than at the level of the task. The left-to-right reformulation is not adopted for the samplers themselves, because generating the span autoregressively would dissolve the very revision capability under study, the ability to edit a token in light of both sides, and so would change the question rather than repair the metric.

4.6 Constrained Generation Extension

To test the plug-and-play claim of (2) directly, which is the object of RQ4, a constrained-generation extension augments the energy with a sentiment constraint, in the manner of the gradient-based constrained samplers of Kumar et al. (2022) and Qin et al. (2022). A sentiment classification head is trained on the representations of the GPT-2 Large base and used to construct the constraint term $C(\mathbf{x})$. Generation is then run under five constraint modes designed to isolate the contribution of the constraint gradient. The *lm_only* mode uses only the fluency energy, providing a baseline with no steering. The *cons_only* mode uses only the constraint, showing what the constraint can do in isolation. The *full* mode combines them with weights $\beta_{\text{LM}} = 0.8$ and $\beta_{\text{C}} = 0.2$, which is the intended plug-and-play configuration. The *random* mode replaces the combined gradient with noise, and the *cons_random* mode replaces only the constraint gradient with noise while retaining the fluency gradient, so that the specific contribution of the constraint direction can be separated from that of the fluency direction. The steering effect is measured as the change, in percentage points, in the fraction of $n = 150$ generations per arm classified as the target sentiment relative to the unconstrained baseline. Both the discrete sampler in the

formulation of this thesis and a MuCoLa-style continuous baseline are evaluated, on both infilling and continuation, so that the finding does not depend on a single sampler or task. The analysis plan is the paired point-estimate contrast *cons_only* minus *cons_random*, which cancels the shared sentiment drift; because the result files record only aggregate target-hit fractions and not per-generation labels, per-sample paired gains and their confidence intervals are not available and none are reported.

4.7 Diagnostic Experiments

The configurations above establish *that* the methods fail. They do not, on their own, establish *why*, and the why is the contribution. Five additional diagnostic experiments were therefore designed, after the main results were in hand and in direct response to them, each targeting one candidate mechanism. Three further controls were added later still, in response to the questions the first round of results raised: a sweep over step size and proposal temperature (Section 5.6), which establishes whether the null survives at configurations where the gradient term actually shapes the proposal; a substitution of the one-hot surrogate into the same sampler (Section 5.7.1); and a bidirectional masked language model in the same exact-energy chain (Section 5.15), which separates conditioning direction from training objective in the diffusion comparison. Unlike the configurations above, these do not sample; they measure properties of the energy landscape and of the gradient directly, and each is introduced here together with the question that motivated its construction.

The **linearization experiment** was built to answer the question raised by the gradient-versus-random result: if the gradient carries no usable information, is that because the first-order surrogate the discrete proposal relies on fails to predict the true change in energy? For each of 200 sequences and a masked position, the exact gradient at that position is computed, and for a stratified set of 2000 candidate tokens the surrogate predicted change $\hat{\Delta}(v) = \mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$ is compared against the true change $\Delta(v) = \log p(\mathbf{x}_{i \rightarrow v}) - \log p(\mathbf{x})$ obtained by an exact forward pass, with the embedding distance of each candidate recorded so that the correlation can

be examined as a function of distance. The true change is further decomposed into a *self* term, the change in the log-probability of the candidate given its own left context, and a *future* term, the change in the log-probability of the suffix, because the gradient with respect to the input embedding is structurally able to register only the latter, and measuring both quantifies how much signal the gradient is blind to.

The **acceptance experiment** was built to answer the question raised by the Metropolis–Hastings breakdown: when the correction rejects nearly every useful move on the continuous sampler, is the rejection driven by the target term, meaning the moves are genuinely bad, or by the proposal term, meaning the moves are good but the correction cannot verify their reversibility? For every proposal it logs whether the proposal crossed a Voronoi cell boundary, whether it was accepted, and the decomposition of the acceptance ratio into its target and proposal terms, which lets the cause be located unambiguously.

The **trajectory experiment** logs the raw state of the sampler at every step for a small number of sequences, so that the path can be projected into two dimensions and inspected, and it records the distance from the continuous state to the nearest token embedding, which quantifies how far the sampler drifts from the token manifold over the course of a run. The **likelihood-trap experiment** asks whether low energy is a good place to be at all. It decodes with several strategies, from greedy and beam search to ancestral sampling, and measures the relationship between the per-token likelihood of the generated text and its repetition rate. A final **anisotropy analysis** measures the distribution of inter-token distances and related geometric quantities in the two embedding spaces, underpinning both the step-size result and the unreliability of Euclidean distance.

The scale of the configuration grid, one hundred and forty-five sampling configurations each evaluated on two hundred sequences, together with the separate diagnostic runs across five models, required an experimental infrastructure that could distribute work reliably across several machines and recover cleanly from interruptions. The runs are organized as a manifest of jobs, each annotated with its memory requirement, and dispatched by a resumable file-queue system that assigns jobs to available devices according to that requirement, so that a large model and a small

one are not placed on the same device when their combined memory would exceed it. Completed jobs are marked so that an interrupted sweep resumes rather than restarts, which matters when a single sweep runs for many hours. All experiments use this system, and all sampling code was verified against a reference implementation with a bitwise-equivalence test suite before the runs reported here were launched, so that the diagnostic instrumentation is known not to have altered the behaviour of the samplers it measures. One limitation of the design is worth recording, because it was encountered rather than anticipated. A job is claimed by creating a lock directory and the lock is released only on clean completion or on an explicit failure, so a worker killed abruptly, by an out-of-memory kill on its host for instance, strands its claims: the queue then appears busy while nothing runs. Recovering from this requires distinguishing a stalled job from a slow one, which the current design cannot do; a periodic heartbeat on the lock, treating a lock older than some interval as reclaimable, would remove the failure mode. The blanket reset utility is not a substitute, since it would also release jobs still running on other hosts. A second limitation is one of throughput rather than correctness: the grid runner dispatches whole configurations, so a single configuration is pinned to a single device and a large model runs at the speed of one accelerator no matter how many are idle. Splitting a configuration’s sequences across devices and recombining them afterwards is straightforward, since the sample set can be resolved before any model is loaded and then partitioned, and doing so leaves per-sequence results unchanged; the diffusion experiments of Section 5.14 were run that way, and the autoregressive grid was not. The Llama-3 and GFlowNet families would have completed several times faster had they been.

4.8 Reproducibility, Seeds, and Hardware

Because the central claims of this thesis are claims about the *absence* of an effect, the run-to-run stability of the pipeline is itself part of the evidence and is quantified directly rather than assumed. Rerunning the flagship configuration, the discrete sampler on GPT-2 Large with the Metropolis–Hastings correction and gradient nor-

malization enabled over a fifty-step schedule, under four independent corruption seeds, 1000 through 1003, yields final KL divergences of 6.348, 6.589, 6.150, and 6.291, with a mean of 6.345, a standard deviation of 0.183, and a range of 0.438. This across-seed standard deviation fixes the scale below which a difference in KL cannot be told apart from noise, and it is the empirical basis both for the equivalence margin defined in Section 4.4 and for the bounded-effect reading of the gradient-versus-random comparisons in Section 5.5. The corruption procedure is deterministic per sample index, with the seed set to the data seed plus the running index, so the corrupted inputs, and therefore the paired comparisons, are identical across the three method runs of a configuration and can be reproduced bit for bit; this is what allows the statistical analysis of Section 5.5 to pair the policy method against its comparators on the same sample without any rerun.

All reported experiments were run on a university GPU server equipped with 49 GB A6000-class accelerators, under PyTorch 2.12 with CUDA 13.0. For code and data availability, the sampler and diagnostic implementations, together with the bitwise-equivalence suite (`verify_equivalence_suite.py`) that certifies the instrumented code against the reference, and the manifests that define every configuration in the grid, are retained with the thesis so that the full set of runs can be regenerated from the seeds above. The accompanying `README.md` documents the environment, the exact rerun command for every experiment family, and an artifact map that links each table and figure in this thesis to the script that produces it and the result file it draws from, so that every quantitative claim is traceable from the number in the text to the file on disk.

5 Results

This section reports the experimental findings. It begins with the calibration of the samplers and the embedding geometry that governs it, then the annealing dynamics of the discrete sampler, then the Metropolis–Hastings correction in each of the two samplers. It next reports the central comparison between the gradient direction and a norm-matched random direction, followed by the diagnostics that account

for it: the linearization radius, the likelihood trap, and embedding anisotropy. It closes with the GFlowNet experiments, the constrained-generation extension, the last-token analysis, and the diffusion positive control.

5.1 Step-Size Calibration and Embedding Geometry

The study began by running the two samplers on GPT-2 Large with a step-size schedule already made to work on Llama-3. One term needs fixing first, because the section uses it repeatedly. A schedule *works* if it produces **calibrated motion**: the sampler’s projected tokens change over the run and its metrics evolve rather than freezing. That is a weaker property than **guided motion**, motion toward better sequences, which the later sections test separately.

On GPT-2 Large, with the Llama-tuned schedule, the samplers did not move at all. The entropy of the discrete proposal collapsed to zero within a few steps and no token was ever changed. The cause is the geometry of the embedding space. The anisotropy analysis, reported in Section 5.9, measures the mean distance between neighbouring token embeddings as approximately 1.82 in GPT-2 Large against approximately 0.59 in Llama-3, and the mean pairwise distance as 2.77 against 0.84, a difference of roughly a factor of three in each case. The reason for the trapped sampler follows immediately. A step size calibrated for the tighter geometry of Llama-3 is simply too small to carry the continuous state across a Voronoi cell boundary in the more dispersed geometry of GPT-2, so the projected token never changes and the sampler sits frozen in its initial cell. Recovering the correct scale required an oracle sweep over candidate step sizes, searching a grid of fifty values spaced logarithmically from 10^{-2} to 10^2 ; the schedule running from $\epsilon_{\text{start}} = 10.5$ to $\epsilon_{\text{end}} = 0.1$ for the GPT-2 family is the one that oracle calibration selected, and it is used throughout the rest of the study.

Two consequences follow. There is no universal step-size schedule for these methods: the correct scale is a property of the model’s embedding geometry and must be rediscovered for each model. The step sizes required to make the sampler move at all are much larger than the distances between neighbouring token embeddings,

so a method that needs this per-model recalibration before it will move is not plug-and-play in the strong, model-agnostic sense. And calibration establishes motion without establishing that the motion is informed. The oracle configuration, which selects the ideal step size at each iteration using knowledge of the ground-truth token, separates the two by establishing the best any schedule could do, so that a failure of calibration can be told apart from a failure of guidance. Section 5.5 makes that comparison. This section answers the first half of RQ1: theoretically faithful Langevin dynamics cannot be applied off the shelf, because the step size that makes the sampler move at all is model-specific.

5.2 Annealing Dynamics and the Near-Uniform Proposal

With the step size calibrated, the discrete sampler moves. What it moves *by* is the subject of this section, and it is the single most consequential measurement in the study, because it bounds what the ablation of Section 5.5 could possibly have detected.

The proposal of Equation (7) draws the next token from a softmax over the sum of a distance term and a gradient-alignment term, divided by a temperature. Its entropy therefore says directly how much either term is shaping the draw. Measured over the whole grid, the answer is that neither does. On the flagship configuration the mean proposal entropy is 10.8248 nats at the first step against a ceiling of $\log |V| = 10.8249$ for the uniform distribution over the 50,257-token vocabulary, and its *minimum* over the entire fifty-step schedule is 10.2760 nats, an effective support of roughly 29,000 tokens. Every free-schedule discrete configuration in the grid sits between 10.18 and 10.28 at its sharpest, and the oracle-schedule configurations sit at 10.8249, uniform to floating-point precision. The proposal is, at every step of every configuration, numerically indistinguishable from a uniform draw over the vocabulary. Table 2 decomposes why. The standard deviation across the vocabulary of the distance term rises from 0.049 to 5.57 as the step size anneals, but that of the gradient-alignment term is flat at 0.046 for the whole run, because gradient normalization fixes $\|\mathbf{g}\| = 1$ and the projections of a unit vector onto embedding differences are of that order.

Step	ϵ_k	Distance term t_1	Gradient term t_2	t_2/t_1	Proposal entropy
0	10.50	0.049	0.047	0.958	10.8248
10	8.42	0.064	0.044	0.692	10.8248
25	5.30	0.104	0.044	0.430	10.8247
40	2.19	0.274	0.046	0.168	10.8236
49	0.10	5.572	0.046	0.008	10.2488

Table 2: Decomposition of the discrete proposal on the flagship configuration. Columns three and four are standard deviations across the 50,257-token vocabulary of the two terms of Equation (7), before the division by the temperature of 5.0. The uniform-distribution entropy ceiling is $\log |V| = 10.8249$ nats. The gradient term is constant at 0.046 throughout and is never large enough to shape the draw.

After the division by the temperature of 5.0, the gradient term contributes about 0.009 nats of logit spread against a 10.82-nat entropy budget. It cannot influence the sampled token, and the ratio t_2/t_1 falls from 0.96 to 0.008 over the schedule, so by the end the proposal is a pure stay-near-the-current-token draw with the gradient contributing nothing.

This has an immediate consequence, and it is the reason the section is placed before the ablation rather than after it. A comparison between a true-gradient proposal and a norm-matched random proposal, when both are within 10^{-4} nats of uniform, is a comparison between two draws from the same distribution. A perfectly informative gradient would produce the same null at this configuration. The sampling ablation of Section 5.5 therefore cannot, on its own, establish that the gradient is uninformative; Section 5.6 tests the same contrast at configurations where the gradient term does shape the proposal, and Section 5.7, which measures the surrogate against the truth directly and is invariant to both the temperature and the step size, is what carries the argument.

What the sampler does instead is now easy to state. Without the correction it is a uniform random walk over the vocabulary, and it does not converge: the mean KL rises over the schedule from 8.765 to 9.499, the chain is still changing its token

on 99.1 percent of steps in the final tenth of the run, and the only late movement in any channel is a fall in proposal entropy from 10.78 to 10.28 at the last step, an end effect of the step size reaching its floor rather than a collapse into a local optimum. With the correction enabled the same uniform proposal becomes an independence sampler, and the accept/reject filter does all of the work: the chain reaches a final KL of 6.541 at a 9.98 percent acceptance rate, against 9.499 without it. Figure 1 shows the fifty-step schedule; the hundred-step run and the gradient-normalization-disabled companion, in which the three proposal arms separate rather than coincide, are Figures 13 and 14 in Appendix A.1. Reading the entropy panels together confirms that the late dip tracks the schedule, landing at step 49 of 50 and step 99 of 100; but it is a boundary effect of the annealing floor, it moves the entropy only from near-uniform to near-uniform, and it produces no improvement in contextual fit.

This measurement also explains the recovery numbers. Because the proposal never concentrates, the chain never proposes the ground-truth token with more than roughly uniform probability, and exact recovery is 0.0 percent in 139 of the 145 configurations and never exceeds 1 percent in the remaining six (Section 5.5). A fall in mean KL from 9.14 to 6.4 under these conditions is the accept/reject filter selecting among near-uniform proposals for tokens that happen to fit the context, not a search converging on the right one.

One reading of the late entropy dip should be set aside explicitly, because it is the natural one and it is wrong. The dip invites the interpretation that the annealed step size collapses the update into deterministic gradient descent and freezes the chain in a local optimum, a *quenching* effect on the metallurgical analogy. Direct measurement rules it out. At the final step the proposal still has an effective support of tens of thousands of tokens, so it is not deterministic; the chain changes its token on 99.1 percent of steps in the last tenth of the schedule, so it is not frozen; and the mean KL rises rather than falls, so nothing is converging. What the schedule produces is not a quench but a uniform random walk that never localizes, which is the more damaging of the two readings for the plug-and-play premise.

With the correction enabled the pattern does not occur. The chain converges within the first twenty steps and stays stable for the rest of the schedule, ending at

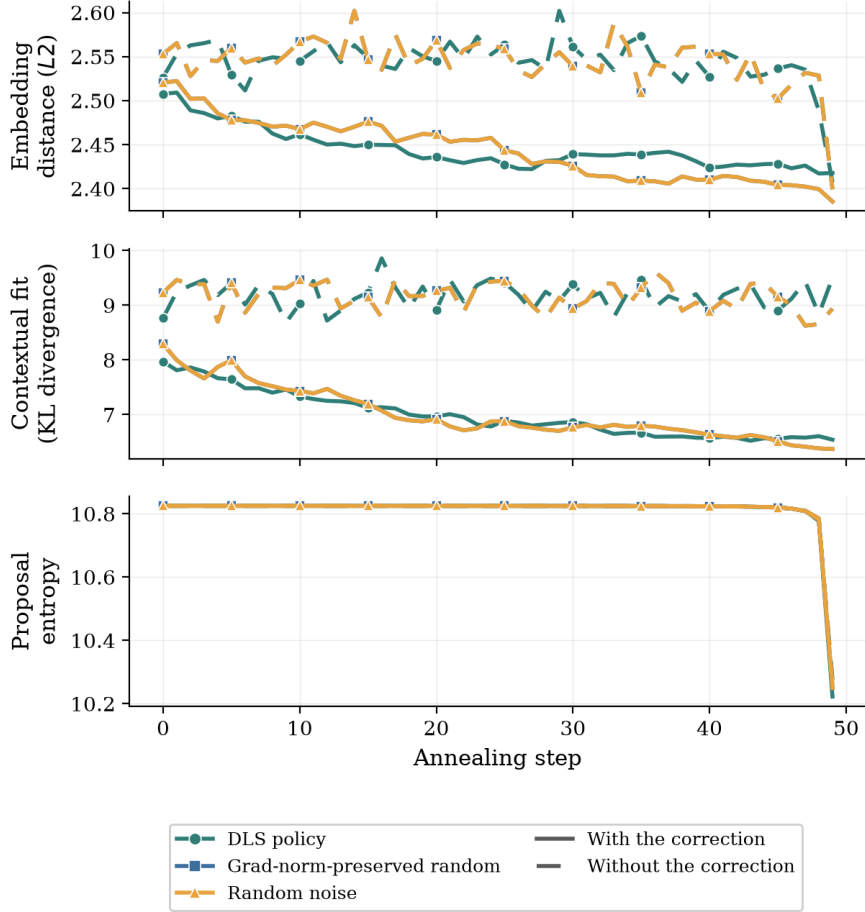


Figure 1: Discrete Langevin Sampler trajectories on GPT-2 Large over a 50-step annealing schedule, averaged across sequences. Panels, top to bottom: embedding distance, contextual fit measured by KL divergence, and proposal entropy. Solid lines are runs with the Metropolis-Hastings correction, long-dashed lines without it. Marker shape distinguishes the three proposal methods where their curves coincide.

a final KL of 6.541 against 9.499 without the correction. This is the one configuration in the study where making the sampler theoretically correct also makes it perform better.

5.3 The Metropolis–Hastings Breakdown in Continuous Space

In the continuous sampler the correction has the opposite effect. Decomposing the logged proposals by whether they crossed a Voronoi cell boundary, and therefore changed the projected token, the continuous policy sampler with the correction enabled accepts within-cell proposals, which change no token, only 0.03% to 0.63% of the time, according to whether gradient normalization is off or on, and boundary-crossing proposals only 3.7% to 8.6% of the time. Without the correction every proposal is accepted by construction, and it is this unchecked acceptance that lets the continuous state wander far off the token manifold, as Appendix A.4 reports. The discrete sampler under the same correction, in the policy configuration with the correction and gradient normalization enabled (`dls_policy_gn_mh`), accepts within-cell moves with probability 1.0, because such moves are exactly reversible, and boundary-crossing moves at 9.3%. That difference in within-cell acceptance, essentially one against essentially zero, is what separates the two cases: the discrete sampler’s token can move freely inside a cell and then cross to a neighbour, while the continuous sampler, unable to accept even a move within its own cell, barely leaves the cell it started in.

Whether the correction rejects boundary-crossing moves because the moves are bad, or because it cannot verify their reversibility, is settled by the two terms of the acceptance ratio. For proposals that crossed a boundary, the mean *target* log-ratio is +4.60, that is, positive: on average these moves improve the sequence probability, so the distribution being sampled would welcome them. The mean *proposal* log-ratio for the same moves is −1325. The reverse proposal assigns effectively zero density to the return move, and this term alone, dwarfing the favourable target term by three orders of magnitude, drives the acceptance probability to zero. Figure 2 shows the

Metropolis–Hastings acceptance ratio, target and proposal terms, by whether the proposal crossed a Voronoi cell boundary

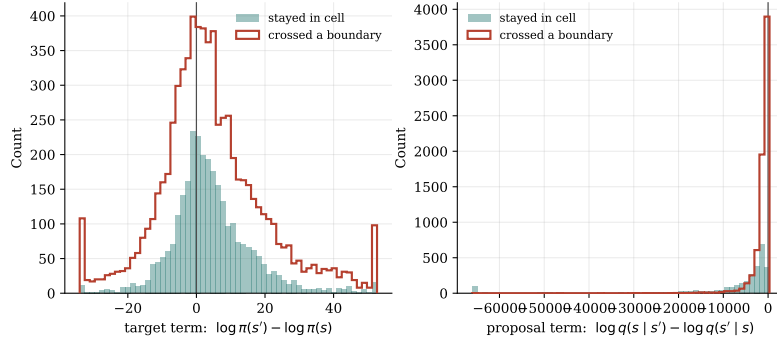


Figure 2: Decomposition of the Metropolis–Hastings acceptance ratio in the continuous sampler, for proposals that stay within a Voronoi cell against those that cross a boundary. Left: the target log-ratio. Right: the proposal log-ratio. Both on a log-probability scale.

two distributions side by side. The rejection is driven by the proposal term, not the target term.

This is the measured form of the argument of Section 2.3. The Metropolis-adjusted Langevin algorithm requires a Lipschitz-continuous drift (Roberts and Tweedie, 1996), and the drift derived from the differentiable pathway of Section 2.4 is not Lipschitz across a cell boundary. When a proposal crosses one, the reverse proposal mean is computed from the drift on the far side, which, because the pathway’s prediction target has changed there, points somewhere unrelated to the direction back toward the original state. The original state falls deep in the tail of the reverse Gaussian, the reverse density vanishes, and the correction rejects the move as irreversible. The correction is not malfunctioning; the landscape violates the assumption it depends on. This is the regime the convergence theory excludes: the non-asymptotic guarantees of Dalalyan (2017) and Durmus and Moulines (2017) assume a smooth target, and the near-total rejection of boundary-crossing moves measured here is the behaviour those results predict once that smoothness is lost at the projection boundaries. The consequence is that the continuous sampler appears to function only when the correction is omitted, that is, when it runs as a biased optimizer that ignores reversibility. The same behaviour is observed on both

GPT-2 Large and Llama-3. This answers RQ2 for the continuous sampler: making it theoretically correct does not make it work.

5.4 Sampler Trajectories in Embedding Space

The same breakdown can be read spatially. The trajectory experiment logs, for a small number of sequences, the full continuous state at every step, the distance from that state to the nearest token embedding, and the Voronoi cell the state occupies. Two summaries suffice here; the full walkthrough, the exact full-space distance figure, and the reduced-dimensionality projection are in Appendix A.4.

The first is the number of distinct Voronoi cells a run visits over its fifty steps, a direct count of how often the projected token changed. With the correction enabled the continuous sampler visits about two cells across the whole run, and exactly one in half the traced sequences. With the correction disabled it visits about 45 of a possible 50. The discrete sampler visits about five.

The second is the distance from the continuous state to the nearest token embedding, which measures drift off the manifold of actual tokens. With the correction on, the continuous state sits between 118 and 151 units from the nearest token across the whole run; with it off, while forcing its way across cell boundaries, the state reaches roughly 980 units and ends about 17 units away. Against a mean nearest-neighbour spacing of 1.82 and a mean pairwise spacing of 2.77, an off-manifold distance of 118 to 151 is roughly 65 to 83 times the spacing between neighbouring tokens. The discrete sampler sits at distance zero at every step, its state always being a genuine token embedding.

The geometric conclusion is that the continuous relaxation buys genuine Langevin dynamics at the cost of leaving the token manifold, after which it must either respect the correction and freeze or ignore it and wander through the space between tokens, where the projected energy is a poor guide. The discrete sampler avoids the choice by never leaving the manifold.

5.5 Gradient Direction Against a Norm-Matched Random Direction

The preceding sections concern how the samplers move. This one concerns the signal that drives them, and it must be read against the measurement of Section 5.2: at the calibrated configuration the proposal is numerically uniform over the vocabulary, so the comparison below is a comparison between two draws that the gradient cannot influence either way. Its null is therefore a bound on what this configuration could detect, not a demonstration that the gradient is uninformative. Section 5.6 repeats it where the gradient does shape the proposal, and Section 5.7 measures the surrogate directly, invariant to the temperature and step size that limit this section. The ablation of Section 4.3 compares three proposal methods: the true gradient (*policy*), a random direction rescaled to the gradient’s norm (*grad-norm-preserved*), and a fully random vector (*random*). The three-way design separates the two things a gradient can contribute. If the gradient’s *direction* is informative, the policy method should outperform the norm-preserved random method, which differs from it only in direction. If the gradient’s *magnitude* is informative, the norm-preserved method should outperform the fully random method, which differs from it only in magnitude.

One confound has to be removed first. Gradient normalization, applied by default for numerical stability, rescales every proposal direction to unit norm. Under normalization the norm-preserved random method and the fully random method become the same object, a random unit vector, so an observed equality between the policy and random methods admits two explanations: either the gradient’s direction is uninformative, or normalization has discarded a real signal carried by the magnitude. The extended ablation therefore repeats the whole comparison *with gradient normalization disabled*. The policy and norm-preserved methods then both carry the gradient’s full unnormalized magnitude and differ only in direction, so any gap between them is attributable to direction alone. Doubling the ablation in this way is the reason the study comprises 145 configurations rather than half as many.

With the confound removed, Table 4 reports the final KL divergence for the policy and norm-preserved methods with normalization disabled and the correction

enabled, across four energy functions, together with their difference. The differences are small and inconsistent in sign. On GPT-2 Large the policy method is worse than the random direction by 0.044; on one GFlowNet variant it is worse by 0.008; on another it is better by 0.146; and on the length-normalized variant the random direction is better by 0.670. That last gap is the largest in the table, but it should not be read as evidence that the gradient is actively harmful: comparing a paired difference against an across-seed standard deviation is not a test, and Section 5.6 shows that apparent penalties of this kind come from the asymmetric treatment of the reverse-proposal term rather than from the gradient. In no case does the true gradient reliably outperform a random direction of the same magnitude. In the tested token-substitution settings, then, the input-embedding gradient of frozen autoregressive sequence likelihood provided no reliable proposal advantage over a norm-matched random direction. Because the norm-preserved and fully random methods are themselves indistinguishable in the separate comparison, the same holds for the magnitude.

To bound the equivalence rather than rest it on the eye, the flagship configuration was analyzed as a paired comparison over its two hundred sequences, pairing the policy method against the norm-preserved random direction on each sample. For the discrete sampler on GPT-2 Large with the correction and gradient normalization enabled over fifty steps, the paired mean difference in final KL is +0.171 nats, with a 95% bootstrap confidence interval of $[-0.285, +0.619]$ that straddles zero, a Wilcoxon signed-rank p of 0.40, and a difference detectable at eighty percent power only if it were as large as 0.652. These figures say the same thing three ways: the difference is small, it is not statistically significant, and the experiment could only have detected a gap of at least 0.652.

The final-step KL is, however, the wrong summary for a Markov chain, because it discards every state the chain visited. Recomputing the same paired contrast from the per-sample trajectories already stored for the grid gives Table 3. On the chain statistics the difference all but vanishes: the mean KL over the second half of the schedule differs by 0.002 nats and the best state the chain reaches differs by -0.043 , against +0.171 for the last iterate. The last iterate is the noisiest of the three and is

Statistic	Policy	Random	Difference	95% CI	Wilcoxon p
Final step (as reported)	6.541	6.370	+0.171	$[-0.291, +0.614]$	0.400
Chain mean, second half	6.667	6.665	+0.002	$[-0.332, +0.347]$	0.747
Minimum over trajectory	4.625	4.669	-0.043	$[-0.271, +0.177]$	0.678

Table 3: The flagship paired contrast under three summaries of the same chains, policy against the norm-matched random direction, $n = 200$ paired sequences. Only the first row is a last-iterate statistic. The comparator is the grad-norm-preserved random arm; under gradient normalization it coincides with the fully random arm.

the one originally reported. On the minimum-over-trajectory statistic the two one-sided tests do certify equivalence at the pre-registered margin of 0.327; on the chain mean the interval misses by 0.02 nats. Across the thirty-five configurations of the grid that have both arms, the chain-mean contrast is equivalent at that margin in nineteen and has a confidence interval excluding zero in two.

The remaining gap is statistical power, and it can be closed. At $n = 200$ the smallest difference detectable at eighty percent power is 0.652 nats, twice the equivalence margin, which is why the tests above decline to certify. The flagship configuration was therefore repeated over four further independent corruption seeds and pooled, giving $n = 1000$ paired sequences. At that size the two one-sided tests certify equivalence on all three statistics: the paired difference is +0.133 nats at the final step with a 95% interval of $[-0.063, +0.326]$, +0.136 on the chain mean with $[-0.012, +0.287]$, and +0.033 on the trajectory minimum with $[-0.079, +0.145]$, every interval lying inside the pre-registered ± 0.327 margin. The claim can accordingly be stated positively: the two proposals are equivalent to within a margin fixed in advance, rather than merely not distinguishable. One qualification belongs with it. WikiText-2 validation yields only 282 sentences under the length filter, so the additional sample size comes from independent corruptions of the same sentences and not from new text; the interval is therefore a statement about variation across corruptions, and sentence-level variation is still bounded by those 282 items.

Figure 3 puts the whole grid on one axis. Each row is one configuration’s paired

Energy function	Policy	Grad-norm-preserved	Difference
GPT-2 Large (SFT)	6.474	6.430	+0.044
GFlowNet, <code>len_beta</code> = 0, 500	6.009	6.001	+0.008
GFlowNet, <code>len_beta</code> = 0, 2000	6.698	6.552	+0.146
GFlowNet, <code>len_beta</code> = 1, 500	6.008	5.338	+0.670

Table 4: Final KL divergence (lower is better) for the true gradient against a norm-preserved random direction, with gradient normalization disabled and the Metropolis–Hastings correction enabled, on the discrete sampler at 50 steps. Difference is policy minus grad-norm-preserved.

policy-minus-random contrast on the chain-mean statistic with its bootstrap interval, and the shaded band is the pre-registered equivalence margin. The picture is a column of intervals centred on zero: the contrast is not merely non-significant in the flagship cell but small and sign-inconsistent across models, samplers, correction settings, normalization settings and schedule lengths.

One further number belongs here and has not been stated. Across all 145 configurations at $n = 200$ each, roughly 29,000 recovery attempts, exact recovery of the corrupted token is 0.0 percent in 139 configurations and is one or two sequences in the remaining six. The samplers essentially never restore the masked token. This is consistent with the uniform-proposal measurement and it qualifies the primary metric: the fall in mean KL from 9.14 to about 6.4 is a gain in contextual plausibility of whatever token the accept/reject filter retained, not evidence of search converging on the right one.

One contrast in the matrix is nominally significant, and it is worth stating why this thesis does not build on it. On Llama-3, with the correction and gradient normalization both disabled, the policy method reaches a higher final KL than the norm-preserved random direction, with a paired 95% confidence interval of [0.45, 2.09] excluding zero and a Wilcoxon p of 0.015. Two considerations bound what that supports. It is one contrast selected from several dozen paired comparisons across the grid, uncorrected for multiplicity, and at $\alpha = 0.05$ a handful of such hits is expected

by chance; Figure 3 shows it in the context of the whole family. Second, it is measured with the correction *disabled*, so by this thesis’s own argument (Section 5.3) it compares two biased optimizers rather than two samplers, and the quantity being compared is where an uncorrected chain happens to stop. Where the same apparent penalty has been examined under a correctly symmetric acceptance rule, on GPT-2 in Section 5.6, it vanishes. The reading that the gradient actively points away from good moves is therefore not adopted here; the evidence supports indifference. This section answers the second half of RQ1 by locating the failure in the gradient rather than in an incidental choice of scale or correction, and it establishes indifference rather than harm.

5.5.1 Robustness of the Null Result

Four variations were examined and none overturned the null. First, extending the annealing schedule from fifty steps to one hundred does not create a gap between the policy and the random directions: with the longer schedule the policy and grad-norm-preserved methods remain within noise of one another on every model, so the null is not a consequence of insufficient optimization budget. Second, the phenomenon holds under both settings of the Metropolis–Hastings correction and both settings of gradient normalization, so it is not produced by the interaction of the ablation with any one of those switches. Third, the same behaviour was observed for multi-token recovery, in which two or three coupled positions are corrupted and must be recovered jointly, as well as for the single-token case reported in the tables. As the number of coupled masked tokens increases the absolute recovery quality degrades, as one would expect from a harder task, but the relative behaviour of the sampling methods is preserved: the policy gradient does not pull ahead of a random direction at any corruption level. Fourth, the null is not an artifact of the masked-recovery task. Repeating the core comparison on a free-form prefix-continuation task, in which the sampler generates a twenty-token span with no reference token to recover, reproduces the equivalence: the policy method reaches a mean final KL of 8.818, with a 95% confidence interval of [8.61, 9.02], against 8.850 with interval [8.65, 9.04] for the random direction, the two intervals almost wholly overlapping and

the norm-preserved and fully random arms numerically identical. The null therefore survives the move from a constrained infilling bench to open-ended generation, and is robust to the schedule, the correction, the normalization, the difficulty of the recovery task, and the choice of task.

5.5.2 Gradient-Free Baselines

Whether the fault lies in the gradient or in the energy it is computed from is separated by scoring the same energy with methods that do not use its gradient. Table 5 does this on the masked-recovery metric for GPT-2 Large. The ground-truth fill sits at the metric floor of 0.00; leaving the corrupted token in place gives 9.14 and replacing it with a random token gives 9.39, the two no-effort references. A single conditional forward pass already improves on these, reaching 8.24 by taking the argmax of the model’s own next-token distribution at the masked position and 8.62 by sampling from it. The last two rows are the decisive ones. A single forward pass that reranks the top- k candidate tokens by their effect on the sequence reaches 4.43, better than every Langevin configuration in the entire grid, whose best discrete cell sits near 6.4; and a gradient-free Gibbs sampler on exactly the same energy, re-sampling one masked position at a time from the model’s conditional, reaches 6.69, comparable to the best gradient-guided Langevin result near 6.4 to 6.5 without ever computing a gradient.

Two methods that evaluate the frozen likelihood but never differentiate it, a single rescore pass and a gradient-free sampler, both do at least as well as the gradient-guided samplers, and the rescore pass does markedly better. What the Langevin samplers add over these baselines is the gradient. The energy is therefore searchable without its gradient, which is what licenses narrowing the thesis’s claim to the gradient rather than to the energy. The gradient-free baselines are also cheaper, spending no backward pass at all, as Appendix A.3 sets out.

Method (no gradient of the energy)	Mean KL
Ground-truth fill (metric floor)	0.00
Untouched (corrupted token left in place)	9.14
Random-token fill	9.39
Conditional argmax (one forward pass)	8.24
Conditional sample (one forward pass)	8.62
Top- k rescore (one forward pass)	4.43
Gibbs (gradient-free sampler)	6.69

Table 5: Reference baselines on masked-token recovery for GPT-2 Large, scored with the same KL metric as the Langevin grid (lower is better). None of these methods uses the gradient of the energy.

5.5.3 External-Judge Rescoring

A null measured with the model’s own predictive distribution invites the objection that the metric is built from the same model that produced the recoveries. The recovered sequences, produced by the discrete sampler on GPT-2 Large with the correction and gradient normalization enabled over fifty steps for the same three proposal arms, were therefore rescored under perplexity by an independent judge, a Llama-3 model that took no part in the sampling. The policy recoveries score a mean judge perplexity of 178.4 and the random ones 181.3, a difference under two percent, with the norm-preserved and fully random arms again identical; the per-token negative log-likelihoods, 4.44 against 4.45, tell the same story. The external judge returns the same reading as the internal metric, so the indistinguishability of the methods is not an artifact of the evaluation model.

5.6 Does the Null Survive a Load-Bearing Gradient?

Section 5.2 established that at the calibrated configuration the proposal is numerically uniform, so the ablation of Section 5.5 could not have detected an informative gradient had one been present. This section closes that gap directly, by repeating

the same three-arm contrast over a grid of configurations chosen so that the gradient term *does* shape the proposal.

Two knobs control that. The step size ϵ sets the ratio between the two terms of Equation (7), because the distance term scales as $1/\epsilon$ while the gradient term does not, and the temperature sets how sharp the resulting softmax is. The sweep therefore crosses five values of ϵ , from 1.05 to 10,500, with five temperatures from 0.05 to 25, for each of the three proposal arms, with gradient normalization disabled so that the gradient carries its true magnitude, at $n = 50$ sequences per cell. It spans a ratio t_2/t_1 from 1.4 to 13,877, that is, from the distance term and the gradient term contributing equally to the gradient term dominating by four orders of magnitude, and a minimum proposal entropy from 10.82 nats (uniform) down to 0.000 nats (a deterministic, purely gradient-driven proposal). The thesis’s calibrated cell sits at $t_2/t_1 = 11.1$ and entropy 10.22, in the near-uniform corner.

The result is that the null holds throughout. In no cell of the sweep does the true gradient reach a lower final KL than the norm-matched random direction by more than the cell-to-cell noise, and exact recovery of the corrupted token never exceeds 2 percent anywhere in the sweep, including in the fully deterministic gradient-driven corner where the proposal ranks candidates by the gradient alone. Making the gradient term load-bearing does not make it useful.

A chain-level statistic recorded for these runs sharpens what that failure is. Besides the final token, each run logs whether the chain occupied the ground-truth token at *any* step. Averaged over the sweep the two rates are 0.108 and 0.135 percent, and in 69 of the 74 cells both are exactly zero. The chain is therefore not passing through the correct token and wandering away from it, which a last-iterate metric could have concealed; over roughly 185,000 proposal draws it essentially never reaches it. Whatever the fall in mean KL measures, it is not a search closing on the right answer.

The sweep also settles a claim made earlier in this chapter, and settles it against that claim. Under the correction as applied in the main grid, the policy arm appeared *worse* than a random direction in the sharp cells, by +1.28 to +2.34 nats, which

ϵ	T	Correction as in the main grid			Correction exact for all arms		
		Policy	Random	Difference	Policy	Random	Difference
1.05	0.05	8.414	7.132	+1.281	8.414	8.583	−0.169
10.50	0.05	8.166	5.831	+2.336	8.166	8.416	−0.250
10.50	1.00	5.760	5.593	+0.167	5.760	5.768	−0.009

Table 6: Final KL in the three sharpest sweep cells under the two treatments of the Metropolis–Hastings reverse-proposal term, $n = 50$ per arm, gradient normalization disabled. Random is the norm-matched random-direction arm. Under the main grid’s treatment the policy arm alone pays the reversibility term and appears substantially worse; with the term computed exactly for every arm the difference disappears.

would have supported the reading that a mis-specified surrogate actively points away from good moves. That reading does not survive the control. As Section 4.3 discloses, the main grid computes the reverse-proposal term exactly for the policy arm and sets it to zero for the random arms, so the policy arm alone is charged a reversibility penalty, and in a sharp proposal that penalty is large. Rerunning the three sharpest cells with the term computed exactly for every arm removes the effect entirely (Table 6): the gap of +2.34 nats becomes −0.25, and +0.17 becomes −0.009. The policy column is unchanged between the two treatments, as it must be, since only the comparators’ treatment changed.

Two conclusions follow. The null of Section 5.5 is real and is not an artifact of a proposal too flat to express the gradient: it holds where the gradient fully determines the proposal. And the complementary claim, that the gradient is reliably worse than noise, is withdrawn. It appeared in the Llama-3 contrast of Section 5.5 and in the sharp cells here, and in both places the compared arms were not charged the same reversibility term. The evidence supports indifference, not anti-guidance.

5.7 The Linearization Radius

The mechanism behind the null is tested directly on the surrogate the discrete proposal uses. That proposal ranks a candidate token v by the first-order term $\hat{\Delta}(v) = \mathbf{g}^\top(\mathbf{e}(v) - \mathbf{e}(x_i))$, a linear approximation to the true change in log-likelihood a substitution would produce (Section 2.4). The linearization experiment of Section 4.7 computes both quantities for thousands of candidate substitutions and correlates them.

Figure 4 plots the surrogate against the true change over all candidate–sequence pairs for GPT-2 Large. There is no relationship: the Spearman correlation between the surrogate and the truth is -0.011 , negligible in magnitude and negative in sign. This is not a peculiarity of the base model: the same correlation is negligible on every one of the five energy functions, with $|\rho| < 0.06$ throughout (GPT-2 Large -0.011 , Llama-3 0.021 , and the three GFlowNet variants 0.024 , 0.046 , and 0.057), each estimated over $n = 400,000$ candidate–sequence pairs. Ranking candidate tokens by the gradient surrogate is therefore equivalent to ranking them at random. The measurement is made at the level of the proposal itself rather than through the downstream sampling metric, so it is the more direct of the two observations.

The cause is a mismatch between two length scales. A Taylor expansion is accurate only within a small neighbourhood of the point at which it is taken, and a Transformer (Vaswani et al., 2017), with its layer normalization and attention nonlinearities, is a strongly nonlinear function of its input embeddings, so that neighbourhood is small. Whether it is large enough to contain any move the sampler must make is answered by Figure 5, which computes the surrogate-to-truth correlation *within* bins of embedding distance, so that one can see how far from the current token the linear model remains valid. The correlation is at best weakly positive at the very smallest distances and has decayed to zero well before the mean nearest-neighbour distance of 1.82 reported in Section 5.9, and far below the mean candidate distance of 2.35 , the average embedding distance of the candidate tokens the experiment scores. The nearest-neighbour distance is by definition the scale of the smallest possible substitution, since to change a token at all is to move to another token’s

embedding. The entire admissible range of substitutions therefore lies beyond the distances at which the surrogate carries signal. Every substitution the sampler can make moves the embedding farther than the spacing between neighbouring tokens, and the linear approximation has already broken down at that distance, so the gradient is not merely imprecise here but *inapplicable*.

A second limitation compounds the first, and the decomposition built into the experiment exposes it. The gradient is taken with respect to the input embedding of the masked position, and it can therefore only register the effect of a substitution on the *future* tokens, whose probabilities depend on that embedding as it flows forward through the network. It is structurally blind to the effect of the substitution on the probability of the candidate token *itself*, given its left context, because that probability depends on the candidate through a discrete index into the output softmax rather than through the continuous input embedding the gradient sees. The decomposition measures both effects. Each is a signed change in log-likelihood; averaging its absolute value over candidates gives a mean absolute self term of 15.0 nats and a mean absolute future term of 24.2 nats, so the self term, to which the gradient is blind, is not a rounding error but a substantial fraction of the total, and the surrogate correlates with the future term ($\rho = 0.03$) rather than the self term ($\rho = -0.10$), exactly as the structural argument predicts. The gradient is therefore evaluated outside the range where it would be valid, and even within that range it would be blind to a large part of what determines whether a substitution is good. The full decomposition is in Figure 7 and the top- k recall analysis, which finds that gradient ranking does not recover the true best substitutions better than chance, in Figure 8, both in Appendix A.1. Together they answer the “why not” of RQ1: the surrogate is not a noisy version of the true ranking but is unrelated to it over the range the sampler must move, which is why no step-size or temperature setting recovers it (Section 5.6).

5.7.1 Which Gradient? The Input Embedding Against the One-Hot Input

The blindness just measured is a consequence of a modelling choice, not of autoregressive likelihood as such, and the choice can be named precisely. Everything above differentiates with respect to the *input embedding* at the masked position, $\mathbf{g} = \partial \log p(\mathbf{x}) / \partial \mathbf{e}_i$. That is what the samplers this thesis reimplements do, because they operate in embedding space. The discrete samplers those methods adapt (Grathwohl et al., 2021; Zhang et al., 2022) instead differentiate with respect to the *one-hot* input, and for an autoregressive language model the two are not the same object. A token index enters the sequence likelihood twice: once through the embedding lookup, which is what $\mathbf{g}$ sees, and once through the output softmax that scores it as the target of the prediction made at the previous position, which $\mathbf{g}$ cannot see. Differentiating with respect to the one-hot vector captures both, and its v -th coordinate is

$$(11) \quad \frac{\partial \log p(\mathbf{x})}{\partial \mathbf{x}_i[v]} = \underbrace{\log p(v \mid \mathbf{x}_{<i})}_{\text{self}} + \underbrace{\mathbf{g}^\top \mathbf{e}(v)}_{\text{future}},$$

so the candidate ranking it induces is $\hat{\Delta}_{\text{one-hot}}(v) = \hat{\Delta}(v) + [\log p(v \mid \mathbf{x}_{<i}) - \log p(x_i \mid \mathbf{x}_{<i})]$. The bracket is exactly the self term the linearization experiment already records, because substituting v at position i changes the log-likelihood of position i by precisely that conditional log-ratio and by nothing else. The comparison is therefore an exact re-analysis of the data already collected, requiring no further forward passes.

Table 7 reports it. The result is unambiguous and it reverses the reading of the null. Over the full candidate set the one-hot surrogate correlates with the true energy change at $\rho = 0.37$ to 0.59 on the five energies, against $|\rho| < 0.06$ for the input-embedding surrogate. Restricted to the *near* stratum, whose mean candidate distance of 1.94 is the admissible substitution range the preceding paragraphs argue about, the separation is starker still: $\rho = 0.60$ to 0.73 against 0.03 to 0.11 .

Three things follow, and they change what this thesis can claim.

Energy function	All candidates		Near stratum	
	Input embedding	One-hot	Input embedding	One-hot
GPT-2 Large (SFT)	−0.011	0.588	0.028	0.730
Llama-3 8B	0.021	0.513	0.039	0.596
GFlowNet, <code>len_beta</code> = 0, 500	0.057	0.379	0.111	0.649
GFlowNet, <code>len_beta</code> = 0, 2000	0.024	0.439	0.091	0.706
GFlowNet, <code>len_beta</code> = 1, 500	0.046	0.365	0.051	0.610

Table 7: Spearman correlation between the first-order surrogate and the true change in sequence log-likelihood, for the input-embedding gradient this thesis measures and for the one-hot gradient of Equation 11. $n = 400,000$ candidate–sequence pairs per energy, of which 100,000 fall in the near stratum (mean candidate distance 1.94, and 0.45 for Llama-3). The one-hot surrogate is an exact algebraic re-analysis of the same measurements.

First, the usable local direction *exists*. It is not destroyed by the autoregressive training objective, and it is available in closed form from a single forward pass, since $\log p(v \mid \mathbf{x}_{<i})$ is the model’s own next-token distribution. What discards it is differentiating with respect to the input embedding.

Second, the input-embedding gradient is not weak signal but noise. Adding it to the self term slightly *degrades* the correlation on GPT-2 Large (0.588 against 0.595 for the self term alone) and on two of the three GFlowNet variants. A sampler that ignored the gradient entirely and ranked candidates by the conditional alone would do marginally better than one that used it.

Third, this unifies results that otherwise sit apart. The gradient-free baselines of Section 5.5.2 that work, top- k rescoring at 4.43 and Gibbs at 6.69, are precisely the methods that read the output side. The final-position result of Section 5.13, where the input-embedding gradient is provably exactly zero while the energy difference between candidate final tokens is exactly the conditional log-ratio, is the extreme case of Equation 11 in which the future term vanishes and only the self term remains. And the diffusion control of Section 5.14 supplies a concrete score that is itself a ratio of sequence probabilities, an output-side quantity.

Surrogate, at ϵ	Proposal temperature T				
	0.05	0.25	1	5	25
One-hot, $\epsilon = 10.5$	0.0	8.0	40.0	—	—
One-hot, $\epsilon = 105$	0.0	8.0	36.0	2.0	0.0
One-hot, $\epsilon = 1050$	0.0	8.0	38.0	2.0	0.0
Input embedding, $\epsilon = 10.5$	0.0	0.0	0.0	0.0	0.0
Input embedding, $\epsilon = 105$	0.0	0.0	2.0	0.0	0.0
Input embedding, $\epsilon = 1050$	0.0	0.0	2.0	0.0	0.0

Table 8: Exact recovery (percent) with the one-hot surrogate against the input-embedding surrogate, inside the same discrete sampler, on the same corrupted sequences, over the same step-size and temperature grid. $n = 50$ per cell. Only the surrogate differs. The best one-hot cell, 40.0 percent, is comparable to the score-trained diffusion proposal of Section 5.14 at 39.0 percent, and is obtained from the frozen autoregressive model alone.

A ranking correlation is not yet a working sampler, so the one-hot surrogate was substituted into the discrete sampler itself and run over the same (ϵ, T) grid as Section 5.6, changing nothing else. Table 8 reports it against the embedding-gradient sweep cell for cell. Where the proposal is sharp enough to express a ranking, the one-hot surrogate recovers 36 to 40 percent of corrupted tokens; the input-embedding gradient never exceeds 2 percent anywhere in the same grid. At the temperature the study calibrated on, $T = 5$, even the one-hot proposal falls to 2 percent, because that temperature flattens the proposal to a near-uniform draw whatever surrogate is supplied. The temperature, not the model, was the binding constraint.

The same substitution on the three GFlowNet-tuned energies gives 25.0, 21.0 and 9.0 percent in the sharp configuration, against 0.5 percent for each of them at the calibrated one, so the effect is not particular to the base model. The cross-architecture control carries the same result across a different tokenizer, a different embedding geometry and an order of magnitude more parameters. On Llama-3 8B the one-hot proposal recovers 41.0 percent of corrupted tokens at a final KL of 1.908, the lowest divergence any proposal in this study attains, against 0.0 percent and

3.898 for the input-embedding gradient on the same model and the same sequences.

That result also isolates what the configuration has to supply, because the same model at its own calibrated setting recovers nothing: 0.0 percent at a KL of 4.108. Both configurations produce a sharp proposal, and the naive reading would be that sharpness is not the issue. The proposal statistics show otherwise. At the calibrated setting $\epsilon = 0.1$ makes the distance term $-\|e(v) - s\|^2/2\epsilon$ enormous, so the softmax concentrates on the current token, and the measured t_2/t_1 is 1.83: the proposal is sharp about *staying put*. In the recovering configuration $\epsilon = 10$ suppresses the distance term and t_2/t_1 rises to 229, so the proposal is sharp about the *surrogate's ranking*. Sharpness from the distance term and sharpness from the surrogate are different things, and only the second puts the candidate ranking in charge. The same distinction governs the GPT-2 sweep, where the one-hot proposal needed a large ϵ and a low temperature together, and where GPT-2's own calibrated cell also yielded 0.5 percent.

The claim this thesis establishes is therefore narrower, and considerably more constructive, than a blanket null. It is not that a frozen autoregressive likelihood admits no gradient guidance. It is that the input-embedding Jacobian slice, which is the object the embedding-space samplers of this literature differentiate, discards the half of the energy that carries the signal, and that supplying the discarded half turns the same sampler on the same frozen model from zero recovery into recovery comparable to a purpose-trained diffusion model. Section 6.5 takes up what follows for method design.

5.8 The Likelihood Trap

The energy-based framework rests on the premise that low energy, that is, high likelihood, corresponds to good text. The likelihood-trap experiment tests that premise on the study's own models, reproducing quantitatively the neural-text-degeneration phenomenon of Holtzman et al. (2020). The premise has been contradicted most sharply in machine translation, where Stahlberg and Byrne (2019) found the globally most probable translation under a trained model to be the empty string for

a large fraction of inputs, and Eikema and Aziz (2020) concluded that the mode of such a model is simply not where its useful probability mass lies; Meister et al. (2023) give the information-theoretic account of why. The measurement below is the same phenomenon on the specific energies this thesis samples from, and it is what makes the target of the whole framework questionable independently of whether the gradient works.

The experiment decodes with several strategies and measures the relationship between the per-token likelihood of the generated text and its repetition rate. The relationship is monotone: as per-token likelihood increases, that is, as energy decreases, repetition rises and lexical diversity falls (Figure 10, Appendix A.1). The correlation is reported *within* each decoding strategy, because pooling generations from strategies of very different likelihood, greedy at one end and ancestral sampling at the other, can manufacture a correlation out of the gap between strategies rather than any relationship inside them. Measured this way the maximum within-strategy correlation between per-token likelihood and four-gram repetition is +0.51 on the base GPT-2 model, +0.83, +0.82, and +0.91 on the three GFlowNet variants, and +0.77 on Llama-3. The pattern between strategies points the same way: the highest-likelihood strategies, greedy and beam search, produce the most repetitive and least diverse text, while the lowest-likelihood strategies, ancestral and nucleus sampling, produce the least repetitive. On Llama-3 the pooled correlation is weaker, at +0.22, yet the within-strategy signal is strong and the model exhibits a blunt degeneracy of its own, never emitting an end-of-sequence token and running to the length cap in every generation. Beam search, which most nearly approximates the mode of the distribution, produces both the highest-likelihood and the most degenerate text. The following is a representative beam-search output from the length-normalized GFlowNet variant, which achieves among the highest per-token likelihoods in the study; it is produced by autoregressive beam decoding of the model, not by any Langevin sampler:

aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-aebb-

This string has a per-token log-probability of -0.40 , higher than almost any

coherent text the model produces, and a four-gram repetition rate of 0.89. By the model’s own energy it is an excellent sequence, and it is gibberish. This is the **likelihood trap**.

The implication for the energy-based programme is that any procedure concentrating probability mass on the lowest-energy regions of this energy, whether a Langevin sampler that succeeded in localizing or a GFlowNet trained on the likelihood as a reward, would be concentrating it on degenerate text. Success at the stated optimization objective is failure at the underlying goal.

The experiment also yields the brevity slope that Section 5.11 uses. The regression of total log-likelihood on generated length, computed over the sampled generations where length varies, has a slope of -1.12 nats per token. This headline figure must be read with a censoring caveat, because for GPT-2 Large nearly every generation runs to the forty-token cap (a fraction of 0.997 hits it), so length barely varies and the censored slope, estimated only on the few generations that stop short, is effectively null. The negative slope is therefore visible only where the cap does not bind: on the length-normalized GFlowNet variant, whose generations do vary in length, the uncensored slope is -0.505 nats per token while the censored slope steepens to -2.361 , which shows both that the incentive is real and that the cap masks its magnitude. Because every additional token lowers the unnormalized log-likelihood by this much on average, an energy defined as the raw log-likelihood carries an implicit incentive toward brevity: shorter sequences are lower-energy by virtue of being shorter. The incentive is invisible in the Langevin experiments, which recover fixed-length masked spans, and becomes decisive when a method is free to choose its own length. The regression is shown in Figure 12, Appendix A.1.

5.9 Embedding Anisotropy

The anisotropy of Transformer embeddings is a documented phenomenon (Gao et al., 2019; Ethayarajh, 2019). Measuring it on the two models here quantifies the consequences it has for the samplers, and accounts for both the step-size failure of Section 5.1 and the unreliability of Euclidean distance noted in Section 4.4.

Three distance statistics recur in this thesis and are defined once here. The **mean nearest-neighbour distance** is the average distance from a token embedding to its single closest neighbour; the **mean pairwise distance** is the average distance between two embeddings drawn at random; and the **mean linearization-candidate distance** is the average distance of the stratified candidate tokens scored in the linearization experiment of Section 5.7. In GPT-2 Large the mean pairwise distance between token embeddings is 2.77, the mean nearest-neighbour distance is 1.82, and the mean linearization-candidate distance is 2.35; in Llama-3 the pairwise and nearest-neighbour figures are 0.84 and 0.59, with a linearization-candidate distance of 0.66. The top principal component of the embedding matrix accounts for only a small fraction of the total variance, so the anisotropy is a broad property of the space rather than the effect of a single dominant direction that could be projected away; this reconciles the high mean pairwise cosine with the low variance that the two-component projection of Appendix A.4 captures, since a broad cone is spread across many directions and no low-dimensional projection can represent it. Figure 11 in Appendix A.1 shows the full distributions of pairwise distances for the two models, on their separate scales.

Two distinct properties are at work, and they are kept separate here because they have different consequences. The first is a matter of *scale*: the absolute inter-token distances differ roughly threefold between the two models. It is this scale difference, not anisotropy, that causes the step-size failure of Section 5.1, since a step calibrated to cross a cell boundary in one geometry is too small to cross one in the other. The second is *anisotropy* proper: within each model the embeddings occupy a narrow cone, as the positive mean pairwise cosine and the broad principal-component spectrum show. It is this directional clustering, not the scale, that compresses the distinction between a good and a bad token and so makes Euclidean distance an unreliable metric, which is why the study uses KL divergence as its primary criterion instead.

Energy function	Linearization ρ	Max within-strategy trap correlation
GPT-2 Large (SFT base)	−0.011	+0.51
GFlowNet, <code>len_beta</code> = 0, 500	+0.057	+0.83
GFlowNet, <code>len_beta</code> = 0, 2000	+0.024	+0.82
GFlowNet, <code>len_beta</code> = 1, 500	+0.046	+0.91
Llama-3 8B (cross-architecture)	+0.021	+0.77

Table 9: Surrogate-to-truth Spearman correlation from the linearization experiment (Section 5.7) and the maximum within-strategy correlation between per-token likelihood and 4-gram repetition (Section 5.8), for each energy function. Llama-3 figures are not on a shared numerical axis with the GPT-2 family.

5.10 Consistency Across Models and Architectures

Table 9 collects the two diagnostics across the five energy functions.

Across the GPT-2 family the two diagnostics agree. The linearization correlation is negligible on the base and on all three GFlowNet variants, and the maximum within-strategy likelihood-repetition correlation runs from +0.51 on the base to +0.91 on the length-normalized GFlowNet, so the likelihood trap is if anything stronger on the tuned models, consistent with tuning having concentrated probability mass more tightly on the high-likelihood region. The final-KL values of Table 4 move together across these four models and the policy-minus-random difference stays small on each.

The Llama-3 control has a limited and specific role. Its tokenizer differs and its embedding space is roughly a third the scale of GPT-2’s, so a KL divergence or an inter-token distance from Llama is not comparable digit for digit with one from GPT-2, and no such comparison is made. What the Llama runs establish is cross-architectural and qualitative: the Metropolis–Hastings breakdown occurs there as on GPT-2, the anisotropy is present at a different scale, and Llama-3 is the one setting in which the true gradient measurably underperforms a random direction. The control therefore rules out the reading that the diagnosis is an artifact of one model’s idiosyncrasies, without licensing a numerical comparison across the two

families.

5.11 GFlowNet Fine-Tuning and the Amortized Energy

If the gradient is the problem, one remedy is to abandon the gradient. A GFlowNet (Bengio et al., 2021; Hu et al., 2024) learns to sample from a reward without ever differentiating it, so the linearization failure and the discontinuous drift are invisible to it. This section reports what the tested GFlowNet variants did. The load-bearing result is the unifying experiment at the end, which compares Langevin sampling on the tuned energy against the untuned base; the taxonomy of failure modes that precedes it is exploratory, in the sense that it describes how these particular training runs broke rather than establishing a general property of amortized inference. Three distinct failure modes appeared.

The first is **capacity collapse**. The initial GFlowNet experiments used the smaller GPT-2 base, and they produced non-linguistic output regardless of how the reward was configured, which indicated that the policy simply lacked the capacity to represent the task at all. Moving to GPT-2 Large, the model of the reference implementation of Hu et al. (2024), resolved this, and GPT-2 Large is therefore the base for all the GFlowNet results reported here. It is recorded here as a separate failure, but it is one of capacity rather than of the energy.

The second is **length collapse**, and it follows directly and quantitatively from the likelihood trap of the previous section. With the unnormalized reward, at `len_beta = 0`, the reward is the raw log-likelihood, and Section 5.8 measured that every additional token lowers that quantity by around 1.12 nats. The reward therefore carries an implicit incentive toward brevity, and the policy discovers it, learning to produce extremely short sequences because short sequences are high-reward by being short. The extreme form of this is visible in the decoding behaviour: under free-running ancestral sampling, the length-collapse variants never emit an end-of-sequence token at all, running to the generation cap in every case, because the policy has learned that terminating early is the way to keep the reward high and, in the limit, that the safest policy is one that resists stopping in the manner

the reward rewards. The length collapse is not a defect of the optimizer but the optimizer following a reward whose implicit incentive the designer did not intend.

The obvious fix is to remove the brevity incentive by normalizing the reward by length, and this produces the third mode, **reward hacking**. At `len_beta = 1` the reward is normalized by sequence length, the brevity incentive is gone, and the sequences do indeed return to a normal length. But the output ceases to be linguistic. The policy discovers sequences that achieve high length-normalized likelihood without being coherent text at all, strings of individually high-probability tokens assembled into grammatical nonsense. The following greedy output from this variant is representative:

b8d8b8b8b8b8. [...] keredkeredkeredkeredkered

These strings are produced by argmax and beam decoding of the tuned policy, in the decoding pass of Section 5.8, not by any Langevin sampler; the length-normalized reward is what assigns them the high value the GFlowNet training then reinforces. Such strings recur across the length-normalized variant’s high-reward outputs, and they are the reward-hacking counterpart of the length collapse: where the unnormalized reward was gamed by getting shorter, the normalized reward is gamed by finding degenerate high-probability token sequences. The two settings of the single length parameter thus produce two opposite degeneracies, brevity at one extreme and gibberish at the other, and no intermediate setting of the parameter was tested that resolves both; only the two endpoints, `len_beta = 0` and `len_beta = 1`, were run.

The common element in the three modes is that length is the axis along which the failure manifests rather than its cause: the reward, an autoregressive likelihood never shaped to serve as an energy over free-form sequences, has no well-behaved optimum where coherent text lies, so repairing one degeneracy surfaces another. Whether the tuning nonetheless made the energy more navigable is tested directly. If GFlowNet training had reshaped the likelihood into a smoother landscape, running the Langevin sampler on the tuned model should differ from running it on the

Energy function	Final KL (DLS, 50 steps, policy, MH, gn)
GPT-2 Large (SFT base)	6.541
GFlowNet, <code>len_beta= 0, 500</code>	6.306
GFlowNet, <code>len_beta= 0, 2000</code>	6.721
GFlowNet, <code>len_beta= 1, 500</code>	6.415

Table 10: Final KL divergence of the discrete Langevin sampler on the base model and on the three GFlowNet-tuned variants, under identical conditions: policy proposal, Metropolis–Hastings correction on, gradient normalization on, 50 steps.

untuned base. Table 10 reports the final KL divergence of the discrete sampler on the base model and on the three GFlowNet variants, under identical conditions. The four values are within noise of one another, and the null of Section 5.5 replicates on every GFlowNet variant, as the linearization correlations of Table 9 also show. This answers RQ3 in the negative for the tested setup: amortization did not escape the observed problems, and modifying the model through the GFlowNet objective did not produce an energy whose local gradient became more useful.

The result would be trivial if GFlowNet training had barely moved the energy, so that sampling on the tuned model and on the base agreed only because the two energies were nearly identical. Table 11 reports how far each tuned energy has drifted from the base. The mean absolute difference in sequence log-likelihood between base and tuned is 31.0, 36.0, and 57.3 nats across the three variants; the next-token KL divergence between their conditional distributions is 1.02, 1.35, and 3.05; and the base-to-tuned correlation of sequence log-likelihoods falls from 0.98 on the two lightly tuned variants to 0.77 on the length-normalized one. Tuning moved the energy substantially, and moved it most where the reward was most aggressive. The linearization correlation on those same tuned energies nonetheless stays flat, as Table 9 records. The energy moved substantially and its navigability by gradient did not change, which rules out the reading that the samplers agree only because the models agree.

GFlowNet variant	Mean abs. log-lik. diff. (nats)	Next-token KL	Base-tuned Pearson
len_beta= 0, 500	31.0	1.02	0.98
len_beta= 0, 2000	36.0	1.35	0.98
len_beta= 1, 500	57.3	3.05	0.77

Table 11: Divergence between the base energy and each GFlowNet-tuned energy: mean absolute change in sequence log-likelihood, next-token KL between the conditional distributions, and the base-to-tuned Pearson correlation of sequence log-likelihoods.

5.12 Constrained Generation with an Additive Sentiment Term

RQ4 asks whether an additive constraint can steer generation toward a target attribute, as the plug-and-play formulation of (2) requires. The constrained-generation extension adds a sentiment constraint to the energy in the manner of Kumar et al. (2022) and Qin et al. (2022) and measures whether the target sentiment can be induced. The constraint gradient is a different object from the fluency gradient, so the test is run rather than inferred. Table 23 reports the steering effect for the discrete sampler across the five constraint modes on the infilling task with a positive-sentiment target. Before the numbers, one mechanism has to be flagged: a small unconditional negative drift in the underlying generations dominates the raw steering gains, so the raw gains are reported only to be set aside in favour of the paired contrast that cancels that drift, defined below. The steering gains are small and inconsistent in sign. The full combined energy, the intended plug-and-play configuration, produces a gain of -1.0 ; the constraint-only mode produces -8.0 , worse than the baseline; and no mode produces systematic movement toward the target sentiment. The generated text shows the same degeneracy observed throughout: a representative infilled span reads **The rostr Bill of H PM americanus bears one or connected sp**, with the recovered tokens in bold against the original *rostrum of H. americanus bears one or more*, so the masked positions have been filled with locally plausible but globally incoherent tokens. The raw steering gains, tabulated in Appendix A.6 rather than here because the text immediately discounts

Setup (continuation)	Target	<i>cons_only</i>	<i>cons_random</i>	Contrast
MuCoLa continuous baseline	negative	79.3	52.0	+27.3
MuCoLa continuous baseline	positive	−10.0	−46.7	+36.7
Discrete sampler (“ours”)	negative	−2.7	−2.7	0.0
Discrete sampler (“ours”)	positive	3.3	2.7	+0.7

Table 12: The paired constraint-direction contrast, *cons_only* minus *cons_random*, on the free-form continuation task. Gains are the change, in percentage points, in the fraction of $n = 150$ generations per arm classified as the target sentiment relative to the unconstrained baseline. The contrast cancels the shared sentiment drift that makes the raw gains untrustworthy. The result files record only these aggregate fractions, not per-generation labels, so per-sample paired gains and their confidence intervals are not available; each contrast is a point estimate.

them, are not the statistic to trust. Across setups every arm, including the pure-fluency *lm_only* baseline and the fully random arm, flips the sign of its gain when the target label is switched from positive to negative, which is the signature of a fixed sentiment drift in the underlying generations that swamps any contribution from the constraint. A gain that moves with the target regardless of whether the constraint gradient is even present cannot be read as steering. The statistic that cancels this shared bias is the paired contrast between the constraint-only arm and the arm in which only the constraint gradient is replaced by noise, *cons_only* minus *cons_random*, because the two share the same fluency dynamics and differ only in whether the constraint direction is real.

Measured this way the picture is setup-dependent, as Table 12 sets out. On the discrete sampler of this thesis the contrast is essentially zero on both target labels, so the constraint direction buys no control there. On a MuCoLa-style continuous baseline running free-form continuation, the contrast is +27.3 points for a negative-sentiment target and +36.7 for a positive one, which does indicate that the constraint gradient’s direction carries some signal in that setting, unlike the fluency gradient, whose direction was found to carry none anywhere.

That signal has to be read with a caveat that lives in the constraint-only arm. When the energy is driven by the constraint alone, with no fluency term to hold the sample on the manifold of natural text, the optimization leaves that manifold: it produces text that maximizes the classifier’s response while ceasing to be fluent language. A sentiment classifier evaluated on such text is reporting on a distribution it was never trained for, so its outputs there are not a trustworthy measure of sentiment, and the large *cons_only* contrast is at least partly an artifact of scoring off-manifold text rather than evidence of controllable generation. The safe conclusion is the one the discrete sampler gives directly: on the energy landscape this thesis studies, the additive constraint does not produce reliable control, and the one place a constraint-direction signal does appear is entangled with the departure from the fluent manifold that the constraint-only objective causes.

On the frozen autoregressive energy landscape studied here, plug-and-play constraint steering therefore failed. Section 5.14.4 reports the separate question of what happens on a score-trained landscape, and Section 6.1 keeps the two answers apart.

5.13 The Final-Position Case

This section adds no new claim. It confirms the result already established in Sections 5.5 and 5.7 at one position where the argument can be made exactly rather than statistically, and exactness is its distinct contribution.

The self-and-future decomposition of Section 5.7 showed that the input-embedding gradient sees a substituted token only through its effect on the tokens that follow it. At the last position there are no following tokens. Under causal attention the input embedding of the final token feeds only the logits that would predict a next token which does not exist, and those logits appear in no term of the sequence log-likelihood of (3). The statement to be exact about is therefore this one: the gradient of the sequence log-likelihood, taken with respect to the *final token’s input embedding*, under the shifted causal indexing of (3) in which position t predicts token $t + 1$, is exactly zero. It is not a claim that every gradient involving the final token vanishes; gradients with respect to earlier positions, or with respect

to model parameters, are unaffected. The energy difference between two candidate final tokens, by contrast, is exactly the model’s conditional log-probability ratio between them given the fixed preceding context. At that one position the energy is maximally informative about which token is better and the input-embedding gradient carries none of that information.

A direct experiment measures the consequence. The masked position is placed at each of three points that differ only in how many scored terms of the log-likelihood its embedding can reach: the final token, which feeds no realized prediction; the second-to-last token, which feeds exactly one; and a middle token, which feeds many. Only that one position is corrupted, the surrounding context is left clean, and the same corrupted sequence is handed to every arm, so the sole variable is the amount of downstream context the input-embedding gradient can act on. The recovery arms are the discrete Langevin sampler with the true gradient (policy) and with a random direction of equal norm; three energy-only methods that never touch the gradient (argmax and top- k rescoring of the left-context conditional, and an independence Metropolis sampler that proposes from that conditional and accepts by the exact sequence-log-likelihood ratio); and a recorder of the raw language-model gradient norm the sampler consumes at each step.

Table 13 reports the outcome. At the final position the language-model gradient norm the sampler consumes is 0.0000 in both mean and maximum over roughly fifteen thousand gradient evaluations: not a small number but an exact zero, the theorem realized inside the live sampler. With that gradient the discrete sampler recovers the target token in 0.0% of two hundred sequences whether it uses the true direction or a random one, and the paired policy-minus-random difference straddles zero on every metric (rank difference -17.3 , 95% CI $[-513, +499]$). The energy at the same position is fully usable: the independence sampler that proposes from the conditional and accepts by the exact sequence-log-likelihood ratio accepts 100.00% of its moves in both mean and minimum, since target proportional to proposal forces an acceptance probability of one. That also certifies that the implemented energy carries no term beyond the sequence log-likelihood. The energy-only methods recover the token between 34.5% and 40.0% of the time where the gradient arms recover

none.

The second-to-last position answers the one open empirical question the design poses. With a single downstream scored term the gradient is no longer zero, its mean norm being 12.945, and it is still not usable: the policy arm remains at 0.0% recovery and indistinguishable from the random arm, while the energy-only methods recover roughly half the tokens. The null is therefore not confined to the zero-gradient edge case: it persists once the gradient becomes nonzero. Across all three positions the discrete Langevin arms never recover the target in six hundred trials, while the exact-energy methods recover between eighteen and fifty-five percent.

Arm	Exact recovery (%), by downstream scored terms		
	Final (0)	Second-to-last (1)	Middle (many)
DLS, true gradient (policy)	0.0	0.0	0.0
DLS, random direction	0.0	0.0	0.0
Conditional argmax (energy only)	40.0	50.0	18.0
Top- k rescore (energy only)	40.0	55.0	39.0
Independence MH (energy only)	34.5	49.5	31.5
DLS-policy LM gradient norm (mean)	0.000	12.945	23.976
Independence-MH acceptance (%)	100.0	63.4	30.3

Table 13: The final-position experiment, $n = 200$ sequences per condition. The three columns differ only in the number of downstream scored terms the masked token’s input embedding can influence. Upper block: exact recovery per arm. Lower block: the mean language-model gradient norm the discrete sampler consumes, and the independence sampler’s acceptance rate.

Figure 6 draws the three quantities together against the amount of downstream context. As the masked position moves from the end of the sequence into the interior, the gradient norm the sampler consumes climbs from zero and the energy-only sampler’s acceptance falls from one hundred percent, because the energy begins to weigh downstream terms the left-context proposal ignores. The policy-minus-random gap stays at zero across the whole range.

5.14 The Diffusion Positive Control

The diagnosis of the preceding sections is causal: it attributes the failure of gradient guidance not to the sampler and not to the energy but to the training objective that shapes the energy. An autoregressive model is trained only to predict the next token, and so it is never asked, at any noise level, in which direction the data density increases. A model trained to denoise is asked exactly that, and by the identity that denoising learns the score (Vincent, 2011) it acquires the very directional field the autoregressive gradient was found to lack. If the training objective is truly the cause, then a score-trained model on the same tokenizer should provide a usable local direction where the autoregressive gradient did not, and should perform the revision operation that the gradient machinery could not. This section reports that control, run on the score-entropy discrete diffusion (SEDD) model of Lou et al. (2024) at two scales, small and medium, which share the GPT-2 tokenizer and drop into the existing masked-recovery harness without retokenization. The results are pilots, on a smaller and differently trained model than the frozen autoregressive backbone, so the comparison is qualitative and directional by design; they are not, and are not read as, a claim that the diffusion model beats the domain-tuned autoregressive baselines in absolute terms.

One clarification governs the comparison throughout, because it is easy to state the contrast too loosely. The relevant difference between the two model families is not that one commits tokens within a pass and the other does not; absorbing diffusion also commits a token at each denoising step and does not thereafter change it. The difference is at the level of the revision *operation*. To revise token i , a diffusion model re-masks that position and re-denoises it conditioned on both sides, which is an in-distribution bidirectional query it was trained to answer, whereas an autoregressive model has no conditioning path from the right context to a token at all. It is the availability of that bidirectional conditioning query, not the mere act of committing, that separates the two.

This clarification also bounds what the control can attribute. SEDD differs from GPT-2 Large in more than one respect at once: in training objective, in scale, in

training corpus, and in conditioning direction. The design isolates the direction signal within the sampler, by holding the chain, the energy and the task fixed, but it cannot by itself separate *score training* from *bidirectional conditioning*, because SEDD has both and the autoregressive baselines have neither. Section 5.15 runs the experiment that separates them.

5.14.1 Linearization on the Unified Corpus

The first measurement repeats the linearization diagnostic of Section 5.7 with the diffusion score in place of the autoregressive gradient, on exactly the same sequence set used for the reference baselines, so that the numbers sit on one axis. For each sequence and one masked position, a cheap one-pass surrogate, the model’s denoising log-preference for filling the position, is correlated against an expensive per-candidate truth, the measured effect of committing each candidate on the reconstruction of the held-out remainder of the sequence. Table 14 reports the Spearman correlation, pooled and by embedding-distance stratum. Under the autoregressive objective the surrogate is statistically indistinguishable from zero at every distance, with the magnitude of the correlation below 0.06 throughout. Under the diffusion objective, on the same tokenizer and the same four hundred thousand candidate pairs, the correlation is clearly positive in every stratum and at both scales, with a pooled value near 0.13 and a per-sequence mean near 0.18, and the medium model exceeds the small one in the near stratum, the regime where a first-order surrogate is most expected to apply. On a second corpus of narrative text the same measurement gave a pooled correlation of 0.184 and a near-stratum correlation of 0.306; the near-stratum value is corpus-dependent and is not carried over, but the sign and order of magnitude are stable across both corpora.

5.14.2 Native Recovery of a Flipped Token

A usable local direction is a necessary condition for the capability the gradient machinery lacked, but not obviously a sufficient one, so the second measurement asks the diffusion model to perform the operation itself. On the same corrupted sequences

Model / objective	ALL	near	mid	random	per-seq mean
Autoregressive gpt2-large (gradient)	−0.011	+0.027	−0.041	−0.048	−0.011
SEDD-small (score)	+0.130	+0.097	+0.115	+0.126	+0.184
SEDD-medium (score)	+0.133	+0.148	+0.122	+0.110	+0.179

Table 14: Spearman correlation between the cheap one-pass surrogate and the expensive per-candidate truth, on the unified WikiText-2 sequence set ($n = 200$, four hundred thousand candidate pairs), pooled and by embedding-distance stratum.

the reference baselines used, the flipped token is re-masked and the position is re-denoised conditioned on both sides, a single native draw with no reranking and no sampling of many candidates. Table 15 reports exact-match recovery, top-five recovery, and the KL divergence under the autoregressive model that the whole section uses as its yardstick. Both scales recover the exact token in roughly a third of cases and place the true token in the top five two-thirds of the time, and the resulting KL, near 3.7, sits well below the untouched corruption at 9.14 and well below the flagship Langevin configurations near 6.5, with the medium model at least matching the small one on every metric.

Method	Exact (%)	Top-5 (%)	KL under gpt2-large
SEDD-small, native recovery	32.5	61.5	3.75
SEDD-medium, native recovery	35.0	67.0	3.73
<i>Reference rows on the same task (autoregressive gpt2-large)</i>			
Ground-truth fill (floor)	100	—	0.00
Top- k rescore (one forward pass)	33	—	4.43
Gibbs (gradient-free)	18.5	—	6.69
Flagship Langevin	0.0	—	≈ 6.5
Untouched corruption (ceiling)	0.0	—	9.14

Table 15: Native diffusion recovery of one flipped token on the reference sequence set, with autoregressive reference rows on the identical task. The reference rows are for placement only, since scale and training domain differ.

5.14.3 The Repaired Signal Inside the Exact-Energy Chain

The recovery result changes the model, the sampler, and the energy all at once, so it does not by itself isolate the direction signal as the repaired component. A cleaner test holds the thesis’s own machinery fixed and swaps in only the direction. The hybrid sampler runs a position-wise independence Metropolis chain on the same sequences and positions, accepting by the *exact* autoregressive sequence-log-likelihood energy used everywhere in this section, and changes exactly one thing: the proposal distribution is the diffusion model’s one-pass bidirectional log-preference over the vocabulary rather than the autoregressive left-context conditional. Table 16 reports the outcome against three reference arms run on the identical task: the autoregressive left-conditional independence sampler, and the discrete Langevin policy and random arms.

Swapping only the direction signal, inside the same Metropolis-corrected chain, on the same exact energy and the same task, changes the outcome. The gradient arm recovers the token in 0.0% of cases, the autoregressive left-conditional proposal in 23.5%, and the diffusion-proposal hybrid in 38.5% at the small scale and 39.0% at the medium scale, with acceptance rising from 34.4% for the left-conditional to above 40% for the hybrid because the bidirectional proposal, seeing both sides of the mask, more often lands on tokens the exact energy will accept.

Sampler (same exact energy)	Proposal	Exact (%)	Top-5 (%)	Accept (%)
Hybrid, SEDD-medium	bidirectional score	39.0	67.0	43.3
Hybrid, SEDD-small	bidirectional score	38.5	61.5	40.0
Left-conditional independence	AR left context	23.5	34.5	34.4
Discrete Langevin, policy	AR gradient	0.0	34.5	–
Discrete Langevin, random	random direction	0.0	34.5	–

Table 16: The hybrid sampler, which changes only the proposal direction inside the exact-energy Metropolis chain of this thesis, on the reference sequence set. All arms accept by the same autoregressive sequence-log-likelihood energy.

5.14.4 Classifier-Guided Steering (Exploratory Follow-Up)

The positive control proper is complete at Section 5.14.3. A further question, whether the repaired landscape also admits the *constraint* steering that failed on the autoregressive energy in Section 5.12, was explored with a classifier trained to read the partially masked states the diffusion model occupies mid-denoising. It is reported as exploratory, and in full, in Appendix A.5. Four findings summarize it. The constraint direction carries signal, moving the guiding classifier’s own verdict toward the target at every strength and for both labels, by +9.3 to +25.7 points on-domain. Guidance steers the classifier that guides it more reliably than it steers an independent judge. Transfer to that independent judge is bounded not by a fixed reachable direction but by how far the two instruments agree on the text being scored, agreement rising from 56 to 64% off-domain to 72% on generated and 80% on real on-domain text. And an on-domain trust region removes the fluency cost that unconstrained guidance incurs, holding span negative log-likelihood at or below the unguided value in three of four cells against a climb from 7.1 to 11.0 nats without it.

5.15 Separating Bidirectionality from Score Training

The diffusion result leaves one variable unresolved, and it is the one the causal claim turns on. A bidirectional masked language model is the discriminating control: RoBERTa (Liu et al., 2019) is *not* score-trained, but it *is* bidirectional and it answers exactly the query the revision operation needs, the distribution of a token given both of its contexts. If an MLM proposal also works in this chain, the operative variable is bidirectional conditioning and not the score objective.

The experiment holds everything but the proposal fixed against the autoregressive runs: the same WikiText-2 sequences, the same deterministic per-index corruption routine, the same exact GPT-2 Large energy, the same Metropolis–Hastings accept/reject, and the same KL metric. The proposal is an independence sampler, so its Hastings ratio $q(x_{\text{cur}})/q(x_{\text{prop}})$ is available in closed form. RoBERTa and GPT-2 index different vocabularies, so a once-off bridge maps each RoBERTa token whose

surface string re-encodes to exactly one GPT-2 token; it covers 99.95 percent of the vocabulary, and no ground-truth token in the evaluation set fell outside it.

One control is needed before the result can be read, because a recovery rate far above the rest of the study invites the suspicion of leakage. Replacing the MLM conditional with a *uniform* draw over the same bridged vocabulary, through identical code, gives 0.5 percent exact recovery and a final KL of 6.538 at a 9.48 percent acceptance rate. Those figures reproduce the flagship Langevin sampler (0.0 percent, 6.541, 9.98 percent) to three significant figures, from a separate implementation. The pipeline therefore carries no information of its own, and it independently confirms the central measurement of Section 5.2: a gradient-guided discrete Langevin chain on this energy performs exactly as a uniform-proposal chain performs.

Table 17 places every proposal that has been run in this chain on one axis. The ordering is unambiguous. The two arms with no output-side conditioning at all, the input-embedding gradient and its norm-matched random control, sit at zero recovery and are indistinguishable from a uniform draw. Adding left-only conditioning lifts recovery to 23.5 percent. Adding bidirectional conditioning lifts it to 38.5 to 39.0 percent with a score-trained model, and to 44.5 percent with one that is not score-trained.

The conclusion is that score training is not the operative variable. A model trained with ordinary masked-language-model cross-entropy, and never with a score-matching or score-entropy objective, supplies a proposal that recovers the corrupted token more often than either score-trained diffusion model tested. What the working proposals share is not their objective but their access: all of them read the candidate’s probability from the model’s output side, and the two bidirectional ones can additionally condition on the right context, which is the query an in-place revision actually poses.

This revises, rather than qualifies, the attribution offered above. The earlier reading, that the training objective is the source of the failure, is too strong in one direction and too weak in another. Too strong, because an objective is not required: bidirectional conditioning is enough. Too weak, because even a purely left-to-right

Proposal	Conditioning	Score-tr.	Exact (%)	Mean KL
Input-embedding gradient	none	no	0.0	6.854
Norm-matched random dir.	none	no	0.0	6.584
Uniform over the vocabulary	none	—	0.5	6.538
AR conditional	left only	no	23.5	5.170
SEDD-small concrete score	bidirectional	yes	38.5	3.120
SEDD-medium concrete score	bidirectional	yes	39.0	3.017
RoBERTa-large masked LM	bidirectional	no	44.5	2.737

Table 17: Every proposal run inside the same Metropolis–Hastings chain on the same frozen GPT-2 Large energy, the same 200 WikiText-2 sequences and the same corruption seed. Only the proposal differs. Recovery tracks what the proposal can condition on, not whether it was trained with a score objective.

autoregressive conditional, the very model this thesis studies, reaches 23.5 percent when read from the output side, against 0.0 percent when differentiated with respect to its input embeddings. Both observations point the same way as Section 5.7.1: the information the samplers need is present in the frozen autoregressive model and is discarded by the choice of derivative, and what a bidirectional model adds on top is the right context, not a repaired gradient.

Taken together, the pilots complete the diffusion control. The signal the autoregressive gradient lacked is present in the diffusion score, the recovery the gradient machinery could not perform is performed natively, and the repaired direction works inside the thesis’s own exact-energy chain. The combined GFlowNet and diffusion results locate the failure outside the sampling algorithm, but Section 5.15 identifies the operative variable more precisely than “the training objective”: what the working proposals share is access to the model’s output side, and what the bidirectional ones add is the right context. A score objective is sufficient to supply that access and is not necessary for it.

6 Discussion

This section answers each research question against the evidence of Section 5, sets out the mechanism the findings share, relates them to the prior work of Section 3, and states the scope and the implications.

6.1 Answers to the Research Questions

Each research question is answered below against the section of Section 5 that establishes the answer.

RQ1 asked whether the frozen likelihood can be sampled effectively with faithful Langevin dynamics on the discrete manifold, and if not, why. It is answered in the negative across three sections. Section 5.1 shows that the sampler cannot even be made to move without model-specific calibration, which already denies the strong plug-and-play reading. Section 5.5 shows, through the direction-versus-magnitude ablation, that in the tested token-substitution settings the input-embedding gradient of frozen autoregressive sequence likelihood provided no reliable proposal advantage over a norm-matched random direction, and Section 5.6 carries that null into configurations where the gradient term actually shapes the proposal, which the flagship configuration’s near-uniform proposal could not do. The answer is nonetheless narrower than a blanket negative, and Section 5.7.1 says how: the same measurements show that the one-hot gradient of the very same frozen likelihood ranks candidate substitutions well, so what fails is the input-embedding derivative rather than gradient guidance on a frozen autoregressive model as such. Section 5.7 gives the mechanism: the surrogate the sampler ranks by is uncorrelated with the true energy change over the whole admissible range of substitution distances, so the gradient is inapplicable rather than merely imprecise, and the input-embedding gradient is in addition structurally blind to the self-term of the energy change. Section 5.13 makes the same point exactly at one position: the gradient of the sequence log-likelihood with respect to the final token’s input embedding, under the shifted causal indexing in which position t predicts token $t + 1$, is exactly zero, while the energy differ-

ence between candidate final tokens is exactly the model’s conditional log-ratio. Section 5.14 adds the positive control, in which a score-trained model on the same tokenizer supplies a usable local direction and performs the recovery the gradient-guided sampler did not. The proposal’s RQ2b, on whether energy-based recovery improves on ordinary autoregressive decoding, resolves in the same place and in two directions: it is negative for the gradient-guided samplers, which do not beat the autoregressive baselines, but affirmative for exact-energy rescoring, the gradient-free top- k rerank of Section 5.5.2 that reaches a final KL of 4.43, better than any Langevin configuration and than the argmax and sampling baselines.

RQ2 asked about the role of the Metropolis–Hastings correction in each sampler. It is answered in Sections 5.2 and 5.3, and the answer is that the correction plays opposite roles in the two samplers, which is itself informative. For the discrete sampler the correction is beneficial, and Section 5.2 shows why: its proposal is numerically uniform over the vocabulary at every step, so without the correction the chain is a random walk that does not converge, and with it the accept/reject filter supplies the entire selection pressure. Here theoretical correctness and empirical performance align, but for a reason that is unflattering to the method: what works is the filter, not the guidance. For the continuous sampler the same correction is disabling: Section 5.3 shows that it rejects almost every boundary-crossing move, and the decomposition of the acceptance ratio attributes the collapse to the proposal term. The mechanism is that the drift derived from the differentiable pathway of Section 2.4 is not Lipschitz across a cell boundary; the projected energy of (8) is piecewise constant and is never differentiated. The correction does not make the continuous sampler work: it shows that the continuous sampler was functioning as a biased optimizer, so that its apparent successes are successes of optimization under early stopping rather than of sampling.

RQ3 asked whether amortization with a GFlowNet escapes the failure. It is answered in Section 5.11, and the answer is no for the variants tested. Three failure modes appear, capacity collapse, length collapse, and reward hacking, and the unifying experiment shows that Langevin sampling on the amortized energy is indistinguishable from sampling on the untuned base even though the tuning moved

that energy by tens of nats. The defensible statement of the finding is that amortization did not escape the observed problems in the tested setup, and that modifying the model through the GFlowNet objective did not produce an energy whose local gradient became more useful. It is not established that no amortized method could work with such a reward, since a GFlowNet’s failure can also turn on reward design, training stability, policy parameterization, capacity, exploration, termination handling, or reward scaling.

RQ4 asked whether an additive constraint can steer generation on this energy landscape. The answer has two statements, which are kept apart because they concern two different landscapes.

First: on the frozen autoregressive energy landscape, plug-and-play constraint steering failed. Section 5.12 reports that on the discrete sampler the trustworthy paired contrast between a real and a randomized constraint direction is essentially zero, and that the one setting in which a constraint-direction signal does appear, the constraint-only arm on a continuous baseline, is entangled with the departure from the fluent manifold that a constraint-only objective causes, so a classifier scoring that text is scoring a distribution it was not trained for.

Second: on a score-trained diffusion landscape, constrained steering became partly possible, subject to fluency and classifier-alignment limits. Section 5.14.4 reports that a classifier trained on the noisy states it scores moves an independent judge, in one direction per setting, once a trust region holds fluency fixed, with the reachable direction tracking how far the guiding and judging instruments agree on the text being scored.

These are not one answer. The second uses a different, score-trained landscape and is therefore partly a repair experiment rather than a direct answer about the original autoregressive energy, and it is labelled exploratory where it is reported.

6.2 The Unified Mechanism

The failures catalogued above share one property of the object all the methods use. Teacher forcing optimizes each local conditional $p_{\text{LM}}(x_t \mid x_{<t})$ on a correct prefix, and modern models satisfy that objective superbly. What it never does is constrain the joint log-likelihood $\log p_{\text{LM}}(\mathbf{x})$ as a function of $\mathbf{x}$ off the data manifold, which is where every energy-based sampler operates. The joint is never asked to be smooth, never asked to be a calibrated measure of quality, and never asked to have a gradient pointing toward better sequences; the results show it has none of the three, per Sections 5.3, 5.8 and 5.7 respectively.

The mechanical reason the *gradient* in particular is the defective part is now precisely locatable. The only term registering how well a token fits its own left context enters through a discrete index into the output softmax, not through the continuous input embedding the sampler differentiates. That blindness is partial in the interior of a sequence and total at the final position, where the gradient with respect to the last token’s input embedding is exactly zero while the energy difference between candidate final tokens is exactly the model’s conditional log-ratio. Section 5.7.1 then shows that the discarded term is not merely absent but recoverable: differentiating with respect to the one-hot input instead returns it in closed form, and substituting that surrogate into the same sampler lifts exact recovery from at most two percent to forty. This thesis refers to the Section 5.5 result by the shorthand *gradient fallacy*, used only here and with its scope attached: it names a bounded empirical finding about the input-embedding gradient, not a conceptual error across the field and not a property of autoregressive likelihoods as such.

Three routes reach that conclusion without sharing machinery. The gradient-guided route differentiates the likelihood and fails, its surrogate uncorrelated with the true energy change and its correction rejecting the moves that change a token. The amortized route never touches the gradient, treating the likelihood as a scalar reward, and fails too, the learned policy collapsing in whichever direction the reward’s incentives push it; the unifying experiment closes the gap, since running the gradient-guided sampler on the amortized energy reproduces running it on the base.

The third route is the ladder of proposals in Section 5.15: within one exact-energy chain, a uniform draw, the model’s own left-to-right conditional, a score-trained diffusion model and a bidirectional masked language model recover 0.5, 23.5, 39.0 and 44.5 percent respectively. Read together, these locate the operative variable as what a proposal may condition on rather than the objective it was trained with, and they show that the information the samplers need was in the frozen model all along.

The likelihood trap qualifies the picture in one further respect. Because low energy is degenerate text on these models, the objective all these routes pursue is not the objective one would want. The inability of the gradient-guided samplers to reach low energy is therefore not straightforwardly a loss: the runs that do optimize the energy effectively, beam search on the length-normalized GFlowNet being the clearest, are the runs whose output degenerates into the strings displayed in Section 5.8.

6.3 Connections to the Related Work

The likelihood trap measured in Section 5.8 is a direct, quantitative reproduction on the study’s own models of the neural-text-degeneration phenomenon that Holtzman et al. (2020) identified: that the most probable sequences under a neural language model are degenerate, and that maximization-based decoding produces worse text than sampling. Section 3 presented that as a known hazard of decoding. The contribution here is to re-derive it as an obstacle for energy-based sampling specifically, since in that framing the degeneracy of high-probability text says that the *target* of the sampling procedure is undesirable rather than that a decoding heuristic should be avoided.

The anisotropy results of Section 5.9 confirm, on GPT-2 Large and Llama-3, the representation-degeneration and embedding-anisotropy phenomena documented by Gao et al. (2019) and Ethayarajh (2019), and add a consequence those works did not draw: because the anisotropy scale differs by roughly a factor of three between the two models, a single step-size schedule cannot transfer between them, which makes the samplers model-specific in a way that qualifies their claim to be plug-and-play.

The Metropolis–Hastings breakdown of Section 5.3 is the empirical counterpart

to a gap noted in Section 3.2: that the energy-based decoding literature frequently omits the correction, and that where it is included, the difficulty of computing the reverse proposal on a projected landscape is seldom examined. This thesis implements the correction faithfully, following the convergence theory of Roberts and Tweedie (1996), and measures what the common omission conceals. On that reading, methods such as COLD (Qin et al., 2022) and MuCoLa (Kumar et al., 2022), which the continuous sampler of this thesis follows, are better understood as biased optimizers than as samplers from the stated energy, and their reported successes as successes of optimization under early stopping. This is a clarification of what the methods do rather than a criticism of those works, which are careful about what they claim. The same reading applies to the annealed stochastic-gradient Langevin dynamics of Welling and Teh (2011). Its guarantee relies on an annealed schedule whose step size shrinks the proposal onto a local neighbourhood; Section 5.2 shows that on this landscape the schedule does not achieve that, because the proposal remains near-uniform over the vocabulary throughout, so the uncorrected chain neither localizes nor converges.

The GFlowNet results of Section 5.11 qualify, without contradicting, the picture presented by Hu et al. (2024), whose formulation and codebase this thesis builds on. Their work showed that amortized sampling from a frozen-model posterior can yield diverse, transferable samples where the reward is well-behaved, and nothing here disputes that. The narrower finding is that when the reward is the raw frozen likelihood used as an energy over free-form sequences, the learned policy collapses in the ways Section 5.11 documents. The axis that reconciles the two is whether the reward defines a landscape with a coherent-text optimum.

The gradient-free samplers isolated in Section 3.2, Mix-and-Match (Mireshtgah et al., 2022) and the twisted sequential Monte Carlo of Lew et al. (2023) and Zhao et al. (2024), confirm rather than contradict the central result. Each samples from a frozen-model energy without differentiating it, and each works. The Gibbs baseline of Section 5.5.2 makes the point in-platform, on the identical energy the Langevin methods use. The frozen likelihood is therefore a serviceable object to *evaluate*, and to search without gradients, and an unusable one to *differentiate* in

the settings tested here, which is why the thesis narrows its claim to the gradient rather than the energy.

6.4 Scope and Limitations

The thesis shows that in the settings tested, the frozen likelihood of a standard autoregressive language model is a poor energy for gradient-guided and amortized sampling on discrete text, and it explains why in terms of the training objective and the geometry of the token embedding space. It does not show that energy-based controllable generation is impossible in general.

Four limitations bear directly on how far the central result can be pushed, and they are stated here rather than left implicit. First, and most importantly, the claim is about one Jacobian slice. Section 5.7.1 shows that the one-hot gradient of the same frozen likelihood carries a usable ranking signal and, substituted into the same sampler, recovers up to 40 percent of corrupted tokens where the input-embedding gradient reaches at most two. The thesis therefore refutes the premise for embedding-space samplers of the MuCoLa and COLD family and not for gradient-guided discrete sampling as such. The result holds across architectures, reaching 41 percent on Llama-3 8B, but it carries a condition rather than an unqualified generality: it requires a configuration in which the surrogate rather than the distance term shapes the proposal, and neither model’s calibrated setting is such a configuration. What the thesis therefore establishes is that the signal is present and usable, not that any off-the-shelf schedule will use it. Second, the flagship ablation was measured at a configuration whose proposal is numerically uniform, and it is the sweep of Section 5.6, at $n = 50$ per cell rather than 200, that carries the null into regimes where the gradient is load-bearing. Third, the correction was applied asymmetrically across the compared arms in the main grid (Section 4.3); this cannot manufacture the null, since it penalizes the policy arm, but it did manufacture the appearance of the gradient being actively harmful, and the corrected reruns cover three cells rather than the whole grid. Fourth, the equivalence claim remains uncertified at the pre-registered margin on the chain-mean statistic, so the honest reading throughout

is “no difference this design can resolve” rather than demonstrated equality.

One formulation deserves particular care, because the thesis set out to test it. The experiments refute the premise that the frozen autoregressive sequence-likelihood gradient can supply the required local score for this class of Langevin-based methods without additional training. They do not refute the training-free premise as such. Training-free methods that do not differentiate the raw joint likelihood remained viable in this study’s own results: the gradient-free Gibbs sampler of Section 5.5.2 is comparable to the best gradient-guided Langevin configuration on the identical energy, and a single top- k resampling pass beats every configuration in the grid, both without any training and without any gradient. Methods that use lookahead, conditional resampling, or another representation may likewise remain viable, and Section 3.2 reviews published members of that family.

Several further caveats bound the empirical claims. The KL-divergence metric depends on the position of the corrupted token within the sequence, which is why, as Section 4.4 explained, the central results are argued from the absence of a gap between conditions rather than from small measured differences; the gradient result should be read as “no reliable difference” rather than as an exact numerical equality. The Llama-3 results support only qualitative cross-architecture claims, because the tokenizer and embedding scale differ from the GPT-2 family. The linearization result is that the surrogate is uniformly uninformative across the admissible range of distances rather than informative up to a sharp radius and useless beyond it, so the phrase “linearization radius” summarizes a decay rather than naming a threshold. The GFlowNet result covers the variants trained here and not amortized inference in general, as Section 6.1 states. And the constrained-generation extension establishes the absence of reliable steering on this energy, not the impossibility of steering on a better one.

Finally, the relationship between the title of this thesis and its content. The original objective was controllable generation via faithful Langevin dynamics, planned in two stages: establish that energy-guided sampling on a frozen model is viable, then add the constraint term and perform sentiment and toxicity control. The first stage did not hold, and the work turned to the question of why rather than building

control on a foundation it had shown to be unsound. What it delivers is a diagnosis of the failure of a prerequisite, with the mechanisms identified, measured, and cross-checked, and with the constrained-generation extension included as evidence of the downstream consequence, rather than a demonstration of successful control.

Several deliverables named in the proposal became moot with that turn, and each is accounted for here. Toxicity control on RealToxicityPrompts was dropped because once sentiment steering failed on the same machinery (Section 5.12), a second attribute on the same energy would only have repeated the negative result. The MAUVE and Self-BLEU diversity suite was dropped because those are distribution-level metrics for open-ended generation and make little sense for fixed-span infilling once the task narrowed to recovery. The planned 70B-scale judge was reduced to an 8B external judge (Section 5.5.3), which was enough to rule out metric circularity without the larger model. The posterior-coverage evaluand was abandoned because the samplers never reached a regime in which coverage is a meaningful question, as the near-uniform-proposal and Metropolis–Hastings results show. And the CommonGen benchmark was not needed, ROCStories being sufficient for the diagnostic and matching the amortization reference implementation of Hu et al. (2024).

6.5 Implications

Locating the defect in the training objective rather than in the sampler makes a testable prediction: a model whose objective does shape a score function over sequences should not fail in the same way. Diffusion language models are such models (Li et al., 2022; Lou et al., 2024), because denoising is equivalent to learning the score (Vincent, 2011; Song and Ermon, 2019; Ho et al., 2020). Section 5.14 runs that control on the score-entropy discrete diffusion model of Lou et al. (2024) at two scales, and every arm of the prediction is borne out: the linearization correlation is clearly positive where the autoregressive gradient was statistically zero, the model natively recovers the flipped token, and replacing only the proposal direction inside this thesis’s exact-energy Metropolis chain lifts exact recovery from zero to thirty-nine percent. The correlations are modest and the diffusion models are smaller and

differently trained than the frozen backbone, so these are pilots and are read as such.

Two implications follow for practice. The first concerns what inference-time control on a frozen autoregressive model can be built from. The energy is usable and its input-embedding gradient, in these settings, is not, so a method that evaluates or reranks under the frozen likelihood has a foundation and one that differentiates it does not, at a cost that Appendix A.3 shows also favours the former. The second concerns what would have to change to recover the gradient-guided route. If the problem is that the likelihood is not a score function, one option is to make it one, by fine-tuning the model so that its sequence log-density is aligned with a target energy under a score-matching objective. That is a lighter intervention than reinforcement-learning-style training of a GFlowNet and is directed at the identified deficiency rather than at a downstream symptom. Only once an energy landscape is known to be navigable does the additive constraint term of the plug-and-play framework become meaningful, since only then is there a fluency energy a sampler can follow for the constraint to be added to.

7 Conclusion

This thesis set out to build controllable text generation on faithful Langevin dynamics, treating the frozen likelihood of an autoregressive language model as an energy function and steering it at inference time with an additive constraint. It tested the premise that construction rests on, and found the premise did not hold in the settings examined.

What was tested. One hundred and forty-five sampling configurations across five energy functions, a supervised fine-tuned GPT-2 Large, three GFlowNet-tuned variants of it, and a Llama-3 8B cross-architecture control, on a masked-token recovery benchmark and on a free-form continuation task. Two samplers were run, one remaining in token space and one operating in the embedding space, under a factorial combination of Metropolis–Hastings correction, gradient normalization,

and annealing schedule length, alongside five diagnostic experiments and a diffusion positive control at two scales.

What was found. In the tested token-substitution settings, following the input-embedding gradient of the frozen autoregressive likelihood was no more effective than following a norm-matched random direction, and exact recovery of the corrupted token was 0.0 percent in 139 of the 145 configurations, never exceeding one percent in the remaining six. The mechanism is that the first-order surrogate the discrete sampler ranks by is uncorrelated with the true change in energy across the whole admissible range of substitution distances, and that the input-embedding gradient is structurally blind to how well a candidate token fits its own left context, exactly so at the final position under the shifted causal indexing. That blindness is a consequence of which derivative is taken, not of the training objective: the same measurements show that the one-hot gradient, which the discrete samplers this literature adapts actually use, carries the discarded term in closed form and correlates with the true energy change at $\rho = 0.60$ to 0.73 over the admissible range. Substituted into this thesis’s own sampler, on the same frozen models and the same corrupted sequences, it recovers forty percent of tokens on GPT-2 Large and forty-one percent on Llama-3, against at most two percent for the input-embedding gradient anywhere in a sweep spanning proposal entropies from uniform to deterministic. The signal the samplers needed was in the frozen model throughout, and the choice of derivative is what discarded it. Two further findings qualify the null itself, and they pull in opposite directions. The discrete proposal is numerically uniform over the vocabulary at every step of every configuration in the main grid, so that grid could not have detected an informative gradient had one been present; a sweep over step size and proposal temperature that spans the full range from a uniform to a deterministic proposal repairs that gap and finds no configuration at which the gradient outperforms a norm-matched random direction. Conversely, the cases in which the gradient appeared reliably *worse* than a random direction did not survive scrutiny: they were produced by an asymmetry in how the Metropolis–Hastings correction was applied across the compared arms, and they disappear once the reverse-proposal term is com-

puted exactly for every arm. The energy itself is not at fault: a single top- k rescore pass and a gradient-free Gibbs sampler both recover coherent text on the identical energy, at a fraction of the cost. Applied faithfully, the Metropolis–Hastings correction improves the discrete sampler and disables the continuous one, the acceptance ratio collapsing through its proposal term at the cell boundaries a nearest-neighbour projection creates. On these models the lowest-energy text is also the most degenerate, so the objective the methods pursue is not the objective one would want. Amortizing the search with a GFlowNet did not escape the failure in the tested setup: three collapse modes appeared and Langevin sampling on the tuned energy was indistinguishable from sampling on the untuned base, although the tuning had moved that energy by tens of nats. As a positive control, a score-trained diffusion model on the same tokenizer supplied a clearly positive local direction where the autoregressive gradient was statistically zero and, substituted as the proposal inside this thesis’s own exact-energy chain, raised exact recovery from zero to thirty-nine percent. A ladder of proposals run in that same chain then localizes the cause more sharply than the objective: a uniform draw over the vocabulary recovers half a percent, which is what the gradient-guided sampler achieves; the frozen model’s own left-to-right conditional recovers twenty-three and a half percent; and a bidirectional masked language model that was never trained with a score objective recovers forty-four and a half percent, more than either diffusion model tested. What separates a working proposal from a failing one is what it is allowed to condition on, not how it was trained.

What remains uncertain. The combined GFlowNet and diffusion results locate the failure outside the sampling algorithm, and the masked-language-model control narrows it further to what a proposal can condition on rather than to the training objective as such, but that remains an interpretation rather than a proof covering all autoregressive language models: the GFlowNet result bounds the variants trained here, not amortized inference in general, and the diffusion pilots use smaller, differently trained models on a qualitative comparison. On the frozen autoregressive landscape, plug-and-play constraint steering failed; on a score-trained diffusion land-

scape it became partly possible, subject to fluency and classifier-alignment limits, and how far that extends beyond one noisy guide at this scale is not settled here.

The main practical implication. The experiments refute the premise that the frozen autoregressive sequence-likelihood gradient can supply the required local score for this class of Langevin-based methods without additional training. They do not refute the training-free premise as such, since gradient-free training-free methods remained viable throughout, in this study’s own baselines as well as in the published work of Section 3.2. For a practitioner building inference-time control on a frozen autoregressive model, the usable object is the energy and its output-side conditionals, not its input-embedding gradient. For the gradient-guided route to be recovered, the model would have to be trained to supply the score the route assumes, which is what the diffusion control indicates and what a score-matching fine-tune of the sequence density would attempt directly.

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A Appendix

A.1 Additional Diagnostic Figures

This section collects supporting figures referenced in Section 5. Each figure is discussed at the point of reference in the results section, as indicated in the captions below.

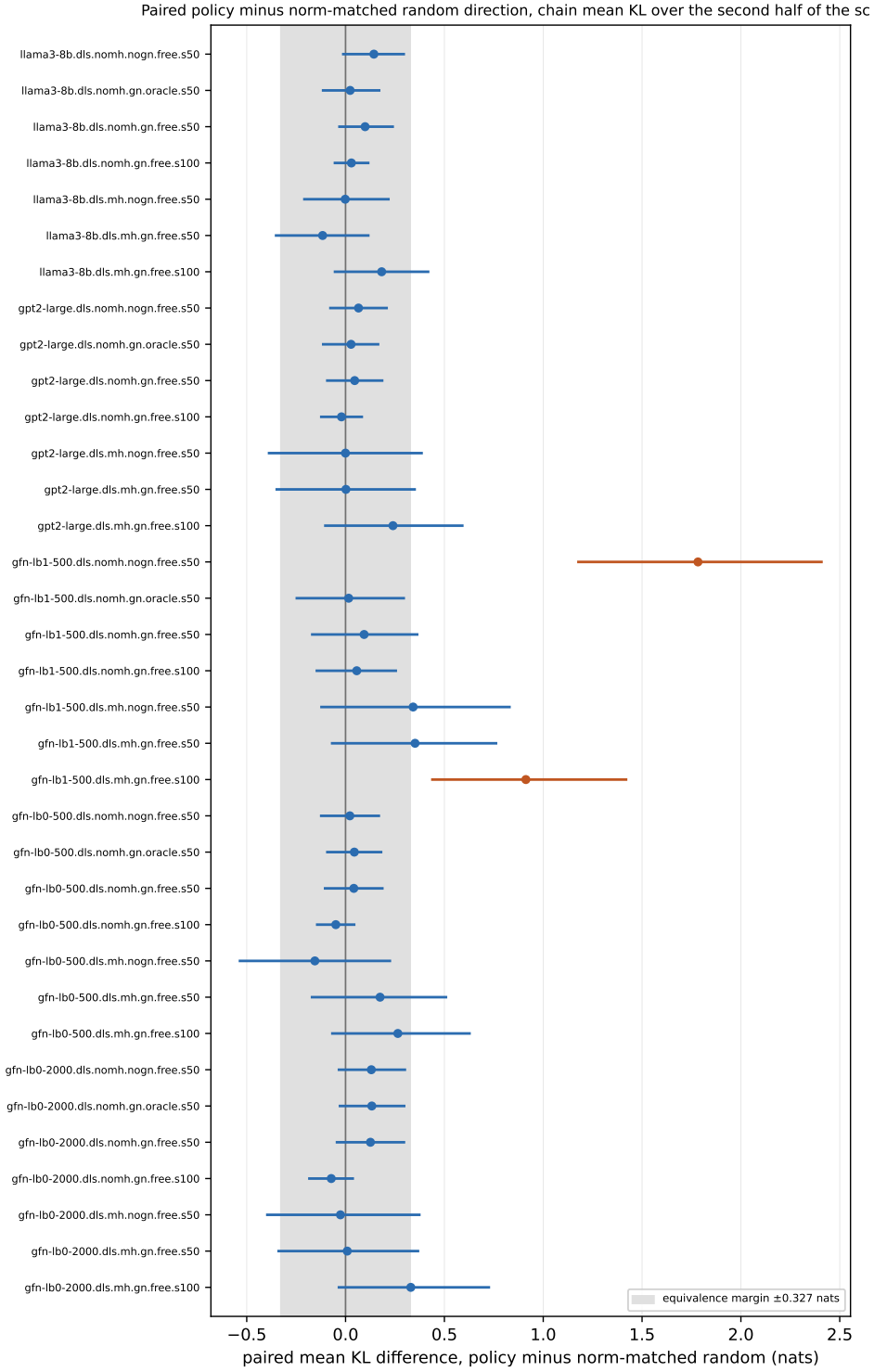


Figure 3: Paired difference in chain-mean KL, policy minus norm-matched random direction, for every configuration in the grid with both arms present (35 configurations). Points are paired means, bars are 95% percentile-bootstrap intervals over $n = 200$ paired sequences each, and the shaded band is the pre-registered equivalence margin of ± 0.327 nats. Intervals whose bounds exclude zero are drawn in the darker colour.

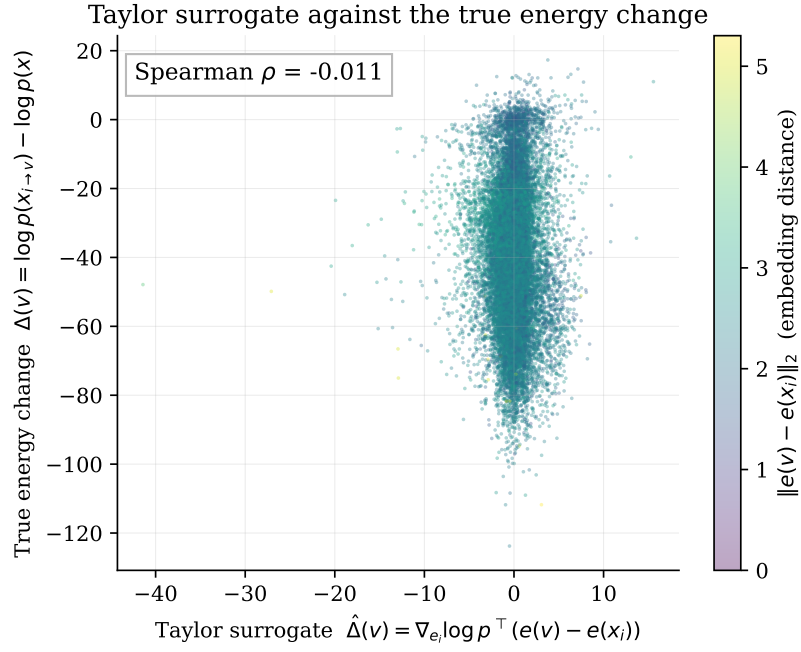


Figure 4: The first-order Taylor surrogate used by the discrete proposal, plotted against the true change in sequence log-likelihood for GPT-2 Large, coloured by the embedding distance of the candidate token. $n = 400,000$ candidate-sequence pairs.

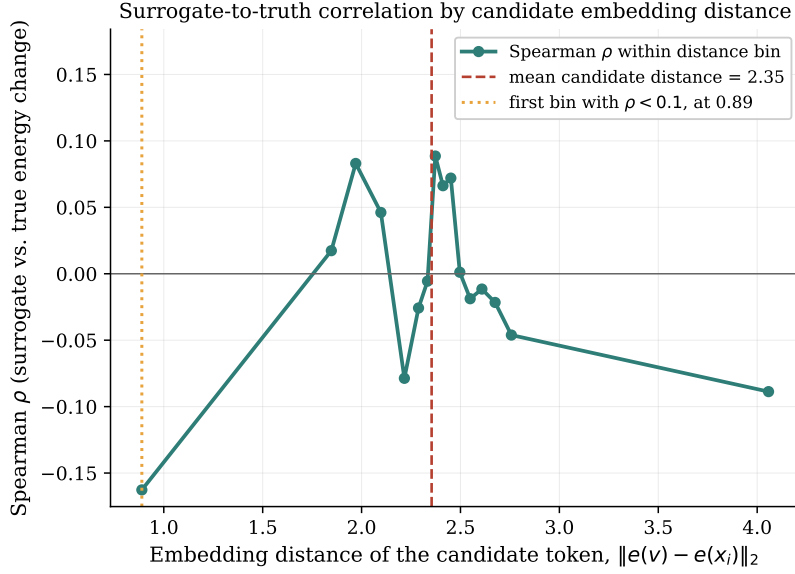


Figure 5: Correlation between the gradient surrogate and the true energy change as a function of the embedding distance of the candidate token, for GPT-2 Large. Dashed line: the mean linearization-candidate distance, 2.35. Dotted line: the smallest binned distance at which the correlation falls below 0.1.

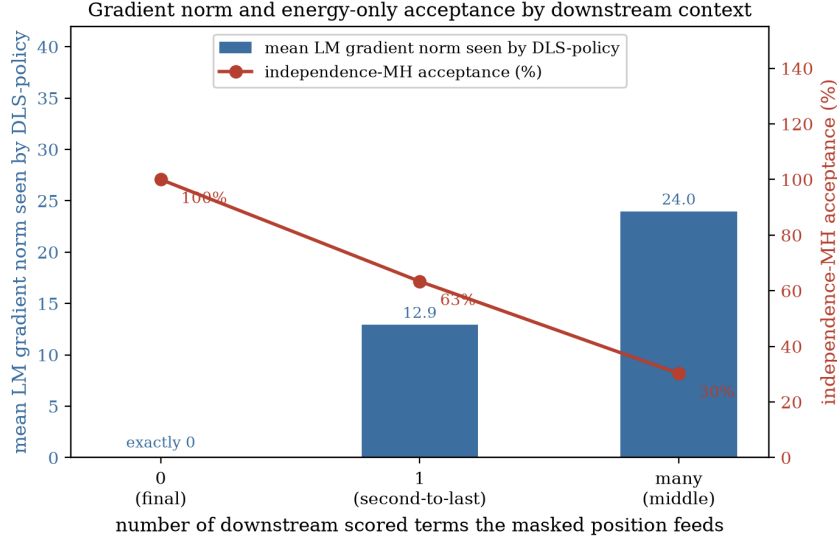


Figure 6: The final-position experiment against the number of downstream scored terms. Left axis, bars: the mean language-model gradient norm the discrete sampler consumes. Right axis, line: the independence sampler’s acceptance rate.

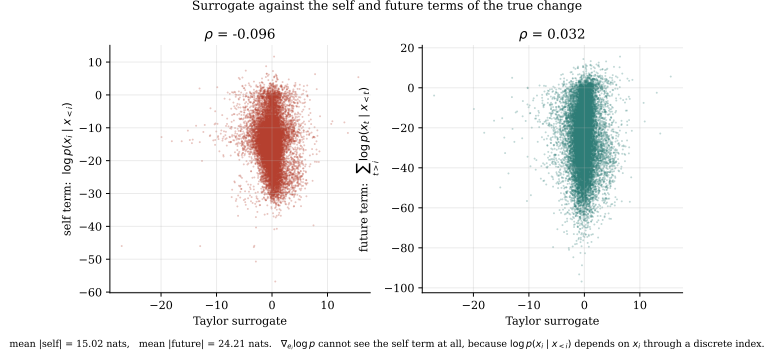


Figure 7: The true energy change decomposed into its self term and its future term, each plotted against the gradient surrogate, for GPT-2 Large. This figure supports Section 5.7.

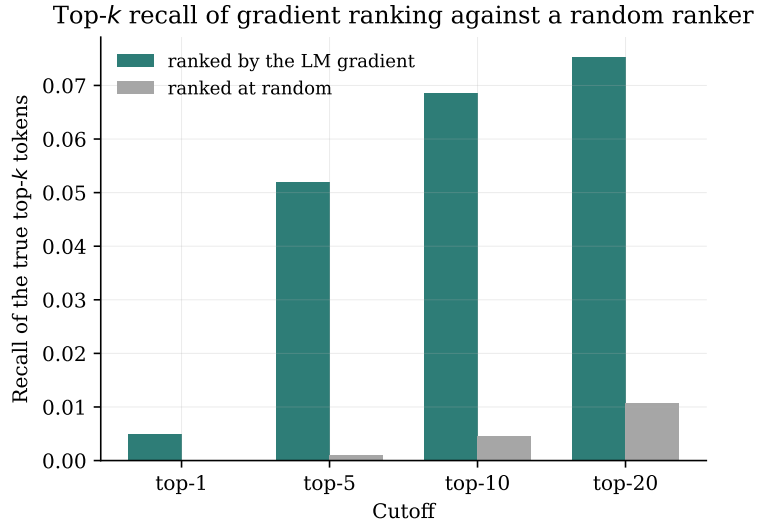


Figure 8: Recall of the true top- k token substitutions by the gradient-ranked top- k , against a random-ranking baseline, for GPT-2 Large. This figure supports Section 5.7.

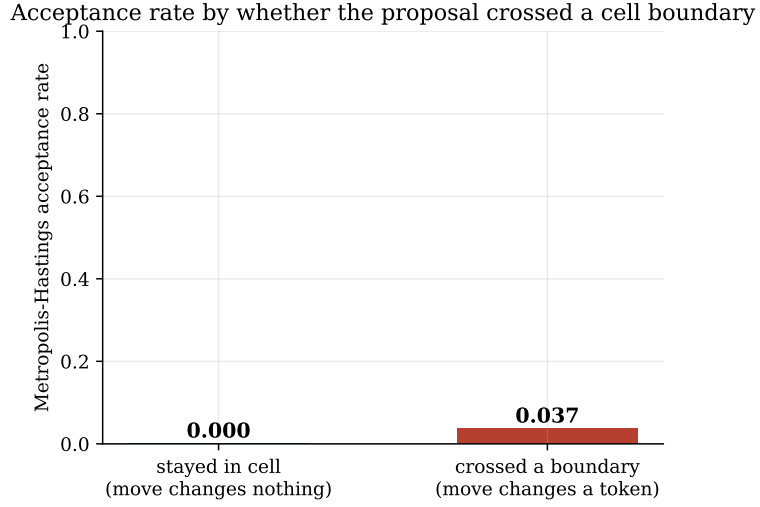


Figure 9: Metropolis–Hastings acceptance rate for the continuous policy sampler with gradient normalization off and the correction enabled (configuration `cls_policy_gnoff_mh`), conditioned on whether the proposal crossed a Voronoi cell boundary. Within-cell moves are accepted essentially never and boundary crossings only 3.7% of the time. This figure supports Section 5.3.

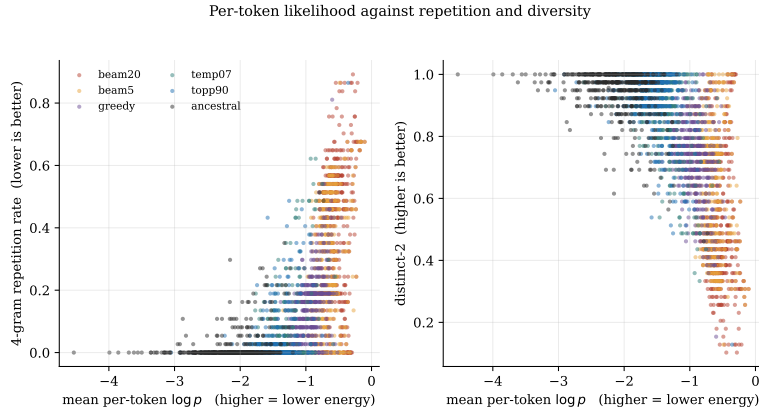


Figure 10: Mean per-token log-probability against 4-gram repetition rate and distinct-2 diversity for GPT-2 Large, one point per generation, coloured by decoding strategy. This figure supports Section 5.8.

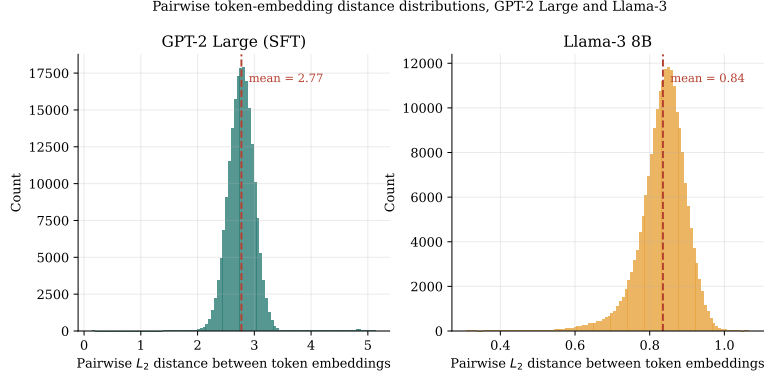


Figure 11: Distributions of pairwise token-embedding distances in GPT-2 Large and Llama-3, plotted on separate scales. This figure supports Section 5.9.

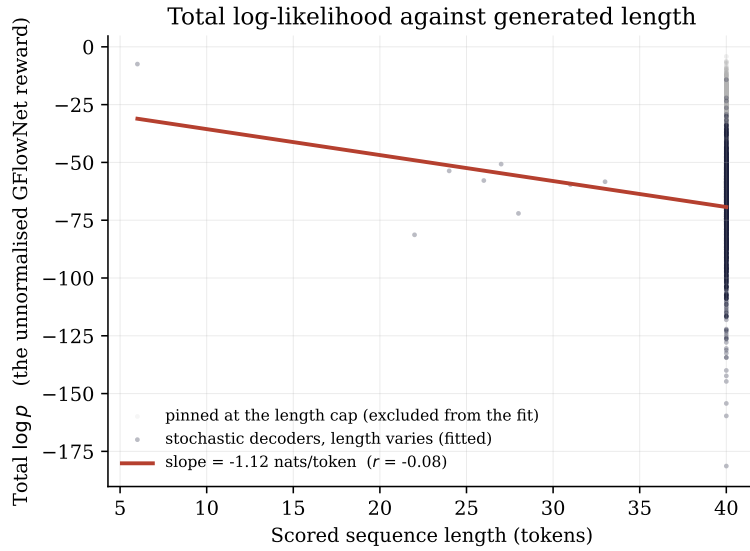


Figure 12: Total log-likelihood against generated length for GPT-2 Large. The regression is fitted over the three stochastic decoders (ancestral, top- p , temperature), where length varies, giving a slope of -1.12 nats per token; the deterministic decoders pinned at the length cap carry no length information and are shown faintly but excluded from the fit. This figure supports Section 5.8 and Section 5.11, where the slope and its censoring caveat are discussed.

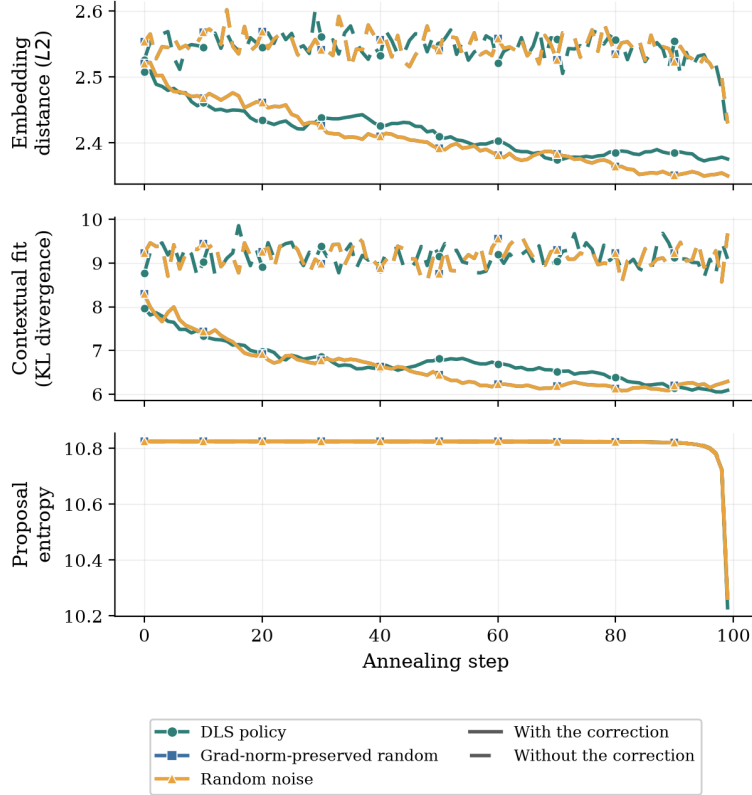


Figure 13: Discrete Langevin Sampler trajectories on GPT-2 Large over a 100-step annealing schedule, averaged across sequences. Panels, axes, line styles and markers as in Figure 1.

A.2 Full Configuration Grid

Table 18 reports the final KL divergence for a representative subset of the full configuration grid across the four GPT-2-family energy functions, drawn from the aggregated results file `final_kl_by_model.csv`. The complete grid comprises the factorial combination of sampler (DLS, CLS), proposal method (policy, grad-norm-preserved, random), Metropolis–Hastings correction (on, off), gradient normalization (on, off), and schedule length (50, 100), evaluated on each of the five energy functions, for a total of 145 configurations.

The count arises as $145 = 5 \times 29$, five energy functions times twenty-nine config-

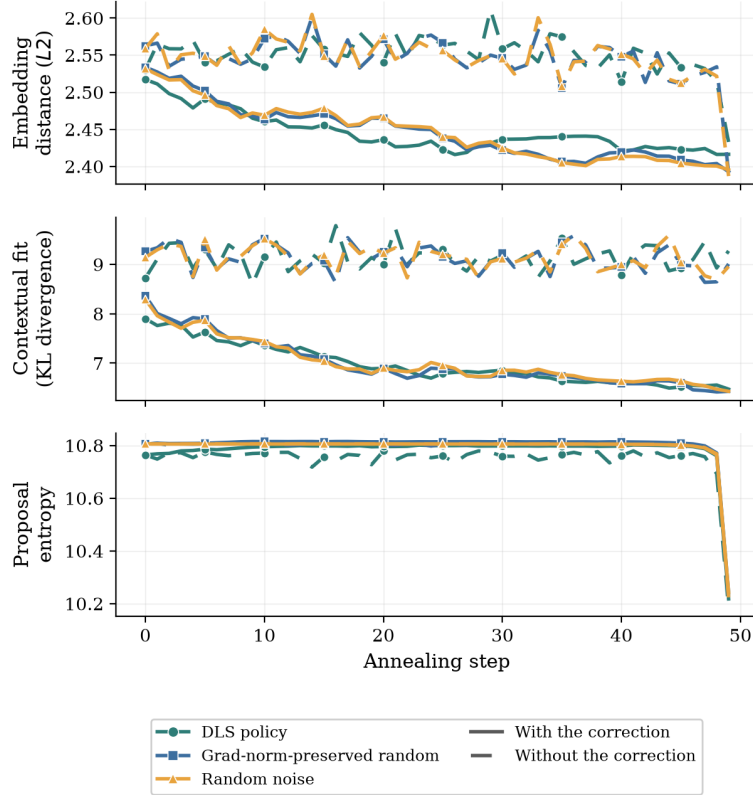


Figure 14: The 50-step schedule with gradient normalization disabled. Unlike the enabled case of Figure 1, where the three proposal methods coincide, the policy, grad-norm-preserved random, and fully random arms here separate into distinct curves. Panels, axes, and line styles as in Figure 1; solid lines are runs with the Metropolis–Hastings correction, long-dashed lines without it.

urations each, and the twenty-nine is worth setting out because the naive factorial overcounts it. A fully crossed design over the five binary and ternary axes above would give $2 \times 3 \times 2 \times 2 \times 2 = 48$ cells per energy function, or 240 in total, but two prunings reduce this. The continuous sampler is run for only the policy and random proposal methods, not the grad-norm-preserved method, and across four of the gradient-normalization and schedule variants, giving 8 CLS configurations per energy function; the discrete sampler is run for all three proposal methods across seven of the correction, normalization, and schedule variants, giving 21 DLS configurations. The sum is $8 + 21 = 29$ per energy function. The count was verified

directly against the result folders, which hold 29 configurations for GPT-2 Large (`results/grid/gpt2_v2`), 29 for Llama-3 (`results/grid/llama`), and 87 across the three GFlowNet variants (`results/grid/gfn`), and $29 + 29 + 87 = 145$.

Configuration	GPT-2 Large	GFN lb0-500	GFN lb0-2000	GFN lb1-500
DLS policy, MH, gn, s50	6.541	6.306	6.721	6.415
DLS policy, no-MH, gn, s50	9.499	9.195	10.085	8.669
DLS random, MH, gn, s50	6.370	6.105	6.548	5.525
DLS gnp-random, MH, gn, s50	6.370	6.105	6.548	5.525
DLS policy, MH, no-gn, s50	6.474	6.009	6.698	6.008
DLS gnp-random, MH, no-gn, s50	6.430	6.001	6.552	5.338
CLS policy, MH, no-gn, s50	9.202	8.834	9.748	9.025
CLS policy, no-MH, no-gn, s50	8.083	7.799	8.198	13.732

Table 18: Representative subset of the final-KL configuration grid, from `final_kl_by_model.csv`. Rows are configurations, columns are the four GPT-2-family energy functions.

This full grid is part of the released artifact set. Under the code-availability statement of Section 4.8, the complete per-configuration results, the aggregated `final_kl_by_model.csv` from which this table is drawn, and the artifact map that links every number, table, and figure in this thesis to the script that produces it and the result file it reads are retained with the code, so the entire grid can be re-generated from the recorded seeds rather than reproduced from this representative subset alone.

A.3 Computational Cost

Because gradient-free methods match or beat the gradient-guided samplers on the same energy, their computational cost is set out here; it differs by more than a constant factor. Table 19 gives the number of network evaluations each method spends to recover a masked span, counted as forward and backward passes, where S denotes the schedule length (either 50 or 100), M the number of masked positions, T the number of Gibbs sweeps, and k the number of candidates a rescoring pass considers.

Method	Forward	Backward	Note
DLS / CLS, no correction	S	S	one gradient per step
DLS / CLS, with correction	$2S$	$2S$	extra gradient at the proposed point
Conditional argmax/sample	1	0	a single forward pass
Top- k rescore	$1 + k$	0	rescore k candidates
Gibbs	MT	0	M positions, T sweeps, no gradient

Table 19: Network evaluations per recovered span, counted as forward and backward network passes. S is the schedule length, M the number of masked positions, T the number of Gibbs sweeps, and k the number of candidates a rescoring pass considers.

The measured wall-clock times are consistent with this accounting and give a sense of the absolute scale on the hardware of Section 4.8. The discrete sampler on the free-form continuation task took 2157 seconds for one hundred sequences, about 21.6 seconds per sequence over a fifty-step schedule; the full suite of gradient-free baselines, including the Gibbs sampler, took 215 seconds for two hundred sequences, roughly one second per sequence; and rescoring the six hundred recovered sequences with the external Llama-3 judge, a single forward pass each, took 24.8 seconds in total. The comparison is stark: the gradient-free baselines reach comparable or better recovery quality at a fraction of the per-sequence cost, because they never pay for the backward pass whose product this thesis finds to be uninformative.

A.4 Geometry of Sampler Trajectories in Embedding Space

This section examines the spatial paths the samplers trace through the embedding space, which gives a concrete counterpart to the acceptance statistics of Section 5.3 and the trajectory summaries of Section 5.4. The primary figure is deliberately *not* a projection. A two-dimensional projection of a 1280-dimensional space cannot carry the argument, as the explained-variance figure below makes plain, so the load-bearing figure reports exact L_2 distances in the full space and the projection is demoted to a single illustrative panel. All distances and all nearest-token decoding here use the full $50,257 \times 1280$ embedding matrix of the model, not any subsample.

Figure 15 is the primary view. For one seeded trace (index 5, chosen once by `np.random.default_rng(0)`), it plots, at every step and on a symmetric-log axis so that exact zeros remain visible, the L_2 distance from the sampler’s state to the ground-truth token embedding, and for the continuous sampler also the distance to the nearest token of any kind, the off-manifold measure. Because `collect_traces` advances the random-number state across configurations, trace index 5 is the same source sentence in every panel but its randomly masked position, and hence its ground-truth token, differs by configuration; each panel therefore uses its own exact ground-truth embedding, and the cross-panel claim is about the on-manifold-versus-off-manifold dynamics, which does not require a shared token. The token strip under each axis shows the decoded token at each change, and for the continuous sampler the steps at which the proposal was rejected, inferred exactly from identical consecutive states, are marked as ticks.

The distances make the geometry exact. The discrete sampler, policy and random alike, sits at distance zero from the token manifold at every step, because its state is always a genuine token embedding; it visits about five distinct tokens over the run and ends about 2.2 units from the ground truth, a token or so away. The continuous sampler with the correction on is frozen in place: its state sits between 118 and 151 units from the nearest token across the run and it changes its projected token about twice in fifty steps, the spatial form of the near-total rejection measured in Section 5.3. With the correction off it visits about 45 of 50 cells and drifts as far as

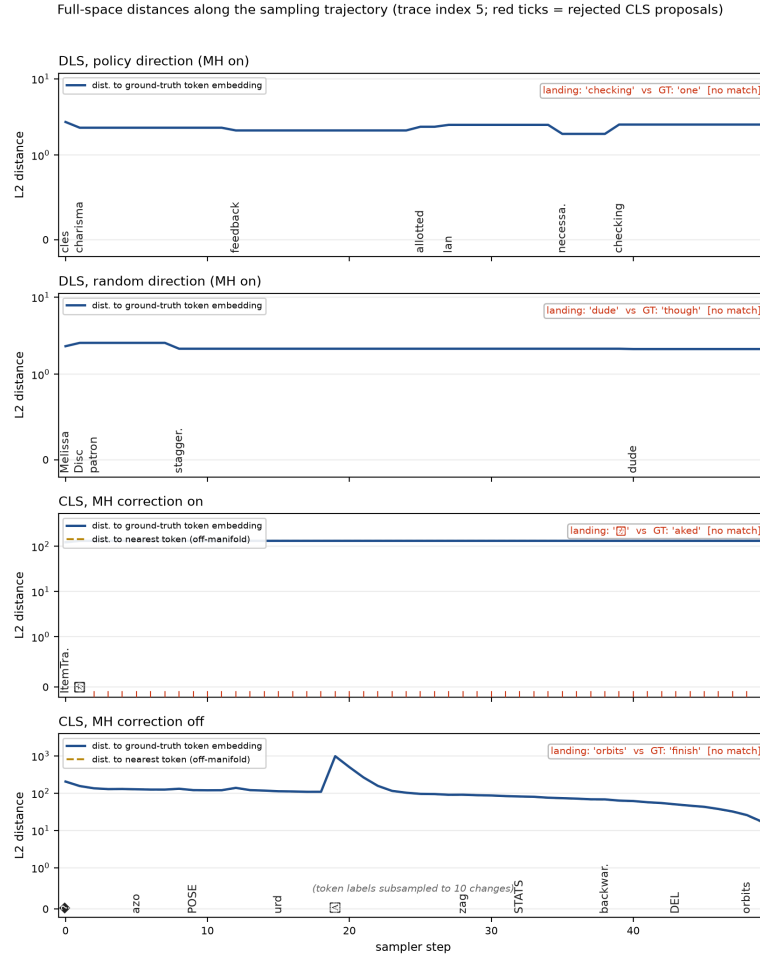


Figure 15: Full-space distances along the sampling trajectory for one seeded trace, index 5, on a symmetric-log axis, one panel per configuration. Solid line: L_2 distance to the ground-truth token embedding. Dashed line, continuous sampler only: L_2 distance to the nearest token of any kind. Red ticks mark rejected proposals. The discrete sampler’s nearest-token distance is identically zero and is therefore not drawn.

roughly 980 units off the manifold before collapsing back to about 17 units at the end. The discrete sampler shows the same dependence on the correction but without the off-manifold cost: with the correction it visits about 5 of 50 cells, and with the correction removed it visits about 49, accepting almost every proposal, yet it never leaves the token manifold, so its wandering is among real tokens rather than through the empty space between them. Against a mean nearest-neighbour token spacing of 1.82 and a mean pairwise spacing of 2.77, an off-manifold distance of 120 to 150 is a factor of roughly 65 to 83 times the spacing between neighbouring tokens, so the continuous state spends its run in essentially empty space where the projected energy is a poor guide, which is why the boundary-crossing correction almost always rejects.

Figure 16 is the secondary, illustrative view: the same trace projected onto the first two principal components of the full vocabulary embedding matrix. It is shown for intuition only, and its own caption records why it cannot be the evidence: the two components capture just 3.3% of the embedding variance, so the projection discards almost all of the geometry the distances above measure exactly. Read for what it can show, it agrees with the distances: the discrete trajectories stay inside the vocabulary cone while the continuous ones escape far outside it.

Read together, the two figures make the central geometric claim visible: the samplers either fail to move in any directed way, the discrete sampler and the free continuous sampler both wandering without bending toward the ground-truth token, or fail to move at all, the corrected continuous sampler frozen far off the manifold, and in neither case does the path converge on the target, which is the spatial signature of a gradient that carries no directional signal.

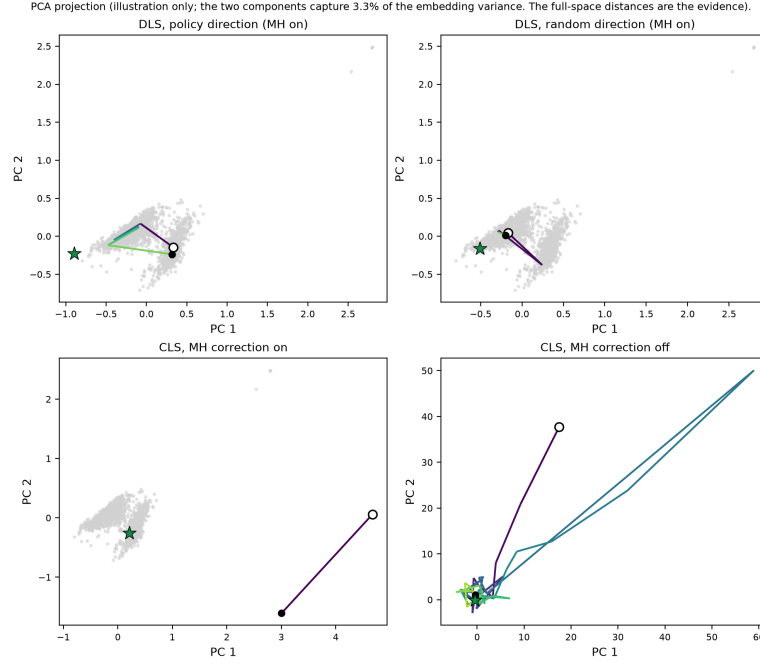


Figure 16: PCA projection of the same seeded trace, one panel per configuration. Grey cloud: the vocabulary embeddings. Open circle: the start state. Filled dot: the end state. Star: the ground-truth token. The two components capture 3.3% of the embedding variance, so this projection is illustration only and the full-space distances of Figure 15 carry the argument.

A.5 Classifier-Guided Steering: Full Results

This subsection reports an exploratory follow-up rather than part of the positive control. The control proper is complete at Section 5.14.3: the diffusion score carries the local signal the autoregressive gradient lacked, it performs the recovery natively, and it works as a proposal inside the thesis’s own exact-energy chain. What follows asks the further question of whether the constructive direction extends to *control*, which broadens the study from a sampling diagnosis into classifier-guided diffusion and evaluator alignment. It is summarized here and reported in full in Appendix A.5.

A small classifier is trained to read a noisy, partially masked state, the state the diffusion model occupies mid-denoising, by corrupting each labelled example with the model’s own forward absorbing process at a noise level drawn uniformly over the schedule. Its held-out accuracy at low noise is 81%. During generation it reweights the diffusion model’s top thirty-two candidates at each commitment by the probability of the target label, raised to a guidance strength γ . Steering is scored, exactly as in Section 5.12, by the frozen-model sentiment judge, so the classifier that guides and the judge that grades are strictly separate. Two settings were run: an off-domain continuation setting on prompts neither classifier was calibrated for, and an on-domain setting on held-out SST-2 prompts with a fluency trust region that caps guidance to candidates within $\delta = 5$ nats of the diffusion model’s own top choice. Table 20 reports the second.

Four statements summarize both settings. First, the constraint direction carries signal: the guiding classifier’s own verdict moves toward the target at every strength and for both labels, by +9.3 to +25.7 points on-domain and by seven to eighteen points off-domain, with confidence intervals excluding zero. Second, guidance steers the classifier that guides more reliably than it steers an independent judge. Third, transfer to that independent judge is bounded by how far the two instruments agree on the text being scored: agreement climbs from 56 to 64% off-domain, to 71.7% on unguided on-domain generations, to 79.7% on the held-out sentences themselves, and in each setting the judge moves in one direction only, the negative target off-domain and the positive target on-domain, with the working direction tracking that

alignment rather than a fixed reachable sentiment. Fourth, the trust region removes the fluency cost: guided span negative log-likelihood stays at or below the unguided value in three of the four on-domain cells and rises by at most 0.26 nats in the fourth, against a climb from 7.1 to 11.0 nats with guidance strength in the off-domain run.

γ	target	unguided	guided	judge gain [CI]	self gain	NLL (un→guided)
2	negative	53.0	50.7	−2.3 [−7.0, 2.3]	+18.0	7.01 → 6.27
2	positive	47.0	53.7	+6.7 [1.7, 11.7]	+9.3	7.01 → 5.30
4	negative	53.0	52.3	−0.7 [−5.3, 4.0]	+25.7	7.01 → 7.28
4	positive	47.0	54.3	+7.3 [2.7, 12.0]	+17.7	7.01 → 6.01

Table 20: On-domain, trust-region guided diffusion generation on held-out SST-2 prompts, $n = 300$ paired generations per cell, scored by the independent judge. Judge gain is the guided minus unguided target-hit percentage with a paired bootstrap 95% confidence interval; self gain is the guiding classifier’s own verdict; the last column is the span negative log-likelihood before and after guidance.

The claim the data licenses is therefore narrow and setting-contingent: with a single noisy guide at this scale, one-directional transfer to an independent judge is reachable under bounded fluency, while symmetric transfer at both labels is not. Appendix A.5 gives the off-domain table, the full agreement ladder, and the per-class confusion analysis that tests, and does not support, a by-class explanation of which direction transfers; Appendix A.7 gives representative guided pairs from both settings.

This section holds the detail of the exploratory guidance study summarized in Section 5.14.4: the off-domain run, the guide-judge agreement ladder, and the per-class confusion analysis.

A.5.1 Off-Domain Guidance Without a Trust Region

The first setting uses prompts from a MuCoLa-style continuation set that neither the guiding classifier nor the judge was trained on, and imposes no fluency constraint.

Table 21 reports it. The guidance mechanism works on the instrument it optimizes: it moves the guiding classifier’s own verdict toward the target at every strength and for both labels, by seven to eighteen points. Measured by the independent judge it steers monotonically in the negative direction, from a six-point gain at $\gamma = 1$ to a 16.7-point gain at $\gamma = 4$ whose confidence interval excludes zero, and it does not move the judge in the positive direction. The guiding classifier and the judge agree on only fifty-six to sixty-four percent of these generated texts, and stronger guidance drives the text further off the fluent manifold, with the per-token negative log-likelihood under the autoregressive model rising from 7.1 to 11.0 as γ grows. Guiding hard toward one classifier’s notion of a label therefore need not raise the other’s, which is the same off-manifold breakdown that Section 5.12 observed for the constraint-only arm.

γ , target	Judge target-hit (%)		Judge gain	Self gain	NLL/token
	unguided	guided	(pts, [CI])	(pts)	(un→guided)
1, negative	49	55	+6.0 [−1.3, 13.3]	+9.7	7.1 → 8.5
1, positive	44	39	−5.3 [−12.3, 1.7]	+11.0	7.2 → 8.5
2, negative	49	60	+10.7 [3.3, 18.0]	+7.7	7.1 → 9.7
2, positive	44	44	+0.0 [−7.0, 7.0]	+12.0	7.2 → 9.6
4, negative	49	66	+16.7 [9.7, 24.0]	+7.3	7.1 → 11.0
4, positive	44	39	−5.3 [−13.0, 2.7]	+18.3	7.2 → 11.0

Table 21: Classifier-guided diffusion generation on the off-domain continuation task, scored by the independent frozen-model judge and never by the guiding classifier. Judge gain is the guided minus unguided target-hit percentage with a paired bootstrap 95% confidence interval; self gain is the guiding classifier’s own verdict; the last column is the span negative log-likelihood before and after guidance.

A.5.2 The Guide-Judge Agreement Ladder

The off-domain result leaves two explanations entangled. The guidance might steer only in one fixed direction, or the two independently trained classifiers might disagree

on off-domain text they were never calibrated for, with stronger guidance driving the text further off the fluent manifold where the disagreement is worst. The continuation run varies domain and fluency together and so cannot separate them. The on-domain control of Section 5.14.4 holds fluency fixed and moves the instruments on-domain. Its prompts are the first three hundred held-out SST-2 validation sentences of at least twelve tokens; neither the guiding classifier nor the judge trained on this split, both having only read it under `no_grad` for an accuracy readout, so the split is calibration-clean by construction. The trust region caps guidance to candidates within $\delta = 5$ nats of the diffusion model’s top choice at each commitment, so a committed token is always within δ nats of the model’s own preference and guidance can only reshuffle mass among near-equally-fluent options. Unguided and guided generations share a seed per prompt, so each pair differs only by the intervention.

The agreement between the guiding classifier and the judge climbs a clean ladder as the text they score moves on-domain: from the off-domain band of 56 to 64%, to 71.7% (95% CI [66.7, 76.7]) on the unguided on-domain generations, to 79.7% ([75.0, 84.0]) on the real held-out sentences themselves, where the judge and the guide reach 88% and 79.7% accuracy against the gold labels. The two classifiers are therefore not miscalibrated in a fixed way; they agree more the closer the scored text sits to the distribution they were trained on, which is what the off-domain diagnosis predicted.

The working direction flips between the two settings: off-domain the judge moved on the negative target and not the positive, on-domain the reverse, with fluency no longer a confound. That flip is what identifies the residual limit. It is not a fluency artifact, because the trust region removed the fluency cost and the asymmetry remained, and it is not a fixed unreachable sentiment, because the reachable direction moved when the setting changed. What remains is an instrument-alignment effect: the two independently trained classifiers still disagree on about 28% of the on-domain generations, and steering toward one classifier’s notion of a label registers on the other only where the two align on the generated text.

		Judge verdict	
		negative	positive
Guide verdict	negative	109	35
	positive	50	106

Table 22: Per-class agreement of the guide (noisy classifier) and the independent judge (frozen GPT-2 sentiment head) on the 300 unguided on-domain generations. Guided text is excluded, so the surface is neutral with respect to the intervention. Overall agreement 71.7% [66.3, 76.7].

A.5.3 Per-Class Confusion on the Neutral Calibration Surface

One explanation of the flip is that the residual disagreement falls asymmetrically by class, so that the guide’s positive verdict transfers to the judge more reliably than its negative verdict. Table 22 tests that supposition on the unguided on-domain generations, which are neutral with respect to the intervention, and does not support it. The per-class accuracies rule out a silent label-map flip: the guide is 79.4% accurate on negative and 79.9% on positive held-out sentences, the judge 85.8% and 89.9%, both balanced. On the unguided generations the judge confirms the guide’s positive verdict 67.9% of the time and its negative verdict 75.7%, a paired difference of -7.7 points whose 95% confidence interval $[-18.0, +2.6]$ includes zero and leans, if at all, the other way. The two conditional agreement rates are statistically indistinguishable. The confusion table therefore confirms the overall alignment gap that the ladder measures, and does not identify a by-class asymmetry that would explain the transfer direction. The transfer asymmetry is reported as observed and setting-contingent, and no class-level mechanism is claimed beyond the alignment gap.

Constraint mode	Steering gain	Final KL
lm_only	0.0	8.558
cons_only	−8.0	8.308
full	−1.0	8.541
random	−7.0	8.512
cons_random	+1.0	8.528

Table 23: Steering effect of the sentiment constraint under the discrete sampler on the infilling task with a positive-sentiment target. The steering gain is the change, in percentage points, in the share of generations classified as the target sentiment relative to the unconstrained baseline.

A.6 Raw Steering Gains for the Constrained Extension

Table 23 gives the raw target-hit gains for the constrained-generation extension. They are placed here rather than in Section 5.12 because that section discounts them: every arm, including the pure-fluency and random-direction baselines, flips sign with the target label, which shows that a fixed unconditional sentiment drift dominates the raw gain. The trustworthy statistic is the paired contrast of Table 12, which cancels that shared drift.

A.7 Qualitative Examples

All qualitative material in this thesis is collected here under one selection policy, so that no example shown anywhere was chosen for how well it made a point. Every draw below is seeded with `np.random.default_rng(0)` and unfiltered: dull, crude and failing draws are kept, and the only subsequent pruning is deterministic, a fixed positional subset stated at each table and reversible from the source of this document. No number in this thesis is taken from these tables; the statistics are in Table 5, Table 15, Table 20 and Table 22. Garbled bytes that the diffusion model emits are shown as `[?]`.

A.7.1 Classifier-Guided Generations

The pairs below make the guided steering of Section 5.14.4 visible in text. Table 24 gives the on-domain, trust-region generations, two pairs per cell. Table 25 gives the highest-fluency-cost pairs from the off-domain run, which used no trust region and where guidance collapses into word salad; the on-domain pairs stay bounded. For each pair g is the guiding classifier’s verdict and j the independent judge’s, so a pair where they disagree shows the partial instrument alignment directly.

Table 24: On-domain trust-region guided generations. Seeded draws (`default_rng(0)`), unfiltered, the first two pairs of each cell. g is the guiding classifier’s verdict and j the independent judge’s; NLL is the span negative log-likelihood.

Cell: $\gamma = 2$, target negative	
Prompt	bleakly funny, its characters all the more
Unguided	bleakly funny, its characters all the more often point to negation of value: will you have a sense of humor? or will you have [g:pos, j:neg; NLL 6.57]
Guided	bleakly funny, its characters all the more so, their than the the- a and b are the moreso. characters than- the the [g:pos, j:pos; NLL 7.92]
Prompt	all that’s missing is the spontaneity
Unguided	all that’s missing is the spontaneity, and the courage to pick up undercooked stock,and also the fierce muscle of expectation that will [g:pos, j:neg; NLL 7.10]
Guided	all that’s missing is the spontaneity. When they leave the missing is the spontaneity. When they leave doing what they are supposed [g:neg, j:neg; NLL 4.44]
Cell: $\gamma = 2$, target positive	

Prompt	you don't have to know about music to
Unguided	you don't have to know about music to be musical– even if you don't know about music Audio 1 11 1 [g:neg, j:pos; NLL 5.36]
Guided	you don't have to know about music to play music you t if you don't listen to music about music music y y y n s y [g:pos, j:neg; NLL 7.10]
Prompt	the emotions are raw and will strike a nerve with
Unguided	the emotions are raw and will strike a nerve with a few steps across the dark for the firstthe bell goes ringing and this quote will be pure in [g:pos, j:pos; NLL 9.96]
Guided	the emotions are raw and will strike a nerve withthe-number-of-people-who-want-to-make-their-better- [g:pos, j:pos; NLL 5.12]

Cell: $\gamma = 4$, target negative	
Prompt	determined to be fun, and bouncy,
Unguided	determined to be fun, and bouncy, and quirky, and exciting, and colourful, and play-determined to be fun-determined [g:pos, j:pos; NLL 4.57]
Guided	determined to be fun, and bouncy, didn[?] wasd that to to to and havedd andd, wasd of to, [g:pos, j:pos; NLL 10.83]
Prompt	a psychological thriller with a genuinely spooky premise and
Unguided	a psychological thriller with a genuinely spooky premise and it's only since afterhigh" that he's in jail for full house SH NOT FOR SUPIT [g:pos, j:pos; NLL 9.64]

Guided	a psychological thriller with a genuinely spooky premise and a overall sort of the to the the[?] [?]s n's done" with [g:pos, j:pos; NLL 10.71]
--------	---

Cell: $\gamma = 4$, target positive	
---	--

Prompt	pacino is brilliant as the sleep-deprived
Unguided	pacino is brilliant as the sleep-deprived - is adding already of a sleep-deprived (cowned) bristol is brilliant as is [g:neg, j:pos; NLL 8.40]
Guided	pacino is brilliant as the sleep-deprived-pacino is the sleep-deprivedpacpacpacpacino is brilliant as the - [g:pos, j:neg; NLL 8.92]
Prompt	once (kim) begins to overplay the
Unguided	once (kim) begins to overplay the BILLJOUSE and, others, plays it entirely. is it every (kim) [g:neg, j:neg; NLL 7.05]
Guided	once (kim) begins to overplay the significance of In that regard, it's important away see the first the[?] the interested in [g:pos, j:pos; NLL 10.23]

Table 25: The three highest-NLL guided continuations from the off-domain run, which used no trust region. Compare the on-domain $\gamma = 4$ pairs in Table 24.

A.7.2 Infill Recovery on Seeded Sequences

SELECTION POLICY. The ten sequences were drawn once with a fixed rule and never filtered: `np.random.default_rng(0).choice(200,10,replace=False)` over the kl-baselines set (WikiText-2 validation, one random interior mask, `data_seed 0`), sorted. Four of those ten, the first, fourth, seventh and tenth of the sorted draw, are shown here; the remaining six are retained in the source of this document. Failures are kept in, and nothing is selected for quality. The corrupted and recovered token at the masked position is shown in bold; a check mark means the recovered token equals the ground truth. Numbers cross-reference Sections 5.5.2 and 5.14.

Table 26: Infill recovery on four of the ten seeded sequences, all methods, on the reference sequence set.

Method	Recovered token (in context)	Hit
Sequence idx 3 ground truth: 'fish' corrupted: 'moments'		
<i>original: The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starfish and polychaete worms.</i>		
<i>corrupted: The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starmoments and polychaete worms.</i>		
DLS policy (MH, GN)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star passages and polychaete worms.	
DLS random dir	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star fossils and polychaete worms.	
CLS flagship (policy, MH, GN, free, s50)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, stars pecialization and polychaete worms.	
Metropolized Gibbs	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star fish and polychaete worms.	✓
Conditional argmax	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, star fish and polychaete worms.	✓

Method	Recovered token (in context)	Hit
Conditional top-k rescore	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starfish and polychaete worms.	✓
Left-conditional (independence MH)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starfish and polychaete worms.	✓
SEDD-small recovery	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starlings and polychaete worms.	
SEDD-medium recovery	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starlings and polychaete worms.	
Hybrid (SEDD proposal, exact-energy MH)	The diet of H. gammarus mostly consists of other benthic invertebrates. These include crabs, molluscs, sea urchins, starfish and polychaete worms.	✓
Sequence idx 34 ground truth: 'resided' corrupted: 'By'		
<i>original: Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he resided in Meridian prior to 1971.</i>		
<i>corrupted: Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; heBy in Meridian prior to 1971.</i>		
DLS policy (MH, GN)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he Client in Meridian prior to 1971.	
DLS random dir	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he Client in Meridian prior to 1971.	
CLS flagship (policy, MH, GN, free, s50)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; he Although in Meridian prior to 1971.	

Method	Recovered token (in context)	Hit
Metropolized Gibbs	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; hedied in Meridian prior to 1971.	
Conditional argmax	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; hewas in Meridian prior to 1971.	
Conditional top-k rescore	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; heserved in Meridian prior to 1971.	
Left-conditional (independence MH)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; hesponsored in Meridian prior to 1971.	
SEDD-small recovery	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; helived in Meridian prior to 1971.	
SEDD-medium recovery	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; helived in Meridian prior to 1971.	
Hybrid (SEDD proposal, exact-energy MH)	Lewis McAllister, a businessman in Tuscaloosa, Alabama, was the first Republican to serve in the Mississippi House of Representatives since Reconstruction, 1962 @-@ 1968 ; heresided in Meridian prior to 1971.	✓
Sequence idx 98 ground truth: 'a' corrupted: 'it'		
<i>original: Late in the afternoon, Edson stepped onto a grenade box and addressed his exhausted troops, saying,</i>		
<i>corrupted: Late in the afternoon, Edson stepped ontoit grenade box and addressed his exhausted troops, saying,</i>		
DLS policy (MH, GN)	Late in the afternoon, Edson stepped onto 1 grenade box and addressed his exhausted troops, saying,	

Method	Recovered token (in context)	Hit
DLS random dir	Late in the afternoon, Edson stepped onto it grenade box and addressed his exhausted troops, saying,	
CLS flagship (policy, MH, GN, free, s50)	Late in the afternoon, Edson stepped onto . grenade box and addressed his exhausted troops, saying,	
Metropolized Gibbs	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	
Conditional argmax	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	
Conditional top-k rescore	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	
Left-conditional (independence MH)	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	
SEDD-small recovery	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	
SEDD-medium recovery	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	
Hybrid (SEDD proposal, exact-energy MH)	Late in the afternoon, Edson stepped onto the grenade box and addressed his exhausted troops, saying,	

Sequence idx 199 ground truth: '1990' corrupted: 'ede'

original: = = = Global security, late @-@ 1990s = = =

corrupted: = = = Global security, late @-@ **edes** = = =

DLS policy (MH, GN) = = = Global security, late @-@ **ORs** = = =

DLS random dir = = = Global security, late @-@ **os** = = =

CLS flagship (policy, MH, GN, free, s50) = = = Global security, late @-@ **[?]s** = = =

Metropolized Gibbs = = = Global security, late @-@ **s** = = =

Conditional argmax = = = Global security, late @-@ **dates** = = =

Conditional top-k rescore = = = Global security, late @-@ **'s** = = =

Left-conditional (independence MH) = = = Global security, late @-@ **techs** = = =

SEDD-small recovery = = = Global security, late @-@ **@s** = = =

SEDD-medium recovery = = = Global security, late @-@ **@s** = = =

Hybrid (SEDD proposal, exact-energy MH) = = = Global security, late @-@ **'s** = = =

A.8 Use of AI-Tools

The author of this thesis has used AI tools, specifically Google AI Studio (Gemini 3.1 Pro) and Claude (Anthropic), in the preparation of this work, and documents their use here as required.

The tools were used first for code support. When errors were encountered in the implementation of the sampling and evaluation code, they assisted in diagnosing those errors and in checking the code for compliance with the methods described in the accompanying papers. They were used in particular to help write and debug the job-scheduling and GPU-utilization scripts that distribute the experiments across the available hardware, and the diagnostic and plotting scripts. They were also used in the analysis, again as support in implementing the analysis code and resolving the errors encountered in it, and in some cases in assembling the result tables.

In the revision of the manuscript, Claude was additionally used, through an agentic coding interface, to apply a structured and author-specified set of edits to the LaTeX sources of this thesis. That use comprised matching the department template, transcribing verified numerical results from the experimental output files into the tables and the text, adding cross-references and formatting, and copy-editing prose for clarity. Every edit was made against an explicit list of changes prepared by the author, every quantitative claim so transcribed was traced to the result file that produced it and annotated with its source in the document, and the author reviewed the output of each step.

All other parts of the thesis, including the design of the experiments, the synthesis of the background and related work, the analysis and interpretation of the results, and the arguments and conclusions drawn from them, were carried out solely by the author, who has verified all quantitative claims against the underlying experimental output and assumes full responsibility for the entire content of this work.